

Covering codes, perfect codes, and codes from algebraic curves

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Covering Codes,
Perfect Codes,
and
Codes from
Algebraic Curves

G.J.M. van Wee

**COVERING CODES, PERFECT CODES,
AND
CODES FROM ALGEBRAIC CURVES**

COVERING CODES,
PERFECT CODES,
AND
CODES FROM ALGEBRAIC CURVES

PROEFSCHRIFT

TER VERKRIJGING VAN DE GRAAD VAN DOCTOR AAN DE
TECHNISCHE UNIVERSITEIT EINDHOVEN, OP GEZAG VAN
DE RECTOR MAGNIFICUS, PROF. DR. J.H. VAN LINT, VOOR
EEN COMMISSIE AANGEWEEZEN DOOR HET COLLEGE VAN
DEKANEN IN HET OPENBAAR TE VERDEDIGEN OP
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Preface

This thesis consists of an introduction and the following eight papers, which were written during the period 1986–1990:

1. G.J.M. VAN WEE, Improved sphere bounds on the covering radius of codes, *IEEE Transactions on Information Theory* **34** (1988), 237–245.
2. A.C. LOBSTEIN AND G.J.M. VAN WEE, On normal and subnormal q -ary codes, *IEEE Transactions on Information Theory* **35** (1989), 1291–1295, and **36** (1990), 1498.
3. G.J.M. VAN WEE, On the non-existence of certain perfect mixed codes, *Discrete Mathematics* **87** (1991), 323–326.
4. G.J.M. VAN WEE, More binary covering codes are normal, *IEEE Transactions on Information Theory* **36** (1990), 1466–1470.
5. G.J.M. VAN WEE, Bounds on packings and coverings by spheres in q -ary and mixed Hamming spaces, to appear in *Journal of Combinatorial Theory (A)* **56** (1991).
6. J.H. VAN LINT, JR., AND G.J.M. VAN WEE, Generalized bounds on binary/ternary mixed packing- and covering codes, to appear in *Journal of Combinatorial Theory (A)* **56** (1991).
7. G.J.M. VAN WEE, G.D. COHEN, AND S.N. LITSYN, A note on perfect multiple coverings of the Hamming space, to appear in *IEEE Transactions on Information Theory* **37** (1991).
8. R. PELLIKAN, B.Z. SHEN, AND G.J.M. VAN WEE, Which linear codes are algebraic-geometric?, to appear in *IEEE Transactions on Information Theory* **37** (1991).

As indicated, each of these papers has been accepted for publication by an international mathematical journal and has already appeared or is about to appear. I have listed the papers in the order in which they were submitted. This order is the same as the order in which the papers appear in the thesis. Each paper, as well as the *Introduction*,

has its own bibliography. I have tried to write the *Introduction* in such a way that it should leave mathematicians who are not familiar with coding theory with a fair impression of the topics dealt with in the papers, and their relation. I do hope, that, to a certain extent, the same holds for those who do not have a mathematical background at all. The *Abstract* gives a very brief outline of the main results. A Dutch translation of the *Abstract* is given at the end of the thesis.

There are several persons to whom I would like to express my gratitude. First of all I thank my supervisor, Prof. dr. J. H. van Lint, for his guidance and support. I see it as an honour to be a member of his research group. I would also like to thank Prof. dr. R. Tijdeman, who was my supervisor at the time when I was a graduating student at the University of Leiden; his advices while writing my first paper are still most valuable to me. Furthermore, my thanks go to my co-authors Antoine Lobstein, Jack van Lint, Jr., Gérard Cohen, Simon Litsyn, Ruud Pellikaan, and Shen Bazhong, for the pleasure I had working with them. Considerable debt I owe to Ruud Pellikaan, for his many advices concerning algebraic geometry. To him and all the other members of the Discrete Mathematics group I owe many thanks for the help, of any kind whatsoever, I have received during my stay among them. Special thanks go to Prof. dr. A. E. Brouwer, Prof. dr. G. D. Cohen, Prof. dr. H. C. A. van Tilborg, and, of course, my supervisor, for their constructive comments on the contents of this thesis. I should not forget Iiro Honkala and Heikki Hämäläinen: I very much appreciated communicating with them about the history of football pool systems.

The manuscripts of the third, the fifth, and the sixth paper were typed by Ms. S. P. M. Michiels-den Ouden and Ms. C. C. H. Welten-Verhulst. I am very much indebted to these two persons for their excellent work. The first four papers in this thesis are reprinted from the journals in which they appeared. I am grateful to the people at IEEE Publishing and at Elsevier Science Publishers for the nice treatment of my papers. For the support I received during my four years of research I thank my employer, the Netherlands organization for scientific research (NWO). Last but not least I thank the Eindhoven student speed skating club ISIS and the Eindhoven student dancing club Footloose for making these four years even more enjoyable.

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ABSTRACT

This thesis consists of eight papers and an introductory chapter. The topics of these papers make part of the mathematical research areas known as combinatorics and algebraic coding theory. The methods and techniques involved are mostly of a combinatorial, algebraic or algebraic-geometrical nature.

The first paper is concerned with lower bounds on the number of codewords in covering codes. We use counting arguments to establish a significant and rather general improvement on the sphere covering bound, which is a well-known lower bound on $K(n, R)$, the minimal number of codewords in a binary code of length n with covering radius R . As a corollary, we obtain a new infinite sequence of exactly determined values of $K(n, R)$:

$$K(2^r, 1) = 2^{(2^r - r)} \quad \text{for } r = 1, 2, \dots$$

Related to this result we state a conjecture concerning perfect codes.

In the second paper we generalize the concepts of a normal and a subnormal code to codes over arbitrary alphabets. We investigate which q -ary codes are normal or subnormal. We prove that a q -ary perfect code is normal if and only if it is binary or trivial. We derive a general lower bound on the number of codewords in a ternary subnormal code.

Partly based on the techniques introduced in the first paper a nonexistence theorem concerning mixed perfect codes is proved in the third paper.

In the fourth paper we solve in the case $R = 1$ the conjecture by Cohen, Lobstein and Sloane that among the codes of length n with covering radius R having $K(n, R)$ codewords there is a normal code. In fact, we prove that *all* such codes are normal in the case $R = 1$. We also prove that every binary code with minimal distance at least twice its covering radius is normal, and we generalize Frankl's construction of abnormal binary codes with covering radius 1 to arbitrary covering radii.

In the fifth and the sixth paper we investigate how to extend the results of the first paper to codes over arbitrary alphabets (q -ary or mixed). Moreover, we give useful modifications of the basic idea, and we consider applications to error-correcting codes

The seventh paper is about so-called perfect multiple coverings (PMC's). These are codes which are perfect in the sense that for fixed numbers r and μ every word is within Hamming distance r from exactly μ codewords. This generalizes the concept of an (ordinary) perfect code, which we get if $\mu = 1$. We give a short analysis of the existence of PMC's, where we mostly restrict ourselves to the case $r = 1$.

Finally, in the eighth paper we investigate which linear codes can be obtained from algebraic curves over finite fields using Goppa's construction. It turns out that this question makes sense only if we impose a certain condition on the degree of the divisor on the curve concerned, since otherwise *all* linear codes can be constructed this way. Imposing this condition, we prove that the only q -ary Hamming codes that are algebraic-geometric in this sense are those with redundancy one or two and the binary $[7,4,3]$ code. Moreover, in the case of the binary $[7,4,3]$ code we prove that the construction, that is to say, the triple (curve, points, divisor), is unique, in a certain sense.

Introduction

Consider the following problem. Suppose we have n football matches, which have not been played yet. Each match has three possible outcomes: the home team wins, loses, or the match ends in a draw. So, in total, there are 3^n possible outcomes. Hence there are also 3^n possible forecasts. Here we mean by a forecast a list of n predicted outcomes, one outcome for each match. Now suppose that people are invited to submit forecasts, as many as they like. For each forecast each participant has to pay a fixed amount of money. In exchange, after the matches have been played, part of the money brought together this way is returned to those participants who had sent in a forecast with only few errors. The amount of money returned for such a forecast depends on the number of errors, and on the total number of forecasts with the same number of errors. This game, which is played in many countries of the world, is called the *football pool* game. Now the question is the following: For given n , what is the minimal number K such that a collection of forecasts exists with the property that, whatever the outcomes of the matches turn out to be, there will always be at least one forecast in this collection with at most one error?

The problem just described is a famous combinatorial problem. It is known as the *football pool problem*. It is a so-called *covering* problem. The adjective “covering” will become clear later, if it is not already. Let us consider the case of the Dutch national football pool. Nowadays this football pool consists of 12 matches, but for a long time this number was equal to 13. There is quite a difference between the cases $n = 12$ and $n = 13$. In the case $n = 12$, at present, we only know that $21329 \leq K \leq 29889$, where the lower bound is taken from Corollary 15 in the fifth paper of this thesis, and the upper bound is from [63]. In the case $n = 13$ the problem has been completely

solved: $K = 59049$; a collection of 59049 forecasts with the desired property exists, but such a collection with less than 59049 elements does not. This remarkable fact is well-known among coding theorists, but it might surprise others. It will be explained in the sequel.

We can state the football pool problem in a more formal and more general way. Let Q be an alphabet consisting of q symbols, where $q \geq 2$. For instance, $Q = \mathbf{Z}_q$, the cyclic group of integers modulo q , or, if q is a prime power, $Q = \mathbf{F}_q$, the finite field with q elements. Let $\mathcal{H} = Q^n$, i.e., the set of n -tuples of elements of Q . The elements of $\mathcal{H}$ are called *words*. Define the metric d on $\mathcal{H}$ as follows: for $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{H}$, $\mathbf{y} = (y_1, \dots, y_n) \in \mathcal{H}$,

$$d(\mathbf{x}, \mathbf{y}) = |\{i | x_i \neq y_i\}|.$$

In words, the distance between two words is the number of positions (coordinates) where they differ. This metric is really a metric, that is to say, it satisfies

- (i) $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$,
- (ii) $d(\mathbf{x}, \mathbf{y}) \geq 0$,
- (iii) $d(\mathbf{x}, \mathbf{y}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{y}$,
- (iv) $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$,

for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{H}$. The metric d is known as the *Hamming distance*, while the metric space $\mathcal{H}$ is called a *Hamming space*. A *code* is any nonempty subset C of $\mathcal{H}$, and its elements are called *codewords*. More in particular, C is called a *block code*, since all codewords have the same number of coordinates. Since we shall not meet any other type of code in this thesis, we shall always leave out the word “block”. Because $|Q| = q$, we say that C is a q -ary code (*binary* if $q = 2$, *ternary* if $q = 3$, etc.). We call n the *length* of C . In what follows, a few other basic concepts from coding theory will be introduced. However, for a real introduction to coding theory the reader is referred to [122] or [109].

The *covering radius* $\text{CR}(C)$ of a code C is defined by

$$\text{CR}(C) = \max\{d(\mathbf{x}, C) | \mathbf{x} \in \mathcal{H}\},$$

where $d(\mathbf{x}, C) = \min\{d(\mathbf{x}, \mathbf{c}) | \mathbf{c} \in C\}$. We define

$$K_q(n, R) := \min\{|C| \mid \text{a } q\text{-ary code } C \text{ of length } n \\ \text{with } \text{CR}(C) = R \text{ exists}\}.$$

Our problem is: Determine $K_q(n, R)$ for all q, n , and R . It is not hard to see that the football pool problem is a restricted case of this problem; namely, when we choose $q = 3$ and $R = 1$. The number $K_3(n, 1)$ is the same as the number K mentioned before. The problem of determining $K_q(n, R)$ is a generalization of this problem, where we can interpret q as the number of possible outcomes of each match, n as the number of matches, and where we want to find the minimal size of a collection of forecasts with the property that, whatever the outcomes of the matches are, there is always at least one forecast in this collection with at most R errors. We interpret a codeword as a forecast, and a collection of forecasts as a code.

Let $\mathbf{x} \in \mathcal{H}$. We define the *sphere* $B_r(\mathbf{x})$ with *radius* r and *center* $\mathbf{x}$ by

$$B_r(\mathbf{x}) = \{\mathbf{y} \in \mathcal{H} | d(\mathbf{x}, \mathbf{y}) \leq r\}.$$

So, equivalently, we could have defined $K_q(n, R)$ to be the minimal number of spheres with radius R needed to 'cover' $\mathcal{H}$:

$$K_q(n, R) = \min\{|C| \mid \bigcup_{\mathbf{c} \in C} B_R(\mathbf{c}) = \mathcal{H}\}.$$

Note that

$$|B_r(\mathbf{c})| = \sum_{i=0}^r \binom{n}{i} (q-1)^i,$$

for every $\mathbf{c}$. This immediately gives us a lower bound on $K_q(n, R)$:

$$K_q(n, R) \geq \frac{q^n}{\sum_{i=0}^R \binom{n}{i} (q-1)^i}. \quad (1)$$

This bound is known as the *sphere covering bound*.

The numbers q, n , and $\text{CR}(C)$ are called *parameters* of C . A fourth parameter is the *minimum distance* $d(C)$. It is defined only if $|C| > 1$:

$$d(C) = \min\{d(\mathbf{x}, \mathbf{y}) \mid \mathbf{x}, \mathbf{y} \in C \text{ and } \mathbf{x} \neq \mathbf{y}\}.$$

We also have the *packing radius*, defined for *any* code by

$$\text{PR}(C) = \max\{e \in \{0, 1, \dots, n\} \mid B_e(\mathbf{b}) \cap B_e(\mathbf{c}) = \emptyset, \\ \text{for all } \mathbf{b}, \mathbf{c} \in C \text{ with } \mathbf{b} \neq \mathbf{c}\}.$$

One easily verifies that

$$\text{PR}(C) \leq \text{CR}(C), \quad (2)$$

for any code C , and that

$$\text{PR}(C) = \left\lfloor \frac{d(C) - 1}{2} \right\rfloor,$$

if $|C| > 1$.

A code with packing radius e is usually called *e-error-correcting*. The reason for this is the following. Suppose that C is a code which is used to transmit information through a noisy channel. Roughly speaking, this means that a *sender* sends (arbitrary) codewords of C through the channel. During transmission errors can occur: If $\mathbf{c}$ is sent by the sender, a possibly different word $\mathbf{y}$ is received by the *receiver* at the other end of the channel. Each time the receiver receives a word, it determines (correctly or incorrectly) which codeword was transmitted. When it receives a word $\mathbf{y}$, it uses the following ‘algorithm’ to output a codeword $\mathbf{c}'$:

Choose a $\mathbf{c}' \in C$ such that $d(\mathbf{y}, C) = d(\mathbf{y}, \mathbf{c}')$.

This decoding strategy is called *nearest neighbour decoding*. Provided all codewords are equally likely to be transmitted, and provided certain assumptions on the channel hold, this strategy is best possible in the sense that it minimizes the probability that the receiver makes a mistake, and is called *maximum likelihood decoding*. Now if $\text{PR}(C) = e$, and if during the transmission of a codeword at most e errors occur (i.e., $d(\mathbf{c}, \mathbf{y}) \leq e$, in the preceding), then the receiver decodes the received signal correctly (i.e., $\mathbf{c}' = \mathbf{c}$).

A code C has $\text{CR}(C) \leq R$ if and only if

$$\bigcup_{\mathbf{c} \in C} B_R(\mathbf{c}) = \mathcal{H}, \quad (3)$$

i.e., if and only if the spheres with radius R centered at the codewords of C cover the whole space. A code C with covering radius R is called *perfect* if, moreover, the spheres $B_R(\mathbf{c})$ in the left-hand side of (3) are disjoint. In other words, a perfect code corresponds to a partition of $\mathcal{H}$ into spheres with radius R . We often speak of a *perfect R -code*. A

code C is perfect if and only if equality holds in (2). Also, for given q, n , and R , a perfect code exists if and only if equality holds in (1).

Trivial examples of perfect codes, for arbitrary q , are the perfect codes with $R = 0$ or $R = n$, corresponding to $C = \mathcal{H}$ and $C = \{\mathbf{c}\}$, for some $\mathbf{c}$, respectively. We shall now give infinitely many nontrivial examples of perfect codes in the case that q is a prime power. Let $Q = \mathbf{F}_q = \{\alpha_0, \alpha_1, \dots, \alpha_{q-1}\}$, where $\alpha_0 = 0$. Define the maps $\pi : \mathbf{F}_q^{(q-1)n} \rightarrow \mathbf{F}_q$ and $T : \mathbf{F}_q^{(q-1)n} \rightarrow \mathbf{F}_q^n$ by

$$\pi(\mathbf{v}) = \sum_{i=1}^{q-1} \alpha_i \sum_{j=1}^n v_{(i-1)n+j}, \quad \text{and}$$

$$T(\mathbf{v}) = (T_1(\mathbf{v}), T_2(\mathbf{v}), \dots, T_n(\mathbf{v})),$$

where

$$T_j(\mathbf{v}) = \sum_{i=1}^{q-1} v_{(i-1)n+j} \quad (1 \leq j \leq n),$$

and $\mathbf{v} = (v_1, v_2, \dots, v_{(q-1)n}) \in \mathbf{F}_q^{(q-1)n}$. For any code $C \subseteq \mathbf{F}_q^n$, define the code $C' \subseteq \mathbf{F}_q^{qn+1}$ by

$$C' := \{(\mathbf{v}|\mathbf{c} + T(\mathbf{v})|\pi(\mathbf{v})) \in \mathbf{F}_q^{qn+1} \mid \mathbf{v} \in \mathbf{F}_q^{(q-1)n} \text{ and } \mathbf{c} \in C\}.$$

We have $|C'| = q^{(q-1)n}|C|$. This construction is due to Vasil'ev [176]. Now define, inductively, the infinite sequence $\{H(r, q)\}_{r=1}^{\infty}$ of codes by

$$\begin{cases} H(1, q) = \{(0)\}, \\ H(r, q) = H(r-1, q)', \quad \text{for } r > 1. \end{cases} \quad (4)$$

Each code $H(r, q)$ is a perfect 1-code of length $(q^r - 1)/(q - 1)$. For the number of codewords we have

$$|H(r, q)| = q^{(q^r - 1)/(q - 1) - r}.$$

The codes $\{H(r, q)\}_{r=1}^{\infty}$ are very well-known, and there are several other ways to construct them than the one described here. The code $H(r, q)$ is generally referred to as the q -ary *Hamming code of redundancy r* , named after R. W. Hamming [64], who discovered these codes in 1950. The existence of these perfect codes implies that

$$K_q\left(\frac{q^r - 1}{q - 1}, 1\right) = q^{(q^r - 1)/(q - 1) - r}, \quad \text{for } r = 1, 2, \dots, \quad (5)$$

if q is a prime power.

Other well-known perfect codes are the *binary Golay code* and the *ternary Golay code*, corresponding to

$$\begin{aligned} K_2(23, 3) &= 2^{11} = 2048, & \text{and} \\ K_3(11, 2) &= 3^6 = 729, \end{aligned}$$

respectively. These codes were discovered by M. J. E. Golay [56] in 1949. If $q = 2$, the *repetition* codes of odd length are perfect codes with covering radius $(n - 1)/2$. These codes consist of two codewords, one with all coordinates equal to 0, one with all coordinates equal to 1.

Perfect codes have received a lot of attention, as the reader can see from this introduction's bibliography. For a survey of the work that has been done, see [116]. In the early seventies van Lint [110] and Tietäväinen [170] showed that, if q is a prime power, the parameters q, n , and R of the perfect codes previously mentioned are the only parameters for which a perfect code exists. In the case that q is not a prime power, it is known that no other perfect codes than the trivial perfect codes (those with $R = 0$ or $R = n$) exist if $R > 2$, see the work by Reuvers [141], Laakso [102], Best [14], [16], and Hong [77].

The determination of $K_q(n, R)$ is a hard combinatorial problem. As for our knowledge of $K_q(n, R)$, the state of the art is that, except for the cases in which a perfect code exists, and some other special cases, only rough lower and upper bounds on $K_q(n, R)$ are known. There are very few cases in which the exact value of $K_q(n, R)$ has been determined. In the first, the second, and the fifth paper of this thesis new lower bounds on $K_q(n, R)$ are obtained. In the first paper we restrict our attention to the case $q = 2$ (and we find new exact values of $K_2(n, R)$). In the fifth paper we consider generalizations of these results to arbitrary q . We pay special attention to the case $q = 3, R = 1$, i.e., the football pool problem. In the second paper we use linear programming in the case $q = 3$.

Now that we have mentioned the football pool problem again, note that

$$K_3((3^r - 1)/2, 1) = 3^{(3^r - 1)/2 - r}, \quad \text{for } r = 1, 2, \dots,$$

by (5). This sequence begins with

$$\begin{aligned} K_3(1, 1) &= 3^0 = 1, \\ K_3(4, 1) &= 3^2 = 9, \\ K_3(13, 1) &= 3^{10} = 59049, \\ &\vdots \end{aligned}$$

This explains the “remarkable” fact that we mentioned at the beginning: It is caused by the existence of a perfect 1-code for $n = 13$.

Again consider a football pool of n matches, but now suppose that for each match in the football pool we exclude one of the three outcomes. That is to say, for each match there is an outcome which we assume to be very unlikely, impossible, actually. For instance, if the home team is in the top of the league and its opponent is somewhere at the bottom, then we may think that the probability that the home team loses is very small. In this setting we can ask the question: What is the minimal number of forecasts that we need to make sure that, whatever the outcomes of the matches are, there is always at least one forecast among them with at most one error, *provided the assumptions made prove to be true*. This number is $K_2(n, 1)$. This indicates that the case $q = 2$ is also important for football pools. By (5) we have $K_2(7, 1) = 16$. A general upper bound on $K_q(n, R)$ is

$$K_q(n + 1, R) \leq qK_q(n, R). \quad (6)$$

The proof is trivial: Let $Q = \{\alpha_0, \alpha_1, \dots, \alpha_{q-1}\}$. Let $C \subseteq Q^n$ be a code with covering radius R . Write down below each other q copies of a list of all the codewords of C . Increase the length of all these $q|C|$ words by one by placing α_0 behind the words in the first copy, α_1 behind the words in the second copy, and so on. The list of words obtained this way constitutes a code of length $n + 1$ with covering radius R . This proves (6). It follows that

$$K_2(8, 1) \leq 2K_2(7, 1) = 32.$$

In fact, it is known that

$$K_2(8, 1) = 32.$$

This was proved in 1969 by Stanton and Kalbfleisch [161], [162], and it took them quite some trouble. In the first paper of this thesis we

prove the following lower bound on $K_2(n, R)$:

$$K_2(n, R) \geq \frac{(n - R + \varepsilon)2^n}{(n - R) \sum_{i=0}^R \binom{n}{i} + \varepsilon \sum_{i=0}^{R-1} \binom{n}{i}}, \quad (7)$$

where

$$\varepsilon = (R + 1) \left\lceil \frac{n + 1}{R + 1} \right\rceil - n - 1.$$

For $R = 1$ this bound reads

$$K_2(n, 1) \geq \begin{cases} 2^n/n, & \text{if } n \text{ is even,} \\ 2^n/(n + 1), & \text{if } n \text{ is odd.} \end{cases} \quad (8)$$

With (5) and (6), this implies that

$$K_2(2^r, 1) = 2^{(2^r - r)}, \quad \text{for } r = 1, 2, \dots \quad (9)$$

The only values of r for which this was already known are 1, 2, and 3. The sequence begins with

$$\begin{aligned} K_2(2, 1) &= 2^1 = 2, \\ K_2(4, 1) &= 2^2 = 4, \\ K_2(8, 1) &= 2^5 = 32, \\ K_2(16, 1) &= 2^{12} = 4096, \\ K_2(32, 1) &= 2^{27} = 134217728, \\ &\vdots \end{aligned}$$

Our proof of (8) also gives an alternative proof of the result by Stanton and Kalbfleisch.

We can interpret (9) in the following way: If $q = 2$ and $R = 1$, and if for a certain length n a perfect code exists, then for length $n + 1$ one cannot do better than the trivial construction (6). We conjecture that this statement is true in general, i.e., for *every* q and R ; see Section VII of the first paper. For $q = 2$ the conjecture holds if and only if

$$K_2(24, 3) = 8192.$$

This has not been solved yet. By (6) and the existence of the binary Golay code, we have

$$K_2(24, 3) \leq 8192.$$

The sphere covering bound (1) gives

$$K_2(24, 3) \geq 7217,$$

but by our bound (7) this can be improved to

$$K_2(24, 3) \geq 8098.$$

So we're close!

We have seen that, by excluding one of the three outcomes of each match, binary codes can be useful for football pools. However, usually not *each* match has the property that at least one of the three possible outcomes is very unlikely, but only a few. We are led to the so-called *mixed codes* (more in particular, “*binary/ternary*” mixed codes): For certain matches we exclude one of the three possible outcomes, for others we don't. Suppose there are b matches of the first category, and t of the second. In terms of codes, we are dealing with a new type of Hamming space: $\mathcal{H} = \mathbf{F}_3^t \mathbf{F}_2^b$. In total, there are $n = t + b$ coordinates: t ternary and b binary. We can define the Hamming distance d on $\mathcal{H}$ in the same way as we did before (and again $\mathcal{H}$ becomes a metric space). The same holds for all the other basic concepts that we introduced, like the covering radius and the packing radius, for instance. A function similar to $K_q(n, R)$ can be defined, but, of course, we have to choose a different notation. We choose $K(t, b, R)$. Lower bounds on $K(t, b, R)$ are discussed in Section V of the fifth paper, and in the sixth paper.

Remember that the covering problem is related to the existence of perfect codes. But what about perfect codes in $\mathcal{H} = \mathbf{F}_3^t \mathbf{F}_2^b$? In the third paper we show that in $\mathcal{H}$ no other perfect codes exist than the trivial ones, for any covering radius R . This result is a corollary of a more general nonexistence theorem. For the general theorem we do not only consider $\mathbf{F}_3^t \mathbf{F}_2^b$, but we allow $\mathcal{H}$ to be *any* mixed Hamming space. That is to say,

$$\mathcal{H} = Q_1 \times Q_2 \times \cdots \times Q_n, \quad (10)$$

where the Q_i are alphabets consisting of q_i symbols. In the third paper we prove that, for a nontrivial perfect code with covering radius R in $\mathcal{H}$ to exist, it is necessary that

$$q_i - q_j \equiv 0 \pmod{R+1}, \quad \text{for all } i \text{ and } j.$$

Earlier results on perfect mixed codes were obtained by Schönheim [146], Herzog and Schönheim [74], [75], Lenstra [105], Heden [66], [67], Lindström [106], and Reuvers [141].

Except for the case $\mathcal{H} = \mathbf{F}_3^t \mathbf{F}_2^b$, not much is known about good covering codes in general mixed Hamming spaces $\mathcal{H}$ as in (10). Nevertheless, these codes, perfect or not, can be of great interest, as was recently shown by Östergård [134]. From a good covering code in $\mathbf{F}_4 \mathbf{F}_2^7$ he obtained, by carrying out certain substitutions, a binary code of length 10 with covering radius 1 and 120 codewords, a new record holder! The mixed code had 60 codewords and covering radius 1, and was found by simulated annealing, a computer method which has been successfully adopted many times already in the search for good covering codes, see [1], [48], [101], and [103]. Östergård's result implies that $K_2(10, 1) \leq 120$, whereas the best lower bound known is $K_2(10, 1) \geq 103$, by (8).

In this introduction we described a covering problem in coding theory and we have seen that this problem can be formulated for various different Hamming spaces $\mathcal{H}$. However, if we go back into history, at the time of its first appearance in scientific literature and when the first results were found, this problem was formulated in a completely different context and in a much more restricted sense. In the mathematical literature, the problem was mentioned for the first time in a paper by Taussky and Todd [163], in terms of abelian groups. We quote from [163]:

"The formulation of the following problems about finite abelian groups was suggested by the consideration of certain industrial problems.

Let G be an abelian group with n base elements

$$g_1, g_2, \dots, g_n,$$

each of order p (not necessarily a prime). Let S denote the set of the $\nu = n(p-1) + 1$ distinct powers of the base elements. Let AB , where A and B are two subsets of G , denote the set of all products ab where $a \in A, b \in B$.

Problem 1. Determine the minimal integer $\sigma = \sigma(n, p)$ for which there exists at least one set H , consisting of σ elements of G , such that $G = HS$.

Problem 2. Determine the cases when H can be a subgroup of G .

Specially interesting cases appear to be those in which the set HS contains all the elements of G exactly once."

After this, the paper continues with a discussion of the case $n = 4, p = 3$, and the case $n = 7, p = 2$. The authors conclude their paper with a list of values of the function σ that they were able to compute, which, in the terminology of this thesis, reads

$$\begin{array}{ll} K_2(2, 1) = 2, & K_3(2, 1) = 3, \\ K_2(3, 1) = 3, & K_3(3, 1) = 5, \\ K_2(4, 1) = 4, & K_3(4, 1) = 9, \\ K_2(5, 1) = 7, & \\ K_2(7, 1) = 16, & \end{array}$$

(where $K_2(3, 1) = 3$ must be a misprint, since $K_2(3, 1) = 2$, by (5)), and the remark that "the general case seems very difficult to handle." Indeed, more than forty years later we know how right they were.

Since the early work by Taussky and Todd, the topic has been discussed in various papers, by many different persons. We shall not give a survey here, but the bibliography at the end of this introduction should give the reader some impression. Over the years, the problem has been reformulated and generalized many times. It is only a development of the last decade that the covering radius of codes has become a hot topic in coding theory. Key papers in this development are the papers by Graham and Sloane [62], Cohen, Karpovsky, Mattson, Jr., and Schatz [34], and Cohen, Lobstein, and Sloane [35]. The interpretation in terms of codes gives rise to a problem which slightly differs from the covering problem: For a given (class of) code(s), determine the covering radius. For instance, the covering radii of Reed-Muller and BCH codes have received a lot of attention, see [13], [19], [49], [130], [133], [136], [143], [145], [154], and [7], [68], [69], [151], [166], [168], [178], respectively.

However, in this thesis this problem is not our primary interest. The main issue of this thesis is the *minimal* cardinality of a code of given length n and given covering radius R . If we are discussing q -ary codes with q a prime power, then it is possible to impose the additional condition that the code must be *linear*, that is to say, it is

a linear subspace of the vector space $\mathbf{F}_q^n$. The first paper also contains some results for this problem.

Although the paper by Taussky and Todd was the first scientific publication on the subject, the covering problem in terms of football pools is much older. In Sweden the football pool game started in 1934. Soon thereafter, the first issues of Swedish magazines about football pools appeared. The same thing happened in Finland in 1940. Certain *systems*, as the codes are often called in the football pool terminology, date back to 1934. These systems probably have Swedish constructors. The binary Hamming code $H(3, 2)$ of length 7 with 16 codewords was known already in 1936! In Finland, the ternary Golay code was already constructed as a football pool system in the winter of 1946/1947 by Juhani Virtakallio, see [81]. This is more than a year earlier than Golay's discovery of this code, see [56] and [165]. Virtakallio published his system in 1947 in the Finnish magazine on football pools *Veikkaaja*. Although we only mention the developments in Finland and Sweden here, it is very well possible that in other countries football pool systems have existed even earlier.

The search for good football pool systems in Finland and Sweden has continued ever since the football pool game started. There even exist special magazines and booklets on football pools in these countries. Two important (weekly) magazines on football pools in Finland and Sweden are *Veikkaus-Lotto* (the old name is *Veikkaaja*), which we mentioned earlier, and *Vi Tippa*, respectively. Some of the systems that were constructed over the years were published in these magazines, or somewhere else, some were not. Many of these systems were invented by people without a mathematical background. The inventor is sometimes even unknown. Most of these results have never found their way to mathematical journals. However, very recently, the best systems known at present have been collected in a paper by Hämäläinen and Rankinen [63], together with several results of their own. All these systems are binary, ternary, or binary/ternary mixed codes. In [63] an updated table of upper bounds on $K(t, b, R)$ is given for $t + b \leq 13$, $R \leq 3$ (including the cases $t = 0$ and $b = 0$).

A new method to construct good binary linear covering codes was introduced by Graham and Sloane [62] in 1986. They showed how two binary linear codes A and B , of lengths n_A and n_B , dimensions k_A and k_B , and covering radii R_A and R_B , respectively, can be combined and result in a new code $A \oplus B$, of length $n_A + n_B - 1$, dimension

$k_A + k_B - 1$, and, provided A and B both satisfy a certain condition—see later—, with covering radius at most $R_A + R_B$. Graham and Sloane called the code $A \oplus B$ the *Amalgamated Direct Sum* (ADS) of A and B . To explain the condition on A and B , we define, for any code C in $\mathbb{F}_2^n$,

$$N(C) := \min_i \{ \max_{\mathbf{x} \in \mathbb{F}_2^n} \{ d(\mathbf{x}, C_0^{(i)}) + d(\mathbf{x}, C_1^{(i)}) \} \},$$

where

$$C_\alpha^{(i)} = \{ (c_1, c_2, \dots, c_n) \in C \mid c_i = \alpha \},$$

for $\alpha \in \mathbb{F}_2$, and where

$$d(\mathbf{x}, \emptyset) := n,$$

by convention. The integer $N(C)$ is called the *norm* of C (Graham and Sloane used a slightly different definition). If

$$N(C) \leq 2CR(C) + 1, \quad (11)$$

then C is called *normal*. Otherwise C is called *abnormal*. The condition on A and B is that these codes must be normal. This condition appears to be a very weak condition. Up to the present no abnormal binary linear code has been discovered. The ADS-construction has proved to be quite successful. Actually, it is one of the best general constructions of covering codes known.

The ADS and the concept of a normal code were extended to non-linear binary codes by Cohen, Lobstein and Sloane [35]. It was shown by Frankl that, in contrast with the linear case, nonlinear abnormal binary codes do exist. His construction of abnormal codes was published in [100]. Frankl constructed abnormal codes with covering radius 1. In the fourth paper we give a generalization of his construction to arbitrary covering radii. These examples are not good covering codes. In fact, among the good binary covering codes no abnormal codes have been discovered yet. On the one hand, this leads to an interesting conjecture, stated in [35]:

$$\begin{aligned} &\text{For all } n \text{ and } R, \text{ among the binary codes of} \\ &\text{length } n \text{ with covering radius } R \text{ and } K_2(n, R) \\ &\text{codewords, there is a normal code.} \end{aligned} \quad (12)$$

On the other hand, the following interesting theoretical problem arises: For a given length and covering radius, find the smallest abnormal covering code. For this see [100] and [82]. The main result of the

fourth paper in this thesis is a proof of conjecture (12) for $R = 1$. A further result in this paper is a proof that all binary codes C with $|C| > 1$ and minimum distance $d(C) \geq 2CR(C)$ are normal. The question "Which codes are normal?" has been discussed in several other papers than the ones already mentioned, see [80], [84], [87], [91], and [99], for example.

A new concept, that of a *subnormal* code, was introduced by Honkala [79]. Subnormal codes are binary codes that satisfy a condition that is slightly weaker than (11). For the precise definition the reader is referred to the second paper of this thesis. Codes which are not subnormal are called *absubnormal*. In [80], by proving that the abnormal codes of Frankl are subnormal, Honkala showed that the class of subnormal codes is strictly larger than the class of normal codes. At present, it is not known whether an absubnormal binary code exists. In [80] it was also shown that the condition on the codes A and B in the ADS construction is actually stronger than strictly necessary. By defining a slightly modified ADS, Honkala showed that it is sufficient that one code is normal and the other subnormal.

So far we have only considered normal and subnormal *binary* codes. The main goal of the second paper is to investigate what happens when we try to generalize to arbitrary q . It turns out that, as opposed to the binary case, normal and subnormal codes are not at all 'normal' if $q > 2$. That is, for $q > 2$ many good q -ary covering codes are abnormal, even absubnormal. This is expressed in particular by the following result, proved in the second paper:

$$\begin{aligned} &\text{A } q\text{-ary perfect code is normal if and only if} \\ &\text{it is binary or trivial} \end{aligned} \tag{13}$$

This immediately disproves the q -ary analogue of (12) for $q > 2$. Our proof of (13) is based on a computation of the norm of an arbitrary q -ary perfect code. In fact, (sub)normal codes tend to be *bad* covering codes. This inspired us to derive a general lower bound on the cardinality of subnormal ternary codes with covering radius 1:

$$|C| \geq 3^n/2n, \quad \text{if } CR(C) = 1 \text{ and } n > 3.$$

For a general Hamming space $\mathcal{H}$, we can reformulate the definition of a perfect code as follows. A code $C \in \mathcal{H}$ is perfect if and only if an $r \in \{0, 1, \dots, n\}$ exists, such that

$$|B_r(\mathbf{x}) \cap C| = 1, \quad \text{for all } \mathbf{x} \in \mathcal{H}.$$

In this case we say that C is a *perfect r -covering* of $\mathcal{H}$. In the seventh paper we consider a generalization: A code C is called a *μ -fold perfect r -covering* of $\mathcal{H}$ if

$$|B_r(\mathbf{x}) \cap C| = \mu, \quad \text{for all } \mathbf{x} \in \mathcal{H}.$$

We speak of *perfect multiple coverings* (PMC's). We restrict our attention to q -ary codes. A natural question to ask is:

$$\text{For which } q, n, r, \text{ and } \mu \text{ does a } \mu\text{-fold perfect } r\text{-covering of } Q^n \text{ exist?} \quad (14)$$

In our paper we give some general constructions of PMC's and we thoroughly analyze (14) in the case $r = 1$. We give a complete solution if q is a prime, and a necessary condition if q is a prime power. In our analysis, the sequence (4) of the q -ary Hamming codes is a basic ingredient. Earlier results on PMC's were obtained only by Clayton [29], [28], as far as we know. A lot of work remains to be done in this area.

There is one paper in this thesis that we have not mentioned yet. This is the eighth paper, the title of which suggests that it has no connection to the other papers. However, the Hamming codes, which we referred to so frequently in this introduction, and which play an important role in many of the other papers, will make their reappearance. As the title indicates, in this paper we use tools from algebraic geometry. Although a brief introduction to the construction of codes from algebraic curves is included in the paper itself, this paper will be difficult to read for those who do not have any knowledge of algebraic geometry. It is not feasible to include a quick course at this point. However, a nice introduction into the subject is provided by [117], and by [61]. For more on algebraic geometry, but not on codes, see [27], [52], [65], and [88].

The idea to construct codes from algebraic curves dates back to 1977 and is due to V. D. Goppa. In [59] Goppa described a relation between algebraic curves and linear codes which would have tremendous impact on coding theory, and which would motivate the study of various new problems in algebraic geometry. Further study of this relation was done by Goppa in [60] and [58]. Coding theory has had ample benefit of this relation. One of the most exciting results in coding theory is based on it. In 1982 Tsfasman, Vlăduț, and Zink gave a sequence of codes over $\mathbb{F}_q$, with q a square, exceeding the Gilbert-Varshamov

bound if $q \geq 49$. They obtained these codes from Shimura curves. Their result came as a great surprise to coding theorists, among whom it was commonly believed that the Gilbert-Varshamov bound was best possible.

The developments during the last decade suggest that algebraic geometry is perhaps more than only a tool for coding theorists. It is very well conceivable that the relation is so strong, that we actually should regard the theory of linear codes as being part of the theory of algebraic curves, or varieties, perhaps. This is emphasized by Theorem 2 of our paper. Goppa's construction involves a nonsingular algebraic curve $\mathcal{X}$ of a certain genus g , defined over $\mathbb{F}_q$, a divisor G on $\mathcal{X}$, and a set D of n $\mathbb{F}_q$ -rational points on $\mathcal{X}$. Our theorem states that, if we do not impose any condition on the degree of the divisor G , then for every linear code C there exists a triple $(\mathcal{X}, D, G)$ such that Goppa's construction yields C .

Unfortunately, Goppa's construction has the property that, when no conditions on the degree of G are imposed, in general nothing can be said about the dimension and the minimum distance of C . This changes if we demand that

$$\deg(G) > 2g - 2,$$

in the case that differentials are used, or, more or less equivalently, that

$$\deg(G) < n,$$

if rational functions are used. In our paper we call codes which can be constructed under this restriction *algebraic-geometric (AG)*. The other linear codes are called *weakly algebraic-geometric (WAG)*. With this extra condition, the situation becomes completely different. In Section IV of the paper we give criteria for codes to be AG. There are various examples of codes which are not AG. For instance, the binary Golay code is not. For the *ternary* Golay code we do not know whether the same holds. Another question is: Which q -ary Hamming codes are AG? In our paper we give a complete answer. The code $H(r, q)$ is AG only in the cases when $r \leq 2$ or $(r, q) = (3, 2)$. In these cases a suitable triple $(\mathcal{X}, D, G)$ is explicitly given. For $(r, q) = (3, 2)$ this required quite some effort. We also discuss the uniqueness of the triple $(\mathcal{X}, D, G)$ in these cases.

An urgent question that arises is: What about the covering radius of algebraic-geometric codes? Except for a lower bound found by Janwa

[91], based on the so-called Supercode Lemma, almost nothing is known about this. Janwa's bound only applies to AG codes (in fact, only to a certain subclass of the class of AG codes). So our paper can be helpful when applying this bound. Furthermore, if algebraic geometry is really so much related to the theory of linear codes, interpreting certain linear codes as codes from algebraic curves may help to find the covering radii of these codes. Again the results of our paper could be useful then. All this is reading tea-leaves, but it is beyond all doubt that the relevance of algebraic geometry for coding theory is growing steadily.

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Improved Sphere Bounds on the Covering Radius of Codes

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Abstract—The sphere bound is a trivial lower bound on $K(n, R)$, the minimal cardinality of any binary code of length n and with covering radius R . By simple arguments it is considerably improved, e.g., $K(n, 1) \geq 2^n/n$ for n even. A table of lower and upper bounds on $K(n, R)$ for $n \leq 33$, $R \leq 10$ is included.

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I. INTRODUCTION

A RECENT paper of Cohen, Lobstein, and Sloane [4] (see also [3] and [12]) deals with $K(n, R)$, the minimal number of codewords in any binary code, not necessarily linear, of length n and with covering radius R . $K(n, R)$ is closely related to $t[n, k]$, the minimal covering radius of any binary *linear* code of length n and dimension k . The latter has been extensively studied by Cohen, Karpovsky, Mattson, and Schatz [2], and by Graham and Sloane [7]. The sphere bound is a trivial lower bound on $K(n, R)$ and $t[n, k]$. In the papers mentioned this bound has hardly been improved.

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In the present paper new lower bounds on $K(n, R)$ are obtained that improve on the sphere bound for most pairs of n and R . The table of $K(n, R)$ ($n \leq 23$, $R \leq 4$) presented in [4] is considerably improved and extended using these bounds. The new bounds also yield some improvements on the table of $t[n, k]$ ($n \leq 64$) that appeared in [7]. Although we concentrate on binary codes, the fundamental ideas of this paper apply to codes over arbitrary alphabets.

The problem of determining $K(n, R)$, not only in the binary case, is called the *covering problem*. Several papers have preceded [4] in the study of the problem: [5], [6], [8]–[10], [13], [15], [16], [19]–[24], and [26]–[28]. Actually, the problem was posed by Taussky and Todd [23] in 1948 (less generally and in a different context). The results that have been established up to the present mainly apply to either ternary codes with covering radius 1 or binary codes. In the former case the problem is known as the *football pool problem*.

Summary of Results

Section II gives the definitions and properties that are basic in this paper. Section III contains a brief survey of known bounds on $K(n, R)$. It is a remarkable fact that several upper bounds in the table of $K(n, R)$ in [4] can be improved simply by carrying out constructions already known: direct sums or amalgamated direct sums of codes.

In Section IV we derive the improved sphere bounds (Theorems 9 and 10). As a special case, we shall find that

$$K(n, 1) \geq 2^n/n, \quad n \text{ even.}$$

This leads to a new sequence of exactly determined values

$$K(2^r, 1) = 2^{(2^r-r)}, \quad r = 1, 2, \dots, \quad (1)$$

partially solving open problem 3 in [2]. Stanton and Kalbfleisch [21], [22] gave a very complicated proof of $K(8, 1) = 32$. For their proof they needed computer assis-

tance. However, $K(8, 1) = 32$ is only a special case of (1), and we shall see that the proof of Theorem 9, of which (1) is an immediate consequence, is surprisingly simple.

A table of lower and upper bounds on $K(n, R)$, for $n \leq 33$, $R \leq 10$, appears in Section V (Table I). It is an improved and extended version of the one given in [4]. Sections VI and VII deal with some consequences of the new bounds. In Section VI, for $n \leq 33$, we list all improvements on the table of $t[n, k]$ in [7] (Theorem 11). In Section VII, the validity of

$$K(n+2, R+1) \leq K(n, R) \quad (n \neq R),$$

conjectured in [4], is discussed in the case $R=1$ (Theorem 12). Section VII concludes with a conjecture concerning perfect codes, binary as well as nonbinary. A list of symbols is given in the Nomenclature at the end of the paper.

II. BASIC CONCEPTS

In this section, let $n \in \mathbb{N}$, $k, R \in \{0, 1, \dots, n\}$, and $x \in \mathbb{F}_2^n$. We define

$$S_R(x) = \{y \in \mathbb{F}_2^n \mid d(x, y) = R\},$$

$$B_R(x) = \{y \in \mathbb{F}_2^n \mid d(x, y) \leq R\},$$

and

$$V_2(n, R) = \sum_{i=0}^R \binom{n}{i}$$

(cf. [11, pp. 32, 55]). Hence

$$B_R(x) = \bigcup_{i=0}^R S_i(x), \quad (2)$$

$$|B_R(x)| = V_2(n, R).$$

$B_R(x)$ is called the *sphere* with *radius* R and *center* x .

For a code $\mathcal{C} \subseteq \mathbb{F}_2^n$, the *covering radius* $\text{CR}(\mathcal{C})$ is defined by

$$\text{CR}(\mathcal{C}) = \max \{ d(x, \mathcal{C}) \mid x \in \mathbb{F}_2^n \}.$$

An $(n, M)R$ code is a binary code of length n , with M codewords and covering radius R . An $[n, k]R$ code is a binary *linear* code of length n and dimension k and with covering radius R . $K(n, R)$ and $t[n, k]$ are defined by

$$K(n, R) = \min \{ M \mid \text{an } (n, M)R \text{ code exists} \},$$

$$t[n, k] = \min \{ t \mid \text{an } [n, k]t \text{ code exists} \}$$

(cf. [2], [4], [7]). The following properties are obvious:

$$\text{if } t[n, k] \leq R, \text{ then } K(n, R) \leq 2^k, \quad (3)$$

$$K(n, R+1) \leq K(n, R) \quad (R \neq n), \quad (4)$$

$$K(n+1, R) \geq K(n, R). \quad (5)$$

III. SURVEY OF KNOWN RESULTS

We list all bounds on $K(n, R)$ already known. In Section V we shall use these bounds to build up Table I.

A. Trivialities

Theorem 1 (Sphere Bound): For all $n \in \mathbb{N}$, $R \in \{0, 1, \dots, n\}$,

$$K(n, R) \geq \frac{2^n}{V_2(n, R)}. \quad (6)$$

Equality holds in (6) if and only if a perfect R -error-correcting code of length n exists. For example,

$$K(2^r - 1, 1) = 2^{(2^r - r - 1)}, \quad r = 1, 2, \dots \quad (7)$$

Theorem 2:

- a) $K(R, R) = 1, \quad R = 1, 2, \dots;$
- b) $K(n, R) = 2, \quad \text{for all } n, R \in \mathbb{N}$
with $R + 1 \leq n \leq 2R + 1;$
- c) $K(n, 0) = 2^n, \quad \text{for all } n \in \mathbb{N}.$

The next theorem is a consequence of the direct sum (DS) construction (see [14, p. 76]).

Theorem 3: For all $n_1, n_2 \in \mathbb{N}$, $R_1 \in \{0, 1, \dots, n_1\}$, $R_2 \in \{0, 1, \dots, n_2\}$,

$$K(n_1 + n_2, R_1 + R_2) \leq K(n_1, R_1)K(n_2, R_2). \quad (8)$$

As a special case of (8), we have the following important rule:

$$K(n+1, R) \leq 2 \cdot K(n, R), \quad (9)$$

for all $n \in \mathbb{N}$, $R \in \{0, 1, \dots, n\}$. The DS of two codes $\mathcal{C}_1$ and $\mathcal{C}_2$ is denoted by $\mathcal{C}_1 \oplus \mathcal{C}_2$.

B. Results of Earlier Work

Theorems 4–8 summarize known results. $A(n, d)$ denotes the maximal cardinality of any binary code of length n and with minimum distance d (cf. [11, p. 54] and [14, p. 523]). A table of $A(n, d)$, for $n \leq 23$, $d \leq 9$, is given in [14, p. 674]. The improvements on this table made by Best [1], Romanov [18], and Tietäväinen [25], do not affect $K(n, R)$.

Theorem 4: Let $n, R \in \mathbb{N}$, $n \geq 2R + 1$.

$$\text{a) } K(n, R) \geq \frac{2^n - A(n, 2R + 1) \binom{2R}{R}}{V_2(n, R) - \binom{2R}{R}};$$

$$\text{b) } K(n, R) \geq \frac{2^n - 2 \cdot A(n, 2R + 1) \binom{2R}{R}}{V_2(n, R) - \frac{3}{2} \binom{2R}{R}}.$$

Theorem 5: For $R = 1, 2, \dots$,

$$\text{a) } K(2R + 2, R) = 4,$$

$$\text{b) } K(2R + 3, R) = 7.$$

Theorem 6: $K(6, 1) = 12$, $K(8, 1) = 32$.

Theorem 7: $K(11,1) \leq 192$, $K(8,2) \geq 9$, $K(9,2) \geq 13$, $K(10,3) \geq 8$.

Theorem 8: For all $n \in \mathbb{N}$,

$$K(2n+1,1) \leq 2^n \cdot K(n,1).$$

Theorem 6 was proved by Stanton and Kalbfleisch [21], [22]. An alternative proof of $K(6,1) \leq 12$ appeared in [4]. Theorem 8 is from a construction found by Mollard, Katsman, and Litsyn (see [4] and [17]). Theorems 4, 5, and 7 were obtained by Cohen *et al.* [4]. $K(7,1) = 7$ was already established in [21], [23].

The bounds given so far, together with the bounds obtained in the next section, are not sufficient to explain Table I (Section V) completely. One other result is needed, a construction known as the amalgamated direct sum (ADS) of two normal codes. The concept of a normal *linear* code was introduced by Graham and Sloane [7]. They also defined the ADS of *linear* codes, a variation on the DS. All this was extended to *nonlinear* codes in [4]. To obtain the upper bounds in Table I, all possible ADS and DS constructions have been checked by computer, thereby improving several upper bounds in the table of $K(n, R)$ in [4]. Details are given in Section V. The ADS of two codes $\mathcal{C}_1$ and $\mathcal{C}_2$ is denoted by $\mathcal{C}_1 \oplus \mathcal{C}_2$. The reader who is not familiar with normal codes and the ADS construction is referred to [4].

IV. IMPROVED SPHERE BOUNDS

In this section the sphere bound (6) will be improved. This is done by estimating, for every code $\mathcal{C}$ having covering radius R , the number of words in $\mathbb{F}_2^n$ that are covered by the spheres with radius R around the code-words of $\mathcal{C}$ more than once. To avoid messy proofs, a counting formalism is introduced first (Definitions 1 and 2 and Lemmas 1–4). We omit the proofs of Lemma 1 and Lemma 2, which follow immediately from Definition 1.

Definition 1: Let $n \in \mathbb{N}$, $R \in \{0, 1, \dots, n\}$, $V \subseteq \mathbb{F}_2^n$, $\mathcal{C} \subseteq \mathbb{F}_2^n$. The quantity $E_{\mathcal{C}}^R(V) \in \mathbb{N} \cup \{0\}$ is defined by

$$E_{\mathcal{C}}^R(V) = \sum_{c \in \mathcal{C}} |B_R(c) \cap V| - \left| \bigcup_{c \in \mathcal{C}} B_R(c) \cap V \right|.$$

Furthermore, if $\mathcal{C} \neq \emptyset$ (i.e., $\mathcal{C}$ is a code), we define

$$E_{\mathcal{C}}(V) = E_{\mathcal{C}}^{\text{CR}(\mathcal{C})}(V).$$

$E_{\mathcal{C}}(V)$ is called the *excess* on V by $\mathcal{C}$.

Lemma 1: Let $n \in \mathbb{N}$, $R \in \{0, 1, \dots, n\}$, and let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code with $\text{CR}(\mathcal{C}) = R$.

a)
$$E_{\mathcal{C}}(\mathbb{F}_2^n) = |\mathcal{C}|V_2(n, R) - 2^n.$$

b) If $\{V_i\}$ is a finite collection of pairwise disjoint subsets of $\mathbb{F}_2^n$, then

$$E_{\mathcal{C}}\left(\bigcup_i V_i\right) = \sum_i E_{\mathcal{C}}(V_i).$$

c) If $V, W \subseteq \mathbb{F}_2^n$ and $V \subseteq W$, then

$$E_{\mathcal{C}}(V) \leq E_{\mathcal{C}}(W).$$

Lemma 2: Let $n \in \mathbb{N}$, $R \in \{0, 1, \dots, n\}$, $V \subseteq \mathbb{F}_2^n$, and $\mathcal{C}, \mathcal{C}' \subseteq \mathbb{F}_2^n$.

$$\text{If } \mathcal{C}' \subseteq \mathcal{C}, \text{ then } E_{\mathcal{C}'}^R(V) \leq E_{\mathcal{C}}^R(V).$$

Definition 2: Let $n \in \mathbb{N}$ and let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code. We define

$$\begin{aligned} Z_i(\mathcal{C}) &= \{z \in \mathbb{F}_2^n \mid E_{\mathcal{C}}(\{z\}) = i\}, \quad i = 0, 1, \dots, \\ Z(\mathcal{C}) &= \bigcup_{i > 0} Z_i(\mathcal{C}) = \{z \in \mathbb{F}_2^n \mid E_{\mathcal{C}}(\{z\}) > 0\}. \end{aligned}$$

If it is clear from the context which code is meant, we simply write Z_i, Z , respectively.

Lemma 3: Let $n \in \mathbb{N}$, $R \in \{0, 1, \dots, n\}$, $z \in \mathbb{F}_2^n$, and let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code with $\text{CR}(\mathcal{C}) = R$.

a) For $i = 0, 1, \dots$,

$$z \in Z_i \Leftrightarrow |B_R(z) \cap \mathcal{C}| = i + 1.$$

$$\text{b)} \quad z \in Z \Leftrightarrow |B_R(z) \cap \mathcal{C}| > 1.$$

Proof: Starting with a),

$$\begin{aligned} z \in Z_i &\Leftrightarrow \sum_{c \in \mathcal{C}} |B_R(c) \cap \{z\}| - 1 = i \\ &\Leftrightarrow |B_R(z) \cap \mathcal{C}| = i + 1. \end{aligned}$$

Then b) follows from a).

Lemma 4: Let $n \in \mathbb{N}$, $R \in \{0, 1, \dots, n\}$, and let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code with $\text{CR}(\mathcal{C}) = R$.

$$\begin{aligned} \text{a)} \quad E_{\mathcal{C}}(V) &= \sum_i i |Z_i \cap V|; \\ \text{b)} \quad |\mathcal{C}| V_2(n, R) - 2^n &= \sum_i i |Z_i|. \end{aligned}$$

Proof: a) We use Lemma 1b) and deduce

$$\begin{aligned} E_{\mathcal{C}}(V) &= E_{\mathcal{C}}\left(\bigcup_i Z_i \cap V\right) = \sum_i E_{\mathcal{C}}(Z_i \cap V) \\ &= \sum_i \sum_{z \in Z_i \cap V} E_{\mathcal{C}}(\{z\}) = \sum_i \sum_{z \in Z_i \cap V} i \\ &= \sum_i i |Z_i \cap V|. \end{aligned}$$

Then b) follows from a), with $V = \mathbb{F}_2^n$, and Lemma 1a).

Lemma 5 is used often in the rest of this section. Moreover, it is useful when counting the number of words in the intersection of spheres. The proof is trivial.

Lemma 5: Let $n \in \mathbb{N}$ and $x, y, z \in \mathbb{F}_2^n$. Then

$$d(x, z) \equiv d(x, y) + d(y, z) \pmod{2}.$$

The statements in Lemmas 6 and 7 are obvious by intuition. Their proofs, however, are not trivial. The proof of Lemma 7 is given in the Appendix. We omit the proof of Lemma 6, which is similar and simpler.

Lemma 6: Let $n \in \mathbb{N}$, $s, t \in \{0, 1, \dots, n\}$, and $x, y, x', y' \in \mathbb{F}_2^n$. Suppose that $d(x', y') \leq d(x, y)$. Then

$$|B_s(x') \cap B_t(y')| \geq |B_s(x) \cap B_t(y)|.$$

Lemma 7: Let $n \in \mathbb{N}$, $r, s, t \in \{0, 1, \dots, n\}$, $w, x, y, w', x', y' \in \mathbb{F}_2^n$. If $d(w', x') = d(w, x)$, $d(w', y') = d(w, y)$, and $d(x', y') \leq d(x, y)$, then

$$\begin{aligned} \text{a)} \quad & |S_r(w') \cap B_s(x') \cap B_t(y')| \\ & \geq |S_r(w) \cap B_s(x) \cap B_t(y)|, \\ \text{b)} \quad & |B_r(w') \cap B_s(x') \cap B_t(y')| \\ & \geq |B_r(w) \cap B_s(x) \cap B_t(y)|. \end{aligned}$$

All preparations have been made now to derive the new bounds (Theorems 9 and 10). The basic idea that leads to Theorem 9 is the observation that, for most n and R , many spheres with radius 1 have to contain words that are covered more than once.

Lemma 8: Let $n, R \in \mathbb{N}$, $n \geq R$, $x \in \mathbb{F}_2^n$, and let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code with $\text{CR}(\mathcal{C}) = R$. If $d(x, \mathcal{C}) = R$, then

$$E_{\mathcal{C}}(B_1(x)) \geq (R+1) \left\lceil \frac{n+1}{R+1} \right\rceil - n - 1. \quad (10)$$

Proof: Let $c \in \mathcal{C}$. If $d(x, c) > R+1$, then $|B_R(c) \cap B_1(x)| = 0$. If $d(x, c) \in \{R, R+1\}$, then $|B_R(c) \cap B_1(x)| = R+1$. Hence from Definition 1,

$$E_{\mathcal{C}}(B_1(x)) \equiv -|B_1(x)| = -n-1 \pmod{R+1},$$

and (10) follows (since $E_{\mathcal{C}}(B_1(x))$ is a nonnegative number).

Theorem 9: For all $n, R \in \mathbb{N}$ with $n > R$, we have

$$K(n, R) \geq \frac{(n-R+\epsilon)2^n}{(n-R)V_2(n, R) + \epsilon \cdot V_2(n, R-1)}, \quad (11)$$

where

$$\epsilon = (R+1) \left\lceil \frac{n+1}{R+1} \right\rceil - n - 1.$$

Proof: Let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code with $\text{CR}(\mathcal{C}) = R$. Put

$$A = \{a \in \mathbb{F}_2^n \mid d(a, \mathcal{C}) = R\}.$$

Of course, we have

$$|A| \geq 2^n - |\mathcal{C}|V_2(n, R-1). \quad (12)$$

From Lemma 4a) and Lemma 8, we obtain

$$\sum_i |Z_i \cap B_1(a)| \geq \epsilon, \quad \text{for all } a \in A. \quad (13)$$

Let $z \in Z$. We estimate $|A \cap B_1(z)|$. Lemma 3b) states that $|B_R(z) \cap \mathcal{C}| \geq 2$. Therefore, codewords $c, d \in \mathcal{C}$ exist such that $c \neq d$, $d(z, c) \leq R$, and $d(z, d) \leq R$. Words in $\mathbb{F}_2^n$ that are within distance less than R from c or d cannot belong to A . Hence

$$|A \cap B_1(z)| \leq n + 1 - \lambda,$$

where $\lambda = |B_1(z) \cap (B_{R-1}(c) \cup B_{R-1}(d))|$. We show that $\lambda \geq R + 1$. If $R = 1$, this is trivial. If $R \geq 2$, $B_{R-1}(c)$ and $B_{R-1}(d)$ need not be disjoint. Put $d_1 = d(z, c)$ and $d_2 = d(z, d)$. One of the following four cases holds (without loss of generality (w.l.o.g.)): 1) $d_1 = d_2 = R$, 2) $d_1 = R - 1, d_2 = R$, 3) $d_1 = d_2 = R - 1$, 4) $d_1 \leq R - 2, d_2 \leq R$. Lemma 5 and Lemma 7b) imply that we may assume that $d(c, d) = 2, 1, 2$ in cases 1), 2), and 3), respectively. By calculating λ in these three configurations of z, c , and d , we find $\lambda \geq 2R - (R - 1) = R + 1$ (in all three cases). Finally, in case 4) we have $B_1(z) \subseteq B_{R-1}(c)$, so $\lambda \geq |B_1(z)| = n + 1 \geq R + 1$. We conclude that

$$|A \cap B_1(z)| \leq n - R, \quad \text{for all } z \in Z. \quad (14)$$

Using (12)–(14) and Lemma 4b), we find

$$\epsilon(2^n - |\mathcal{C}|V_2(n, R-1)) \leq \epsilon|A|$$

$$\begin{aligned}
&\leq \sum_{a \in A} \sum_i i |Z_i \cap B_1(a)| \\
&= \sum_i i \sum_{a \in A} |Z_i \cap B_1(a)| \\
&= \sum_i i \sum_{z \in Z_i} |A \cap B_1(z)| \\
&\leq \sum_i i(n-R) |Z_i| \\
&= (n-R)(|\mathcal{C}|V_2(n, R) - 2^n).
\end{aligned}$$

This yields

$$|\mathcal{C}| \geq \frac{(n-R+\epsilon)2^n}{(n-R)V_2(n, R) + \epsilon \cdot V_2(n, R-1)}.$$

The bound (11) improves on the sphere bound (6) whenever $\epsilon > 0$. This is the case if and only if $n \not\equiv -1$ (modulo $R+1$). The new bound is quite good. $K(9, 2) \geq 13$ (see Theorem 7) was obtained in [4] by linear programming. However, (11) yields $K(9, 2) \geq 14$. A striking improvement is $K(24, 3) \geq 8098$, for the sphere bound only yields $K(24, 3) \geq 7217$, and we know that $K(24, 3) \leq 8192$ (see Table I). We conjecture that $K(24, 3) = 8192$ (see the Conjecture in Section VII).

In the case $R=1$, Theorem 9 establishes a remarkably sharp result.

Corollary 1:

- a) $K(n, 1) \geq 2^n/n$, if n is even; (15)
- b) $K(2^r, 1) = 2^{(2^r-r)}$, $r = 1, 2, \dots$

Proof: a) Take $R=1$ in Theorem 9. b) From a) we have

$$K(2^r, 1) \geq 2^{(2^r-r)},$$

and from (9) and (7)

$$K(2^r, 1) \leq 2 \cdot K(2^r - 1, 1) = 2^{(2^r-r)}.$$

It should be mentioned here that a very short proof of the particular bound (15) can be given, based on the idea of Lemma 8. Corollary 1b) was already known for $r = 1$ (trivial, see Theorem 2b)), $r = 2$ (sphere bound), and $r = 3$ (Stanton and Kalbfleisch [21], [22], see Theorem 6). As remarked in the Introduction, our proof in the case $r = 3$ is much simpler.

Instead of estimating excess on spheres with radius 1, we shall now take spheres with radius 2. This leads to the bound in Theorem 10, which fills up some of the gaps $n \equiv -1$ (modulo $R + 1$) of the previous bound. The proof is similar but a little more complicated. In Lemma 9 and Theorem 10 we use the conventions

$$\begin{aligned} \binom{n}{k} &= 0, & \text{if } n < k, \\ S_k(x) &= B_k(x) = \emptyset, & \text{if } k < 0, \end{aligned}$$

and

$$V_2(n, k) = 0, \quad \text{if } k < 0.$$

Lemma 9: Let $n, R \in \mathbb{N}$, $n \geq 2R$, $x \in \mathbb{F}_2^n$, and let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code with $\text{CR}(\mathcal{C}) = R$. If $d(x, \mathcal{C}) \geq R - 1$, then

$$E_{\mathcal{C}}(B_2(x)) \geq \binom{R+2}{2} \left\lfloor \frac{\binom{n-R+1}{2}}{\binom{R+2}{2}} \right\rfloor - \binom{n-R+1}{2}. \quad (16)$$

Proof: There are two possibilities: A) $x \notin Z$, and B) $x \in Z$.

Case A: $x \notin Z$. Let $c \in \mathcal{C}$. If $d(x, c) > R + 2$, then $|B_R(c) \cap B_2(x)| = 0$. If $d(x, c) \in \{R + 1, R + 2\}$, then $|B_R(c) \cap B_2(x)| = \binom{R+2}{2}$. Finally, if $d(x, c) \in \{R - 1, R\}$, then $|B_R(c) \cap B_2(x)| = 1 + (n - R + 1)R + \binom{R}{2}$. Since $x \notin Z$, there is exactly one codeword $d \in \mathcal{C}$ with $d(x, d) \in \{R - 1, R\}$, and all other codewords are at distance greater

than R from x . Hence from Definition 1,

$$E_{\mathcal{C}}(B_2(x)) \equiv 1 + (n - R + 1)R + \binom{R}{2} - V_2(n, 2) \pmod{\binom{R+2}{2}}. \quad (17)$$

Work out the right-hand side of (17) to obtain

$$\begin{aligned} E_{\mathcal{C}}(B_2(x)) &\equiv -\frac{1}{2}(n - R + 1)(n - R) \\ &= -\binom{n - R + 1}{2} \pmod{\binom{R+2}{2}}, \end{aligned}$$

and (16) follows.

Case B: $x \in Z$. Then codewords $c, d \in \mathcal{C}$ exist such that $c \neq d$, $d(z, c) \leq R$, and $d(z, d) \leq R$. Put $V = B_2(x) \cap B_R(c) \cap B_R(d)$. We show that $|V| \geq 1 + R^2$. Put $d_1 = d(z, c)$, $d_2 = d(z, d)$. Since $d(x, \mathcal{C}) \geq R - 1$, one of the following three cases holds (w.l.o.g.): 1) $d_1 = d_2 = R$, 2) $d_1 = R - 1$, $d_2 = R$, or 3) $d_1 = d_2 = R - 1$. Lemmas 5 and 7b) imply that we may assume that $d(c, d) = 2R, 2R - 1, 2R - 2$ in cases 1), 2), and 3), respectively. By calculating $|V|$ in these three configurations of z, c , and d , we find $|V| \geq 1 + R^2$ in cases 1) and 2), and $|V| \geq 1 + n + (R - 1)^2 \geq 1 + R^2$ in case 3). Now from Lemma 1c), Lemma 2, and Definition 1, we obtain

$$E_{\mathcal{C}}(B_2(x)) \geq E_{\{c, d\}}^R(V) = 2|V| - |V| = |V| \geq 1 + R^2.$$

Observe that

$$1 + R^2 \geq \binom{R+2}{2} - 1 \geq \binom{R+2}{2} \left[\frac{\binom{n - R + 1}{2}}{\binom{R+2}{2}} \right] - \binom{n - R + 1}{2}.$$

Thus we see that (16) holds in case B as well.

Theorem 10: For all $n, R \in \mathbb{N}$ with $n \geq 2R$, we have

$$K(n, R) \geq \frac{\left(V_2(n, 2) - \frac{1}{2}(R+2)(R-1) + \epsilon \right) 2^n}{\left(V_2(n, 2) - \frac{1}{2}(R+2)(R-1) \right) V_2(n, R) + \epsilon \cdot V_2(n, R-2)},$$

where

$$\epsilon = \binom{R+2}{2} \left\lfloor \frac{\binom{n-R+1}{2}}{\binom{R+2}{2}} \right\rfloor - \binom{n-R+1}{2}.$$

Proof: Let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be a code with $\text{CR}(\mathcal{C}) = R$. Put

$$A = \{a \in \mathbb{F}_2^n \mid d(a, \mathcal{C}) \geq R-1\}.$$

Consequently,

$$|A| \geq 2^n - |\mathcal{C}| V_2(n, R-2). \quad (18)$$

From Lemmas 4a) and 9, we obtain

$$\sum_i |Z_i \cap B_2(a)| \geq \epsilon, \quad \text{for all } a \in A. \quad (19)$$

Let $z \in Z$. We estimate $|A \cap B_2(z)|$. Lemma 3b) states that codewords $c, d \in \mathcal{C}$ exist such that $c \neq d$, $d(z, c) \leq R$, and $d(z, d) \leq R$. Put $\mu = |B_2(z) \cap (B_{R-2}(c) \cup B_{R-2}(d))|$. Then

$$|A \cap B_2(z)| \leq V_2(n, 2) - \mu.$$

We show that $\mu \geq (1/2)(R+2)(R-1)$. If $R=1$ or $R=2$, this is trivial. Now suppose that $R \geq 3$. Put $d_1 = d(z, c)$ and $d_2 = d(z, d)$. One of the following four cases holds

(w.l.o.g.): 1) $d_1 = d_2 = R$, 2) $d_1 = R - 1$, $d_2 = R$, 3) $d_1 = d_2 = R - 1$, or 4) $d_1 \leq R - 2$, $d_2 \leq R$. Lemmas 5 and 7b) imply that we may assume that $d(c, d) = 2, 1, 2$ in cases 1), 2), and 3), respectively. By calculating μ in these three configurations of z , c , and d , we find $\mu \geq 2 \binom{R}{2} - \binom{R-1}{2} = (1/2)(R+2)(R-1)$ (in all three cases). Finally, in case 4) we apply Lemma 6 and find

$$\begin{aligned} \mu &\geq |B_2(z) \cap B_{R-2}(c)| \geq 1 + (n - R + 3)(R - 2) + \binom{R-2}{2} \\ &\geq 1 + (R + 3)(R - 2) + \binom{R-2}{2} \geq (1/2)(R + 2)(R - 1). \end{aligned}$$

We conclude that

$$|A \cap B_2(z)| \leq V_2(n, 2) - \frac{1}{2}(R + 2)(R - 1) \quad (20)$$

for all $z \in Z$.

In the same way as in the proof of Theorem 9, we obtain

from (18)–(20) and Lemma 4b),

$$\begin{aligned} &\epsilon(2^n - |\mathcal{C}|V_2(n, R - 2)) \\ &\leq \epsilon|A| \\ &\leq \sum_{a \in A} \sum_i i |Z_i \cap B_2(a)| \\ &= \sum_i i \sum_{a \in A} |Z_i \cap B_2(a)| \\ &= \sum_i i \sum_{z \in Z_i} |A \cap B_2(z)| \\ &\leq \sum_i i \left(V_2(n, 2) - \frac{1}{2}(R + 2)(R - 1) \right) |Z_i| \\ &= \left(V_2(n, 2) - \frac{1}{2}(R + 2)(R - 1) \right) (|\mathcal{C}|V_2(n, R) - 2^n). \end{aligned}$$

This yields

$$|\mathcal{C}| \geq \frac{\left(V_2(n, 2) - \frac{1}{2}(R+2)(R-1) + \epsilon \right) 2^n}{\left(V_2(n, 2) - \frac{1}{2}(R+2)(R-1) \right) V_2(n, R) + \epsilon \cdot V_2(n, R-2)}.$$

It does not seem very likely that taking spheres with radius greater than 2 will yield drastic improvements on Table I. Moreover, the estimation of excess on a sphere with radius r , as done in Lemma 8 ($r=1$) and Lemma 9 ($r=2$), rapidly becomes more complicated as r increases. The case $R=1$ of Theorem 10 is given in the next corollary.

Corollary 2: Let $n \in \mathbb{N}$. If $n \equiv 2$ (modulo 3), then

$$K(n, 1) \geq \frac{(V_2(n, 2) + 2)2^n}{V_2(n, 2)(n+1)}.$$

V. TABLE OF $K(n, R)$

Table I gives the best lower and upper bounds on $K(n, R)$ known at present, for $n \leq 33$, $R \leq 10$.

Key to Table I

This key indicates the simplest proof of a given result, not necessarily the earliest. Here, the sequences

unmarked, a, b, c, d, e, f, g, h

and

unmarked, p, q, r, s, t

of lower and upper bounds, respectively, are thought of as being of decreasing simplicity. An entry such as $K(18, 3) = e \ 280 - 768 \ q$ indicates that $280 \leq K(18, 3) \leq 768$, and that $K(18, 3) \geq 280$ and $K(18, 3) \leq 768$ are established by the bounds corresponding to e and q , respectively. The bounds in Theorem 4 (marked c and d) have been applied only

for $R=1$, $n \leq 19$, and for $2 \leq R \leq 4$, $n \leq 23$, since in the table of $A(n, d)$ in [14] upper bounds on $A(n, 2R+1)$ are given only for these values of n and R .

Lower bounds:

unmarked: (5) or Theorem 2.

a: Sphere bound (Theorem 1).

b: Cohen *et al.* [4] (Theorem 5).

c: Cohen *et al.* [4] (Theorem 4a)).

d: Cohen *et al.* [4] (Theorem 4b)).

e: Theorem 9.

f: Theorem 10.

g: Cohen *et al.* [4], by linear programming (Theorem 7).

h: Stanton and Kalbfleisch [21], by linear programming (Theorem 6).

Upper bounds: All upper bounds are constructive.

unmarked: A linear code from [7]. For example, from the table of $t[n, k]$ in [7] we have $t[27, 9] \leq 7$. Using (3), we obtain $K(27, 7) \leq 2^9 = 512$. Sometimes (4) has been used too.

p: A DS construction (9).

q: An ADS construction (see Section III-B). Every code entered into the table has been checked to be normal (for *linear* codes this had already been done by Graham and Sloane [7]). All codes marked *q* in the table can be constructed by taking one term in the ADS equal to one of the following codes: $(3, 2)1$, $(5, 7)1$, $(6, 12)1$ or $(11, 192)1$. For example,

$$(18, 768)3 = (6, 12) \dot{+} (13, 128)2.$$

r: Mollard, Katsman, Litsyn [4], [17] (Theorem 8).

s: Cohen *et al.* [4], a piecewise constant code (Theorems 5, 6).

t: Cohen *et al.* [4] (Theorem 7).

VI. LINEAR CODES

From (3) we see that improved lower bounds on $K(n, R)$ may affect lower bounds on $t[n, k]$. Actually, from Table I and (3) we obtain several improvements on the table of $t[n, k]$ given in [7]. For example, $K(32, 7) \geq 1137 > 2^{10}$ implies $t[32, 10] \geq 8$, whereas the best lower bound on $t[32, 10]$ known up to the present was 7, established by the sphere bound. Theorem 11 lists all improvements for $n \leq 33$.

Theorem 11:

$$\begin{aligned} t[23, 6] &= 7 & t[25, 9] &= 6 \\ t[26, 12] &= 5 & t[28, 6] &\geq 9 \\ t[29, 8] &\geq 8 & t[30, 7] &\geq 9 \\ t[32, 10] &\geq 8 & t[33, 6] &\geq 11. \end{aligned}$$

VII. CONJECTURES

Cohen *et al.* [4] made the following conjecture:

$$K(n+2, R+1) \leq K(n, R), \quad \text{for } n \neq R. \quad (21)$$

They studied the validity of (21) in the cases $R=1$ and $R=2$. For example, they proved that

$$K(n+2, 2) \leq K(n, 1),$$

$$\text{for } n = 2, 3, \dots, 8, 10, 11, \dots, 15 \text{ and } n \geq 28.^1$$

From Table I we see that this can be strengthened.

Theorem 12: $K(n+2, 2) \leq K(n, 1)$ for all $n \in \mathbb{N}$, $n \geq 2$, $n \neq 9$.

To conclude, we shall make a conjecture concerning perfect codes. Without difficulty, the definitions in Section II can be generalized for arbitrary alphabets, consisting of $q \geq 2$ elements. In particular, for $q, n \in \mathbb{N}$, $q \geq 2$, and $R \in \{0, 1, \dots, n\}$, $V_q(n, R)$ and $K_q(n, R)$ are defined by

¹In [3, p. 27] they have improved this to: for all $n \in \mathbb{N}$, $n \geq 2$, $n \neq 9, 16$.

$$V_q(n, R) = \sum_{i=0}^R \binom{n}{i} (q-1)^i$$

(cf. [11, p. 55]), and

$$K_q(n, R) = \min \{ M \mid \text{a } q\text{-nary } (n, M) R \text{ code exists} \}.$$

TABLE I
BOUNDS ON $K(n, R)$ FOR $n \leq 33$ AND $R \leq 10$

n	Part 1		
	$R=1$	$R=2$	$R=3$
1	1		
2	2	1	
3	2	2	1
4	a 4	2	2
5	b 7 s	2	2
6	h 12 s	b 4	2
7	a 16	b 7 q	2
8	e 32	g 9–12 q	b 4
9	c 54–64	e 14–16	b 7 q
10	e 103–128	e 21–32	g 8–12 q
11	f 176–192 t	c 32–56 q	c 10–16
12	e 342–384 p	e 61–96 q	e 17–32
13	c 598–768 p	e 96–128	e 25–56 q
14	e 1171–1536 p	c 157–256	e 38–64
15	a 2^{11}	e 307–512	c 58–128
16	e 2^{12}	e 509– 2^{10}	e 111–256
17	f 7377– 2^{13}	c 859–1536 q	e 175–448 q
18	e 14564– 2^{14}	e 1692–3072 p	e 280–768 q
19	c 26216– 2^{15}	e 2889– 2^{12}	f 462– 2^{10}
20	e 52429– 2^{16}	f 5041– 2^{13}	e 889– 2^{11}
21	a 95326– 2^{17}	e 9893– 2^{14}	e 1468– 2^{12}
22	e 190651– 2^{18}	e 17261– 2^{15}	e 2443– 2^{12}
23	f 352049– $3 \cdot 2^{17}$ r	f 30613– 2^{16}	a 2^{12}
24	e 699051– $3 \cdot 2^{18}$ p	e 60350– 2^{17}	e 8098– 2^{13}
25	a 1290556– $3 \cdot 2^{19}$ p	e 107032– $3 \cdot 2^{16}$ q	e 13784– 2^{14}
26	e 2581111– $3 \cdot 2^{20}$ p	a 190651– $3 \cdot 2^{17}$ p	e 23600– 2^{15}
27	a 4793491– $3 \cdot 2^{21}$ p	e 380221– $3 \cdot 2^{18}$ p	a 40623– 2^{16}
28	e 9586981– $3 \cdot 2^{22}$ p	e 683042– 2^{20}	e 80563– 2^{17}
29	f 17977788– $3 \cdot 2^{23}$ p	a 1231356– 2^{21}	e 140212– 2^{18}
30	e 35791395– $3 \cdot 2^{24}$ p	e 2457076– 2^{22}	e 245091– 2^{19}
31	a 2^{26}	e 4459987– 2^{23}	f 433660– 2^{20}
32	e 2^{27}	f 8165162– 2^{24}	e 854890– 2^{21}
33	a 252645136– 2^{28}	e 16207424– 2^{25}	e 1513112– 2^{22}

TABLE I
(Continued)

R	Part 2		
	R = 4	R = 5	R = 6
4	1		
5	2	1	
6	2	2	1
7	2	2	2
8	2	2	2
9	2	2	2
10	b 4	2	2
11	b 7 q	2	2
12	7-12 q	b 4	2
13	d 9-16	b 7 q	2
14	a 12-32	7-12 q	b 4
15	e 21-56 q	e 8-16	b 7 q
16	e 31-64	e 11-32	7-12 q
17	e 46-128	f 15-56 q	7-16
18	e 69-224 q	e 26-64	e 10-32
19	c 106-384 q	e 38-64	f 14-56 q
20	e 201-512	e 55-128	a 18-64
21	e 316-2 ¹⁰	e 82-256	e 32-64
22	e 501-2 ¹¹	f 126-512	e 46-128
23	e 802-3584 q	f 201-2 ¹⁰	e 67-224 q
24	a 1296-2 ¹²	e 361-1792 q	e 99-384 q
25	e 2533-2 ¹²	e 564-3072 q	e 147-512
26	e 4167-2 ¹³	e 890-2 ¹²	f 223-2 ¹⁰
27	e 6901-7·2 ¹¹ q	e 1415-2 ¹²	f 357-2 ¹¹
28	e 11502-3·2 ¹³ q	f 2290-2 ¹³	e 642-2 ¹²
29	f 19506-2 ¹⁵	f 3789-7·2 ¹¹ q	e 1001-2 ¹²
30	e 37973-2 ¹⁶	e 7126-3·2 ¹³ q	e 1572-2 ¹³
31	e 64469-2 ¹⁷	e 11690-2 ¹⁵	e 2487-7·2 ¹¹ q
32	e 109982-2 ¹⁸	e 19287-2 ¹⁶	e 3962-3·2 ¹³ q
33	e 188484-3·2 ¹⁷ q	e 31999-7·2 ¹⁴ q	e 6353-2 ¹⁵

Conjecture: Let $q, n \in \mathbb{N}$, $q \geq 2$, $R \in \{0, 1, \dots, n\}$. If a perfect R -error-correcting q -nary code of length n exists, then

$$K_q(n+1, R) = q \cdot K_q(n, R) = \frac{q^{n+1}}{V_q(n, R)}. \quad (22)$$

In other words, for length $n+1$ we cannot do better than taking the DS $\mathcal{C} \oplus \mathbb{Z}_q$ (compare (9)), where $\mathcal{C}$ is a perfect R -error-correcting code of length n (over the cyclic group $\mathbb{Z}_q$, for convenience).

TABLE I
(Continued)

n	$R = 7$	Part 3			$R = 10$
		$R = 8$	$R = 9$		
7	1				
8	2	1			
9	2	2	1		
10	2	2	2		1
11	2	2	2		2
12	2	2	2		2
13	2	2	2		2
14	2	2	2		2
15	2	2	2		2
16	$b\ 4$	2	2		2
17	$b\ 7\ q$	2	2		2
18	$7-12\ q$	$b\ 4$	2		2
19	$7-16$	$b\ 7\ q$	2		2
20	$e\ 9-32$	$7-12\ q$	$b\ 4$		2
21	$e\ 12-56\ q$	$7-16$	$b\ 7\ q$		2
22	$f\ 17-64$	$e\ 8-32$	$7-12\ q$		$b\ 4$
23	$f\ 23-64$	$e\ 11-56\ q$	$7-16$		$b\ 7\ q$
24	$e\ 39-128$	$f\ 16-64$	$e\ 8-32$		$7-12\ q$
25	$e\ 56-224\ q$	$f\ 21-64$	$e\ 10-56\ q$		$7-16$
26	$e\ 81-384\ q$	$f\ 28-128$	$e\ 13-64$		$e\ 8-32$
27	$e\ 118-512$	$e\ 48-224\ q$	$f\ 19-64$		$e\ 10-56\ q$
28	$e\ 175-2^{10}$	$e\ 68-384\ q$	$f\ 25-128$		$e\ 12-64$
29	$f\ 266-1792\ q$	$e\ 98-512$	$f\ 34-224\ q$		$e\ 16-64$
30	$e\ 395-2^{11}$	$e\ 142-2^{10}$	$e\ 58-384\ q$		$f\ 23-128$
31	$f\ 631-2^{11}$	$e\ 209-1792\ q$	$e\ 82-512$		$f\ 31-224\ q$
32	$e\ 1137-2^{12}$	$e\ 310-2^{11}$	$e\ 118-2^{10}$		$f\ 41-384\ q$
33	$e\ 1767-2^{13}$	$f\ 467-2^{11}$	$e\ 171-1792\ q$		$e\ 71-512$

In the following cases, (22) is known to be valid:

- $-q \geq 2$, $n = 1, 2, \dots$, $R = 0$ (obvious);
- $-q \geq 2$, $n = R$, for $R = 1, 2, \dots$ (obvious);
- $-q = 2$, $n = 2R + 1$, for $R = 1, 2, \dots$
(Theorem 5a));
- $-q = 2$, $n = 2^r - 1$, $R = 1$, for $r = 1, 2, \dots$
(Corollary 1b));
- $-q = 3$, $n = 4$, $R = 1$
(Kamps and van Lint [10]);

we have

$$K_3(5, 1) = 27 = 3 \cdot K_3(4, 1).$$

Note that for $q = 2$ the crucial question is

$$K(24, 3) = 8192?$$

NOMENCLATURE

- $\mathbb{N}$ $\{1, 2, \dots\}$.
- $\mathbb{F}_2$ Field of two elements, 0 and 1.
- $|A|$ Cardinality of the set A .
- x Element of $\mathbb{F}_2^n$.
- x_i i th coordinate of x .
- $d(x, y)$ Hamming distance between x and y .
- $\omega(x)$ Hamming weight of x .
- O $(0, 0, \dots, 0)$.
- $[a]$ $\min \{n \in \mathbb{Z} | n \geq a\}$.

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APPENDIX

Proof of Lemma 7: a) For $a, b, c \in \mathbb{F}_2^n$, $|S_r(a) \cap B_s(b) \cap B_t(c)|$ is determined only by $d(a, b)$, $d(a, c)$, and $d(b, c)$. Hence we may assume that $w = w' = O$ and $x = x'$. Put $d = d(x, y)$, $d' = d(x, y')$. Lemma 5 implies that $d - d' = 2g$, for some $g \in \mathbb{N} \cup \{0\}$. We partition the set $\{1, 2, \dots, n\}$ of coordinates into four subsets: $N_{\alpha\beta} = \{i | x_i = \alpha, y_i = \beta\}$ for $\alpha, \beta \in \mathbb{F}_2$. Observe that $|N_{10}| \geq g$ and $|N_{01}| \geq g$. For example, $d' \geq \omega(y') - \omega(x) = \omega(y)$

$-\omega(x) = |N_{01}| - |N_{10}| = d - 2|N_{10}|$, so $|N_{10}| \geq (1/2)(d - d') = g$. Now choose g coordinates belonging to N_{10} and g coordinates belonging to N_{01} . We may assume that y' is the word obtained from y by changing y in these $2g$ coordinates (for the word thus obtained has the right weight and the right distance to x). In short, we have g coordinates i with

$$x_i = 1 \quad y_i = 0 \quad y'_i = 1 \quad (23)$$

and g coordinates j with

$$x_j = 0 \quad y_j = 1 \quad y'_j = 0 \quad (24)$$

and $d(y, y') = 2g$.

Put $I = S_r(O) \cap B_s(x) \cap B_t(y)$ and $I' = S_r(O) \cap B_s(x) \cap B_t(y')$. To prove a), we have to show that $|I'| \geq |I|$. It suffices to consider the case $g = 1$. For a complete proof use induction.

Now suppose $g = 1$. Put $e = y' - y$. So $\omega(e) = 2$. The two nonzero coordinates of e are a coordinate i satisfying (23) and a coordinate j satisfying (24). Define the map $\varphi: I \rightarrow \mathbb{F}_2^n$ by

$$\begin{aligned} \varphi(a) &= a, & \text{if } a \in I' \\ \varphi(a) &= a + e, & \text{if } a \notin I'. \end{aligned}$$

We show that 1) $\varphi(I) \subseteq I'$, and 2) φ is injective. This implies $|I'| \geq |I|$.

1) Let $a \in I$. If $a \in I'$, then $\varphi(a) = a \in I'$. If $a \notin I'$, then $\varphi(a) = a + e$, and we have $d(\varphi(a), y') = d(a + e, y + e) = d(a, y) \leq t$, so $\varphi(a) \in B_t(y')$. Furthermore, $a \in I - I'$ implies $a_i = 0, a_j = 1$. Consequently, $d(\varphi(a), x) = d(a + e, x) = d(a, x) - 2 \leq s - 2 \leq s$, and $\omega(\varphi(a)) = \omega(a + e) = \omega(a) = r$. Therefore, $\varphi(a) \in B_s(x)$ and $\varphi(a) \in S_r(O)$. We conclude that $\varphi(a) \in I'$.

2) Suppose that $a, \tilde{a} \in I, a \neq \tilde{a}$, and $\varphi(a) = \varphi(\tilde{a})$. If $a, \tilde{a} \in I'$ or $a, \tilde{a} \notin I'$, we immediately have a contradiction. The remaining case is $a \in I', \tilde{a} \notin I'$ (w.l.o.g.). Then $a = \tilde{a} + e$, and we have $d(a, y) = d(\tilde{a} + e, y) = d(\tilde{a}, y') > t$. This contradicts $a \in B_t(y)$. Part b) follows immediately from a) and (2).

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On Normal and Subnormal q -ary Codes

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Abstract—We extend to the q -ary case the notions of a normal code, a subnormal code, and the amalgamated direct sum construction. These concepts were defined in the binary case to investigate problems related to the covering radius of codes. Many nonbinary covering codes appear to be absubnormal. In particular, we prove that every nonbinary nontrivial perfect code is absubnormal. We also include some linear programming lower bounds on ternary codes with covering radii 2 or 3.

I. INTRODUCTION

Let $K(n, R)$ denote the minimal number of codewords in any binary (linear or nonlinear) code of length n and covering radius R . (In other words $K(n, R)$ is the minimal number of spheres of radius R necessary to cover F_2^n .) The distance we consider here is the Hamming distance. The function K has received much attention recently (see, for instance, [2], [3], [5]–[8], [17]).

In [4], Graham and Sloane introduced the notion of a binary linear *normal* code, and the *amalgamated direct sum* of two binary linear codes. This work was extended to the nonlinear case in [2]. The amalgamated direct sum construction, which works best when applied to normal codes, offered several constructive improvements on upper bounds for $K(n, R)$.

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Binary (linear or nonlinear) *subnormal* codes were introduced by Honkala in [6] and were studied in more detail in [7], giving alternative constructions for upper bounds on $K(n, R)$. A code that is not normal or subnormal is called *abnormal* or *absubnormal*, respectively.

In the binary case it has been shown that any linear code with length $n \leq 14$ or dimension $k \leq 5$ or minimal distance $d_{\min} \leq 5$ or covering radius $R \leq 2$ is normal [2]–[4], [12]; so far, no linear abnormal code is known. Also, no (linear or nonlinear) absubnormal code is known.

It is known that abnormal nonlinear codes do exist. Examples of such codes were given by Frankl (see [12]). However, these examples were shown to be subnormal by Honkala [7].

In Section II we extend these concepts to any q -ary (linear or nonlinear) code. We give a few examples of normal q -ary codes (Lemmas 1–3). In Section III we investigate which q -ary perfect codes are (sub)normal: all binary or trivial perfect codes are normal, but all other q -ary perfect codes are abnormal and even absubnormal!

In Section IV we pay special attention to ternary covering codes. We derive an explicit lower bound on the size of a subnormal ternary code with covering radius 1. We include a table of lower and upper bounds on $K_3(n, R)$, the minimal number of codewords in any ternary code of length n and covering radius R , for $n \leq 13$, $R \leq 3$, known to us. We improved some of the known lower bounds by linear programming.

II. NORMAL AND SUBNORMAL q -ARY CODES

Let a $(q, n, M)R$ code be a (linear or nonlinear) code of length n , with M elements and covering radius R , over an alphabet consisting of q (≥ 2) symbols. For the alphabet we take the finite field $\mathbb{F}_q$ of q elements if q is a prime power, or the integers modulo q otherwise. We shall always denote the alphabet by F_q .

The following is a straightforward generalization of the definition of a binary normal code given in [2].

Let C be a $(q, n, M)R$ code. For $i = 1, \dots, n$ and $a \in F_q$, let $C_a^{(i)} = \{(c_1 \dots c_n) \in C | c_i = a\}$ and

$$N^{(i)} = \max_{x \in F_q^n} \left\{ \sum_{a \in F_q} d(x, C_a^{(i)}) \right\},$$

with the convention that $d(x, \emptyset) = n$; $N^{(i)}$ is called the *norm of C with respect to coordinate i* . If, for some i , $N^{(i)} \leq qR + q - 1$, then C is said to be *normal*, and such a coordinate i is called *acceptable*.

We now generalize the definition of the amalgamated direct sum of two normal codes: let A be a $(q, n_A, M_A)R_A$ normal code with last coordinate acceptable, and let B be a $(q, n_B, M_B)R_B$ normal code with first coordinate acceptable. Assume $A_a^{(n_A)}$ and $B_a^{(1)}$ are nonempty for all a in F_q . Then the *amalgamated direct sum* of A and B is given by $C = A \oplus B = \bigcup_{a \in F_q} \{(v, a, w) | (v, a) \in A, (a, w) \in B\}$. Theorem 1 is a generalization of [2, theorem 11].

Theorem 1: C is a code of length $n = n_A + n_B - 1$, containing $\sum_{a \in F_q} |A_a^{(n_A)}| \cdot |B_a^{(1)}|$ elements. C has covering radius $R_C \leq R_A + R_B$, and if $R_C = R_A + R_B$, then C is normal, with coordinate n_A acceptable.

Proof: Let z be an arbitrary element of F_q^n : $z = (x, u, v)$ with $x \in F_q^{n_A-1}$, $v \in F_q^{n_B-1}$, $u \in F_q$. For $s, t \in F_q$ let $\Delta_{s,t} = 0$ if $s = t$, 1 otherwise. Now

$$\begin{aligned} q \cdot d(z, C) &\leq \sum_{a \in F_q} d((x, u, v), C_a^{(n_A)}) \\ &= \sum_{a \in F_q} \{d((x, u), A_a^{(n_A)}) + d((u, v), B_a^{(1)}) - \Delta_{u,u}\} \\ &= \sum_{a \in F_q} d((x, u), A_a^{(n_A)}) \\ &\quad + \sum_{a \in F_q} d((u, v), B_a^{(1)}) - (q-1). \end{aligned}$$

Thus, because A is normal with coordinate n_A acceptable and B is normal with coordinate 1 acceptable,

$$\begin{aligned} q \cdot d(z, C) &\leq (qR_A + q - 1) + (qR_B + q - 1) - (q - 1) \\ &= q(R_A + R_B) + q - 1. \end{aligned}$$

This proves that for all z in F_q^n , $d(z, C) \leq R_A + R_B$. Hence $R_C \leq R_A + R_B$. Now if $R_C = R_A + R_B$, we have, for all z in F_q^n , $\sum_{a \in F_q} d(z, C_a^{(n_A)}) \leq qR_C + q - 1$; thus C is normal and coordinate n_A is acceptable, which completes the proof. $\square$

We now extend to the q -ary case the definition of a binary subnormal code given in [6]. Let C be a $(q, n, M)R$ code. C is said to be *subnormal* if there is a partition of C into q subsets C_a ($a \in F_q$) such that, for all x in F_q^n , $\sum_{a \in F_q} d(x, C_a) \leq qR + q - 1$. (We allow C_a to be empty, putting $d(x, \emptyset) = n$.) Such a partition is called *acceptable*.

Clearly, any normal code is subnormal. Theorem 2 generalizes [6, Theorem 8].

Theorem 2: If C is a $(q, n, M)R$ subnormal code with an acceptable partition without the empty set, then for every natural number p there is a $(q, n + pq, M)R + (q - 1)p$ code.

Proof: $C = \bigcup_{a \in F_q} C_a$, with $\sum_{a \in F_q} d(y, C_a) \leq qR + q - 1$ for all y in F_q^n . For every a in F_q let J_a^k be the vector of length k containing k times a , and let $C_p^* = \{(c, J_a^{pq}) | c \in C_a, a \in F_q\}$. C_p^* is a code of length $n + pq$, containing M words. Let x be any vector in F_q^{n+pq} : $x = (v, w)$ with $v \in F_q^n$ and $w \in F_q^{pq}$, w containing $p_a := p + r_a$ times the symbol a , for $a \in F_q$; we have $-p \leq r_a \leq (q - 1)p$, and $\sum_{a \in F_q} r_a = 0$. Suppose that for all $a \in F_q$, $d(v, C_a) \geq R + r_a + 1$. Then $\sum_{a \in F_q} d(v, C_a) \geq qR + q$, which is a contradiction. Thus there must be some a such that $d(v, c) \leq R + r_a$, with $c \in C_a$. Now $(c, J_a^{pq}) \in C_p^*$, and $d(x, (c, J_a^{pq})) \leq R + r_a + (pq - p_a) = R + (q - 1)p$. This establishes the desired result that C_p^* has covering radius less than or equal to $R + (q - 1)p$. In fact, it is clear that the covering radius of C_p^* is exactly $R + (q - 1)p$. $\square$

Before we present Theorem 3, which combines Theorems 1 and 2 and is a generalization of [7, Theorem 9], we state the following.

Lemma 1: For all q and all n , the q -ary repetition code of length n is normal.

Proof: The repetition code is $C_{\text{rep}} = \bigcup_{a \in F_q} \{J_a^n\}$, with J_a^n the all- a vector of length n . Let w be any vector in F_q^n , containing p_a times the symbol a . Let $p = \max\{p_a | a \in F_q\}$. Then $p \geq \lfloor n/q \rfloor$ and $d(w, C_{\text{rep}}) = n - p \leq n - \lfloor n/q \rfloor$ and so C_{rep} has covering radius $R \leq n - \lfloor n/q \rfloor$. Taking w with $p = \lfloor n/q \rfloor$ shows that $R = n - \lfloor n/q \rfloor$. Now, $q \cdot \lfloor n/q \rfloor \leq n + q - 1$ and so $qR + q - 1 \geq (q - 1)n$. For arbitrary $w \in F_q^n$ however we have $\sum_{a \in F_q} d(w, J_a^n) = \sum_{a \in F_q} (n - p_a) = (q - 1)n$, which proves that C_{rep} is normal. $\square$

We now give a slight modification of the amalgamated direct sum construction, as in [7].

Let A be a $(q, n_A, M_A)R_A$ normal code with the last coordinate acceptable, and let B be a $(q, n_B, M_B)R_B$ subnormal code with partition $B = \bigcup_{a \in F_q} B_a$ acceptable. Assume $A_a^{(n_A)}$ and B_a are nonempty for all $a \in F_q$. Let $C_a = \{(v, w) | (v, a) \in A, w \in B_a\}$. Then the *amalgamated direct sum* of A and B is given by $C = A \oplus B = \bigcup_{a \in F_q} C_a$.

Theorem 3: C is a code of length $n = n_A + n_B - 1$, containing $\sum_{a \in F_q} |A_a^{(n_A)}| \cdot |B_a|$ elements. C has covering radius $R_C \leq R_A + R_B$, and if $R_C = R_A + R_B$, then C is subnormal with $\bigcup_a C_a$ an acceptable partition.

Proof: Let z be an arbitrary element of F_q^n ; $z = (u, v)$ with $u \in F_q^{n_A-1}$ and $v \in F_q^{n_B}$. For $t \in F_q$ let $\Delta_t = 0$ if $t = 0$ and $\Delta_t = 1$ otherwise. Then

$$\begin{aligned}
 q \cdot d(z, C) &\leq \sum_{a \in F_q} d((u, v), C_a) \\
 &= \sum_{a \in F_q} \{d((u, 0), A_a^{(n_A)}) - \Delta_a + d(v, B_a)\} \\
 &= \sum_{a \in F_q} d((u, 0), A_a^{(n_A)}) + \sum_{a \in F_q} d(v, B_a) - (q - 1) \\
 &\leq (qR_A + q - 1) + (qR_B + q - 1) - (q - 1) \\
 &= q(R_A + R_B) + q - 1.
 \end{aligned}$$

Thus $R_C \leq R_A + R_B$. Now if $R_C = R_A + R_B$, we have, for all z in F_q^n , $\sum_{a \in F_q} d(z, C_a) \leq qR_C + q - 1$; thus C is subnormal and $\bigcup_a C_a$ is acceptable, which completes the proof. $\square$

Theorem 3 contains Theorem 1, because any normal code is subnormal, and also Theorem 2, because the repetition code is normal and can be taken in the amalgamated direct sum construction as code A , with length $n_A = pq + 1$ (and covering radius $R_A = (pq + 1) - [(pq + 1)/q] = p(q - 1)$ (see proof of Lemma 1)). Note that in Theorem 1, as well as in Theorem 3, we have $|A \oplus B| \leq |A||B|/q$ (after a suitable permutation of the alphabet in the n_A th column of A , if necessary). It is easy to add more examples of normal codes to the repetition codes of Lemma 1.

Lemma 2: For every q -ary code C the code $C \oplus F_q := \{(c, a) | c \in C, a \in F_q\}$ is normal, with the last coordinate acceptable.

Proof: Let C have length n and covering radius R . Let $D = C \oplus F_q$. D has covering radius R , and for every $w \in F_q^{n+1}$,

$$\sum_{a \in F_q} d(w, D_a^{(n+1)}) \leq R + (q - 1)(R + 1) = qR + q - 1. \quad \square$$

We conclude this section with one less trivial example of a normal covering code. A code like the $(3, 9, 1458)R = 1$ code of Kamps and van Lint [10], which is still the best known ternary code with $n = 9$, $R = 1$, can be constructed in general as follows. Let C be a $(q, n, M)R = 1$ code. Then $C' := \{(x, x + c, y) | x \in F_q^n, c \in C, y \in F_q, y \neq \sum_{i=1}^n x_i\}$ is a $(q, 2n + 1, (q - 1)q^n M)R = 1$ code.

The following result is due to J. H. van Lint, Jr. We omit the proof, which is straightforward.

Lemma 3: C' is normal, with the last coordinate acceptable.

III. ABSUBNORMAL CODES

A q -ary code C with covering radius R is called *perfect* if there are no two different codewords with mutual distance smaller than $2R + 1$. Examples include the *trivial* perfect codes, i.e., those with $R = 0$ or $|C| = 1$, q -ary Hamming codes (q a prime power), or the binary and ternary Golay codes (see [14, ch. 6]). In this section we show that, in contrast with [4, cor. 8], perfectness and subnormality do not, in general, go hand in hand.

Theorem 4: The norm of a perfect R -error-correcting q -ary code with respect to any coordinate is qR if $|C| = 1$ and $qR + q - 1 + (q - 2)R$ otherwise.

Proof: Let C be a perfect R -error-correcting q -ary code of length n . The case $|C| = 1$, i.e., $R = n$, is obvious. Assume $|C| > 1$, i.e., $n \geq 2R + 1$, and consider for instance the first coordinate. For an arbitrary $x \in F_q^n$, there is a (unique) codeword c such that $d(x, c) \leq R$. Let $d := d(x, c)$.

First consider the case $x_1 = c_1$. For any $a \in F_q \setminus \{c_1\}$ a $y^a \in F_q^n$ exists such that $y_1^a = a$, $d(c, y^a) = R + 1$ and $d(x, y^a) = R + 1 - d$.

A (unique) $b^a \in C$ exists such that $d(b^a, y^a) \leq R$. Since $d(c, b^a) \geq 2R + 1$, we necessarily have $d(b^a, y^a) = R$ and $d(c, b^a) = 2R + 1$. Moreover, $b_1^a = a$ and $d(x, b^a) = 2R + 1 - d$. Note that $d(x, b^a) = d(x, C_a^{(1)})$. Hence

$$\begin{aligned} \sum_{a \in F_q} d(x, C_a^{(1)}) &= d(x, c) + \sum_{a \in F_q \setminus \{c_1\}} d(x, b^a) \\ &= d + (q - 1)(2R + 1 - d) \\ &= qR + q - 1 + (q - 2)(R - d) \\ &\leq qR + q - 1 + (q - 2)R, \end{aligned}$$

with equality if x is a codeword, i.e., $d = 0$.

Since $C_a^{(1)} \neq \emptyset$ for all a in F_q (as shown previously), the sum $\sum_{a \in F_q} d(x, C_a^{(1)})$ does not depend on the first coordinate of x . If

$x_1 \neq c_1$, denote by x' the word obtained from x by changing its first coordinate to c_1 . Then $d(x', c) = d - 1 \leq R$ and $x'_1 = c_1$, hence

$$\sum_{a \in F_q} d(x, C_a^{(1)}) = \sum_{a \in F_q} d(x', C_a^{(1)}) \leq qR + q - 1 + (q - 2)R.$$

This proves that $N^{(1)}(C) = qR + q - 1 + (q - 2)R$. $\square$

Corollary 1: A q -ary perfect code is normal if and only if it is binary or trivial.

We strengthen this result in the following theorem.

Theorem 5: If C is a subnormal $(q, n, M)R$ code with minimum distance d , then $d \leq (q/(q-1)) \cdot R + 1$. In particular, the nonbinary nontrivial perfect codes (for which $d = 2R + 1$) are absubnormal.

Proof: If $C = \bigcup_{a \in F_q} C_a$ is acceptable, then for any codeword c , $qR + q - 1 \geq \sum_{a \in F_q} d(c, C_a) \geq 0 + (q-1)d$; this is also true in cases where $C_a = \emptyset$, because $d(c, \emptyset) = n \geq d$. The result follows.

Example: The q -ary Hamming codes with parameters $(q, n = (q^r - 1)/(q - 1), M = q^{n-r})R = 1$ ($r > 1$, $q > 2$ a prime power) are absubnormal linear perfect codes.

As in [17], we denote by $K_q(n, R)$ the minimal cardinality of a q -ary code of length n with covering radius R . A $(q, n, M)R$ code is called *optimal* if $M = K_q(n, R)$. Conjecture *D* in [2] states that for given n and R it is always possible to find a normal code among the binary optimal covering codes. Theorem 5 and any choice of the parameters of the Hamming codes just mentioned can be used to disprove the q -ary generalization of this conjecture, even when we replace "normal" by "subnormal." It also shows that [7, Theorem 11], which states that any binary optimal code with covering radius 1 is subnormal, cannot be generalized to the q -ary case. $\square$

IV. THE TERNARY CASE

Let C be a q -ary (sub)normal code of length n with covering radius R . The argument in the proof of Theorem 5 shows the existence of (a number of) intersections of the spheres with radius R around the codewords of C , at least when $q > 2$ and $0 < R < n$. With this knowledge it is obviously possible to derive a lower bound on the cardinality of C which is better than the well-known trivial *sphere covering bound*

$$|C| \geq \frac{q^n}{\sum_{i=0}^R \binom{n}{i} (q-1)^i}. \quad (1)$$

We illustrate this by working out the idea in the case $q = 3$, $R = 1$. In this case the sphere covering bound reads $|C| \geq 3^n / (1 + 2n)$. We need a lemma to take care of a minor detail.

Lemma 4: If a subnormal $(q, n, M)R$ code C has an acceptable partition with an empty set, then $n \leq R + q - 1$.

Proof: Choose an x in F_q^n such that $d(x, C) = R$. Suppose $C = \bigcup_{a \in F_q} C_a$ is an acceptable partition of C with at least one empty set. Then $qR + q - 1 \geq \sum_{a \in F_q} d(x, C_a) \geq n + (q - 1)R$ and hence $n \leq R + q - 1$. $\square$

Theorem 6: For every subnormal ternary code C of length $n > 3$ with covering radius 1, we have $|C| \geq 3^n / 2n$.

Proof: Let $C = C_0 \cup C_1 \cup C_2$ be an acceptable partition. By Lemma 4 there is no empty set. For every c in C there is a codeword $c' \neq c$ such that $d(c, c') \leq 2$. For example, if $c \in C_0$ then $5 = 3 \cdot 1 + 2 \geq d(c, C_0) + d(c, C_1) + d(c, C_2) = d(c, C_1) + d(c, C_2)$.

Now let C' be a maximal 1-error-correcting subcode of C , i.e., for every pair $a, b \in C'$ ($a \neq b$) we have $d(a, b) \geq 3$ and for every $b \in C \setminus C'$, $d(b, C') \leq 2$. One of the following two cases holds: $|C'| \leq |C|/2$ or $|C \setminus C'| < |C|/2$.

In the first case every codeword in C' covers at most $1 + 2n$ words and every codeword $b \in C \setminus C'$ covers at most $1 + 2n - 2$ words not already covered by a codeword in C' , since the sphere with radius 1 and center b intersects in at least two points with a sphere with radius 1 around a codeword in C' . Consequently, $3^n \leq |C'| \cdot (1 + 2n) + |C \setminus C'| \cdot (2n - 1) = |C| \cdot (2n - 1) + 2|C'| \leq |C| \cdot 2n$.

In the second case we get the same result $3^n \leq |C| \cdot 2n$ if we follow the lines of the proof of the first case with C' and $C \setminus C'$ interchanged. Thus in both cases $3^n \leq |C| \cdot 2n$. $\square$

A. Upper Bounds

$K_3(n, R)$ is, in terms of a football pool, the minimal number of forecasts necessary to guarantee, for n matches (having three possible outcomes), that at least one forecast has at most R errors. Table I gives lower and upper bounds for $K_3(n, R)$,

TABLE I
BOUNDS FOR $K_3(n, R)$, $1 \leq n \leq 13$, $1 \leq R \leq 3$

n	$R = 1$	$R = 2$	$R = 3$
1	1		
2	3	1	
3	5	3	1
4	9	3	3
5	^a 27	^b 6-8 ^c	3
6	57-73 ^f	^b 11-17 ^c	^b 5-6 ^c
7	146-186 ^f	^b 24-35 ^c	^b 7-12 ^c
8	386-486 ^f	51-81 ^c	^b 13-27
9	1036-1458 ^d	121-222 ^c	24-54 ^c
10	2812-3645 ^e	294-567 ^c	51-108 ^c
11	7703-10935	729	114-243 ^c
12	21258-29889 ^c	1839-2187	260-729
13	59049	4704-6561	607-1215 ^c

Key to Table

Lower bounds: unmarked (1) or obvious

^a [11]

^b Section IV-B

Upper bounds: unmarked Hamming or Golay code or obvious

^c [5]

^d [10]

^e [1]

^f [13].

$1 \leq n \leq 13$, $1 \leq R \leq 3$, collected from, [1], [5], [9]-[11], [13] and Section IV-B of this correspondence. In Fig. 1 we list the words of $C_{4,9}$, the ternary Hamming code of length 4 with nine elements and covering radius 1 (see Section III). This code is *unique* up to permutations and translations (see [9]).

To use the amalgamated direct sum construction and improve on the upper bounds on $K_3(n, R)$, one would like to find normal or subnormal codes. For instance, normality of $C_{4,9}$ would imply (taking $C_{4,9} \oplus C_{4,9}$) that $K_3(7, 2) \leq 27$, thus improving on the entries in Table I. However, in the previous section we already saw that, although linear, $C_{4,9}$ is abnormal and even absubnormal. So far, the only ternary optimal normal codes we have found are the trivial perfect codes, the repetition codes, and a $(3, 5, 27)R = 1$ code (see Lemma 2).

```

0 0 0 0
0 2 2 2
0 1 1 1
2 0 2 1
2 2 1 0
2 1 0 2
1 0 1 2
1 2 0 1
1 1 2 0

```

Fig. 1. $C_{4,9}:(3,4,9)R=1$ linear code.

Nonoptimal normal or subnormal codes also can be of interest: for instance, finding a $(3,4,11)R=1$ subnormal code would give, by Theorem 2 with $p=1$, a $(3,7,11)R=3$ code, improving $K_3(7,3) \leq 12$. By Theorem 6 there is no chance of finding a subnormal $(3,4,M)R=1$ code with $M < 11$.

B. Lower Bounds

In the binary case numerous techniques, simple or sophisticated, have been investigated with the specific aim of improving the sphere covering bound (1) on $K(n, R)$ (see, for instance, [2, sec. II], [6], [7], [15]–[17]). Here, in the ternary case, we simply use the so-called *linear programming bound* (see [2, theorem 9] and [15]); let C be a ternary code of length n , and for $i=0, \dots, n$ let A_i denote the number of codewords with weight i .

If C has covering radius 1, then, since every vector of weight i must be within distance 1 from the code, we have

$$2(n-i+1)A_{i-1} + (i+1)A_i + (i+1)A_{i+1} \geq 2^i \binom{n}{i},$$

for $i=0, \dots, n$

(with the convention that $A_j = 0$ when $j < 0$ or $j > n$).

If C has covering radius 2, then

$$\begin{aligned}
 & 4 \binom{n-i+2}{2} A_{i-2} + 2i(n-i+1) A_{i-1} \\
 & + \left[1+i + \binom{i}{2} + 2i(n-i) \right] A_i + (i+1)^2 A_{i+1} \\
 & + \binom{i+2}{2} A_{i+2} \geq 2^i \binom{n}{i}, \quad \text{for } i = 0, \dots, n.
 \end{aligned}$$

If C has covering radius 3, then

$$\begin{aligned}
 & 8 \binom{n-i+3}{3} A_{i-3} + 4(i-1) \binom{n-i+2}{2} A_{i-2} \\
 & + \left[2(n-i+1) \left[i + \binom{i-1}{2} \right] + 4(i-1) \binom{n-i+1}{2} \right] A_{i-1} \\
 & + \left[2i^2(n-i) + 1+i + \binom{i}{2} + \binom{i}{3} \right] A_i \\
 & + \left[(1+i) \left[1+i + \binom{i}{2} \right] + 2(n-i-1) \binom{i+1}{2} \right] A_{i+1} \\
 & + (1+i) \binom{i+2}{2} A_{i+2} + \binom{i+3}{3} A_{i+3} \geq 2^i \binom{n}{i}, \\
 & \text{for } i = 0, \dots, n.
 \end{aligned}$$

To find a lower bound on the cardinality of C , we minimize $\sum_{i=0}^n A_i$ subject to these linear inequalities.

For $R = 2$ and $n = 5, 6$, or 7 , and for $R = 3$ and $n = 6, 7$, or 8 , this rules out the codes with 5, 10, 23, 4, 6, and 12 words, respectively, showing that $K_3(5, 2) \geq 6$, $K_3(6, 2) \geq 11$, $K_3(7, 2) \geq 24$, $K_3(6, 3) \geq 5$, $K_3(7, 3) \geq 7$, and $K_3(8, 3) \geq 13$. These results improve the sphere covering bound.

V. OPEN PROBLEMS

1) Find ternary, optimal or nonoptimal, normal or subnormal codes improving, by the amalgamated direct sum construction, on the upper bounds on $K_3(n, R)$ (cf. Section IV-A).

2) What can be said about the general q -ary case? Little is known about the function $K_q(n, R)$ (see [1], [9], [15], or [16]): a simple lower bound is the sphere-covering bound (1); simple upper bounds can be derived using inequalities such as $K_q(n_1 + n_2, R_1 + R_2) \leq K_q(n_1, R_1) \cdot K_q(n_2, R_2)$ (obvious) or $K_{rs}(n, R) \leq s^{n-1} K_r(n, R)$ (see [1]). When $R = 1$, exact values are known for $n = 2$, $n = 3$ (see [9]), or when q is a prime power with $n(q-1) + 1$ a power of q .

Notes Added in Proof

- 1) The result, mentioned in the Introduction, that binary linear codes with minimum distance $d_{\min} \leq 5$ are normal, has not (yet) been established. X. Hou (University of Chicago) has shown that the proof in [12] is incorrect.
- 2) For open problem 2), see G. J. M. van Wee, "Bounds on packings and coverings by spheres in q -ary and mixed Hamming spaces," *J. Combin. Theory (A)*, to appear.

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Note

On the non-existence of certain perfect mixed codes

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Abstract

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It is shown that if a nontrivial perfect mixed e -code in $Q_1 \times Q_2 \times \dots \times Q_n$ exists, where the Q_i are alphabets of size q_i , then the q_i , e and n satisfy certain divisibility conditions. In particular, all differences $q_i - q_j$ must be divisible by $e + 1$.

1. Introduction

Let n be a positive integer and for $i = 1, 2, \dots, n$ let Q_i be an alphabet consisting of $q_i \geq 2$ symbols. The elements of the Cartesian product $V := Q_1 \times Q_2 \times \dots \times Q_n$ are called *words*. V is a metric space by means of the *Hamming distance* d , which for any two words $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$ is defined by $d(x, y) = |\{i \mid x_i \neq y_i\}|$. Let e be a nonnegative integer. $B_e(x)$ denotes the *sphere* with *radius* e and *center* x : $B_e(x) = \{y \in V \mid d(x, y) \leq e\}$. A *mixed code* is a nonempty subset of V . A *mixed perfect e -code* is a mixed code C such that the spheres $B_e(c)$ form a partition of V , where c runs through C . Examples of mixed perfect codes are those with $e = 0$ or $e = n$. We call these perfect codes the *trivial* ones.

In the special case that all q_i are the same, q say, one speaks of a q -ary code. The existence and nonexistence of q -ary perfect codes has been thoroughly

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studied by various authors. For a survey and many references, see van Lint [7]. One could also consult [8, Ch. 6]. Examples of nontrivial q -ary perfect codes are known only when q is a prime power. In the following we shall abbreviate 'nontrivial mixed perfect e -code' by e -code. We shall use the adjective *proper* if we want to indicate that not all q_i are the same.

The earliest publication on mixed codes and the configurations which we call proper e -codes, seems to be the conference paper by Schönheim [10]. Later on, proper e -codes have been studied by Herzog and Schönheim [3–4], Lenstra [5], Heden [1–2], Lindström [6] and Reuvers [9]. Except for [1] and [9] the papers mentioned mostly deal with the case that the q_i are all powers of the same prime and $e = 1$. By Theorem 1 in [4] a 1-code can be constructed from an abelian group that has a certain partitioning property. As a consequence of Theorem 2 in [4], which gives sufficient conditions for a class of abelian groups to have that partitioning property, one can easily find many proper 1-codes. We give an example: Let q be a prime power and let $\mathbb{F}_q$ be the finite field of size q . Let α and m be integers with $m > \alpha \geq 2$ and put $n = 1 + (q^m - q^\alpha)/(q - 1)$. Then there exists a 1-code in $\mathbb{F}_{q^\alpha} \times \mathbb{F}_q^{n-1}$, which moreover is a subgroup of $\mathbb{F}_{q^\alpha} \times \mathbb{F}_q^{n-1}$ (viewed as an additive group).

In [1] and [9, Ch. 6] proper e -codes are treated with e and the q_i arbitrary. Reuvers [9] proved the nonexistence of certain proper 2- and 3-codes. It is not known to the author whether proper e -codes exist with $e > 1$.

In this note we show that for an e -code to exist, the q_i , e and n must satisfy certain divisibility conditions (Theorem 1). This leads to Corollary 2, which for instance immediately rules out the a priori possible existence of proper e -codes with e arbitrary and $q_i \in \{2, 3\}$ for all i . Some of the arguments used in Section 2 are related to [11, Section IV].

2. Divisibility conditions

Theorem 1. *Suppose an e -code in V exists. Then for $r = 1, 2, \dots, e + 1$ and for all subsets A of $\{1, 2, \dots, n\}$ of size $n - e + r - 1$ we have*

$$\sum_{R \subseteq A: |R|=r} \prod_{j \in R} (q_j - 1) \equiv 0 \pmod{\binom{e+r}{r}}. \quad (1)$$

Proof. Suppose that C is an e -code in V . Let $r \in \{1, 2, \dots, e + 1\}$ and let A be a subset of $\{1, 2, \dots, n\}$ of size $n - e + r - 1$. Let $c \in C$ and let $x \in V$ be such that $\{i \mid x_i \neq c_i\} = \{1, 2, \dots, n\} \setminus A$. Hence $d(c, x) = e - r + 1$ and $B_{r-1}(x) \subseteq B_e(c)$. Now one easily verifies that

$$|B_r(x) - B_e(c)| = \sum_{R \subseteq A: |R|=r} \prod_{j \in R} (q_j - 1).$$

Since C is perfect, we have for every $b \in C \setminus \{c\}$, $B_r(x) \cap B_e(b) \in \{0, (e-r)'\}$. By using the perfectness of C again, (1) follows. $\square$

Remark. For q -ary e -codes Theorem 1 reads

$$\binom{n-e+r-1}{r}(q-1)^r \equiv 0 \pmod{\binom{e+r}{r}},$$

for $r = 1, 2, \dots, e+1$. This is a generalization of Theorems (4.3) and (5.4) in [7].

From Theorem 1 we derive the following (weaker) condition.

Corollary 2. *A necessary condition for an e -code in V to exist, is that all differences $q_k - q_l$ are divisible by $e+1$.*

Proof. Choose $r = 1$ in Theorem 1. Since $0 < e < n$ there exist subsets A_1, A_2 of $\{1, 2, \dots, n\}$ of size $n - e$ such that $A_1 - A_2 = \{k\}$, $A_2 - A_1 = \{l\}$. By (1) we have for $i = 1, 2$:

$$\sum_{j \in A_i} (q_j - 1) \equiv 0 \pmod{(e+1)}.$$

Subtraction of one equation from the other gives the desired result. $\square$

If not all q_i are the same, Corollary 2 gives an upper bound on e : for every e -code in V we have

$$e \leq -1 + \min\{q_k - q_l \mid 1 \leq k, l \leq n, q_k \neq q_l\}.$$

In particular, if $q_k - q_l = 1$ for some k and l , there are no nontrivial perfect codes in V at all (see the example at the end of the Introduction).

3. An application

Some of the proofs of the nonexistence theorems for mixed perfect codes that were found by Reuvers [9, Ch. 6] can be much simplified with the aid of Corollary 2 of this note. We give one example. Somewhat differently stated, the following theorem was proved in [9].

Theorem 3. *If $q_2 = q_3 = \dots = q_n = 2$, then a proper 3-code in V does not exist.*

In the proof of this theorem the following lemma, found by Heden [1], was used.

Lemma 4. *Let p be a prime. If an e -code in V exists and p divides q_i for some i , then p divides at least $n - e + 1$ of the numbers q_i .*

We now give an alternative proof of Theorem 3, which is much shorter than the one given in [9].

Proof of Theorem 3. If a proper 3-code in V exists, then by Lemma 4, $q_1 = 2^r$ for some $r \geq 2$. By Corollary 2 we then would have that 4 divides $2^r - 2$, which is false. $\square$

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More Binary Covering Codes are Normal

GERHARD J. M. VAN WEE

Abstract—It is shown that every optimal binary code with covering radius $R=1$ is normal. This (partly) proves a conjecture of Cohen, Lobstein, and Sloane. It is also proved that codes with minimal distance $2R$ or $2R+1$ are normal. Finally, a generalization of Frankl's construction of abnormal codes is given.

I. INTRODUCTION

Recently much research has been done on the problem of determining which (linear or nonlinear) codes are normal (or subnormal) (see [1]–[5], [8]–[10], [12], and [16]). In this correspondence some further progress towards the solution of this problem is made.

The concept of normality was introduced in [3] for binary linear codes and generalized in [2] for arbitrary binary codes. For the q -ary case, see [12]. In the same papers it was shown that normal codes are of interest for at least one reason: two normal covering codes can be combined by the so-called amalgamated direct sum (ADS) construction, yielding a relatively small covering code of greater length.

A slightly different and weaker property that a binary code can have is to be *subnormal*. This notion was introduced in [4] and discussed more thoroughly in [5]. Every normal code is subnormal, but not the other way around, as shown in [5].

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Although these concepts can be generalized for nonbinary codes (see [12]), in the rest of this correspondence we restrict our attention to the binary case only. Codes that are not normal or not subnormal are called *abnormal* or *absubnormal*, respectively.

Nonlinear abnormal binary codes have been shown to exist by Frankl (see [10, Th. 19]). It is still unknown whether linear abnormal binary codes exist. The same holds for (linear or nonlinear) absubnormal binary codes.

We shall use the following notation. Let $x = x_1 x_2 \cdots x_n$, $y = y_1 y_2 \cdots y_n \in \mathbb{F}_2^n$. The vector space $\mathbb{F}_2^n$ is a metric space with *Hamming distance* $d(x, y)$ defined by $d(x, y) = |\{i | x_i \neq y_i\}|$. The *support* of x , $\text{supp}(x)$, is the set $\{i | x_i \neq 0\}$. The *weight* of x , $wt(x)$, is equal to $|\text{supp}(x)|$. Furthermore, we define $S_r(x) = \{y \in \mathbb{F}_2^n | d(x, y) = r\}$, and $B_r(x) = \{y \in \mathbb{F}_2^n | d(x, y) \leq r\}$. For a code $C \subseteq \mathbb{F}_2^n$ with $|C| > 1$, the *minimal distance* $d_{\min}(C)$ is equal to $\min\{d(u, v) | u, v \in \mathbb{F}_2^n \text{ and } u \neq v\}$. The *covering radius* of C , $CR(C)$, is defined by

$$CR(C) = \max\{d(x, C) | x \in \mathbb{F}_2^n\}.$$

An $(n, M)R$ code is a binary code of length n , with cardinality M and covering radius R . We define

$$K(n, R) = \min\{M | \text{an } (n, M)R \text{ code exists}\}.$$

Let us now recall from [2], [5], [9], or [12] the following definitions. Let C be a binary code of length n with $CR(C) = R$. For $i = 1, 2, \dots, n$ and $a \in \mathbb{F}_2$ let $C_a^{(i)} := \{c \in C | c_i = a\}$ and set $N^{(i)}(C) := \max\{d(x, C_0^{(i)}) + d(x, C_1^{(i)}) | x \in \mathbb{F}_2^n\}$. We use the convention that $d(x, \emptyset) = n$. $N^{(i)}(C)$ is called the *norm* of C with respect to the i th coordinate. If $N^{(i)}(C) \leq 2R + 1$, then coordinate i is called *acceptable*. A code that has at least one acceptable coordinate is called *normal*.

Remark 1: This definition of “acceptable” differs from the one in [2] and [9], but it is in accordance with [5] and [12]. In addition, one could define the *norm* $N(C)$ of C as the minimum of the $N^{(i)}(C)$, but this is done differently in [2], [5] and [9]; in these papers the norm is defined in such a way that a code has more than one norm. Throughout this correspondence the definitions of “acceptable” and “norm” as previously given will be assumed.

When it is clear which code is meant, one simply writes $N^{(i)}$, N , respectively.

To conclude, we repeat from [4] the definition of a subnormal code. An $(n, M)R$ code C is called *subnormal* if it has a subcode C_1 , such that for all $x \in \mathbb{F}_2^n$, we have $d(x, C_1) + d(x, C \setminus C_1) \leq 2R + 1$ (again with the convention $d(x, \phi) = n$).

II. OPTIMAL COVERING CODES WITH COVERING RADIUS ONE ARE NORMAL

In view of the ADS construction, mentioned in the Introduction, it is of particular interest to know which good (that is: with few elements) covering codes are normal. An $(n, M)R$ code is called *optimal* if $M = K(n, R)$. This section is concerned with the following three conjectures:

- a) $K(n+2, R+1) \leq K(n, R)$, for all $n > R$,
- b) for all fixed n and R , among the optimal $(n, M)R$ codes there is a subnormal one,
- c) for all fixed n and R , among the optimal $(n, M)R$ codes there is a normal one.

Conjectures a) and c) were made in [2]. Conjecture b) is a weaker version of Conjecture c). Moreover, as shown in [5], b) implies a). Conjecture a) is known to be valid in the case $R = 1$ (see [1], [2, Th. 34], [6] and [17, Th. 12]). An alternative proof of this fact has been given by Honkala [5, Th. 11], who did this by proving the validity of b) for $R = 1$.

We shall now prove that c) holds for $R = 1$ (Corollary 1). In fact, a stronger result is established. Of course, Corollary 1 also gives an alternative proof of a) and b) in the case $R = 1$.

For a code $C \subseteq \mathbb{F}_2^n$ we already introduced the notation $C_0^{(i)}$ and $C_1^{(i)}$. In addition, for $a \in \mathbb{F}_2$ and $n > 1$, we denote by $C_a^{(i)''}$ the set of words in $\mathbb{F}_2^{n-1}$ obtained from $C_a^{(i)}$ by deleting the i th coordinate.

Theorem 1: Let C be an optimal $(n, M)R = 1$ code with $n \geq 3$. Then for every coordinate i and $a \in \mathbb{F}_2$, $C_a^{(i)''} \neq \phi$ and $CR(C_a^{(i)'}) \leq 2$.

Proof: Again we use the convention $d(x, \phi) = m$ for all $m \in \mathbb{N}$ and $x \in \mathbb{F}_2^m$. Suppose that for some a and i $C_a^{(i)} = \phi$ or

$CR(C_a^{(i)}) > 2$. Without loss of generality $i = 1$ and $a = 0$. We simply write C'_0 and C'_1 instead of $C_0^{(1)}$ and $C_1^{(1)}$, respectively. For every $x \in \mathbb{F}_2^{n-1}$ with $d(x, C'_0) \geq 2$ we have $d(0x, C_0) \geq 2$, hence $d(0x, C_1) \leq 1$ and $d(x, C'_1) = 0$, i.e., $x \in C'_1$.

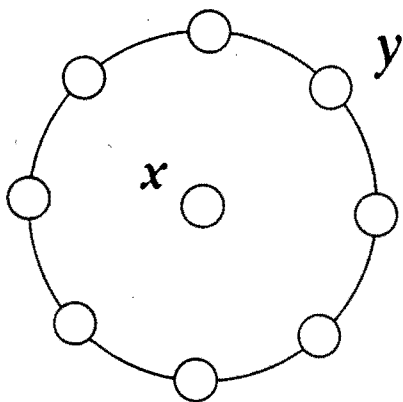
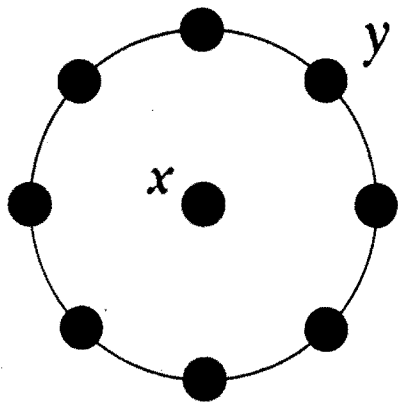
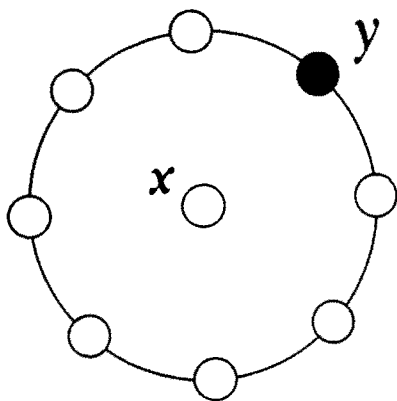
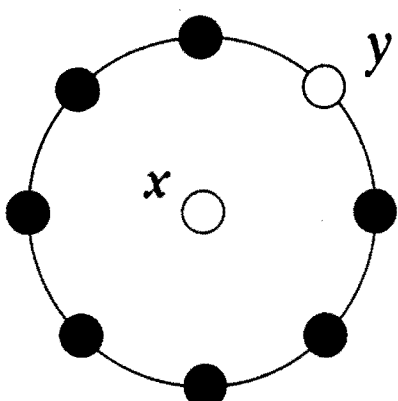
By our assumption, $C'_0 = \emptyset$ or an $x \in \mathbb{F}_2^{n-1}$ exists with $d(x, C'_0) \geq 3$. In any case, $d(y, C'_0) \geq 2$ for all $y \in B_1(x)$, hence $B_1(x) \subseteq C'_1$. Choose a fixed $y \in S_1(x)$. Now let $D := \{0y\} \cup C \setminus \{1x, 1y\}$. This "local change" of C is illustrated by Figs. 1(a), 1(b) (before the change) and 2(a), 2(b) (after the change), where the case $n = 9$ is considered. Small circles indicate words in $\mathbb{F}_2^{n-1}$, and a particular circle has been filled with black if the corresponding word is contained in the set mentioned in the figure caption. We have $|D| < |C|$ and claim that $CR(D) = 1$. Since C is optimal, this gives a contradiction and the theorem follows.

Since, e.g., $1x \notin D$, $CR(D) \geq 1$. We are left with the task to prove that for every $z \in \mathbb{F}_2^n$ there is a $d \in D$ with $z \in B_1(d)$. Let $z \in \mathbb{F}_2^n$. Since $CR(C) = 1$, we are in at least one of the following three cases.

- 1) $z \in B_1(c)$ for some $c \in C \setminus \{1x, 1y\}$. Then $c \in D$ and we are done.
- 2) $z \in B_1(1x) \cup B_1(1y)$ and $z_1 = 0$. Then $z = 0x$ or $z = 0y$ and hence $z \in B_1(0y)$.
- 3) $z \in B_1(1x) \cup B_1(1y)$ and $z_1 = 1$. Put $z = 1w$ with $w \in \mathbb{F}_2^{n-1}$. Then $w \in B_2(x)$. There are three subcases to consider.
 - a) $w = x$. Then choose any $v \in S_1(x) \setminus \{y\}$. We are in case 1 for $c := 1v$.
 - b) $w \in S_1(x)$. Then either $w = y$ and hence $z \in B_1(0y)$, or $w \neq y$ and hence $w \in C_1 \setminus \{x, y\}$ and $z \in D$.
 - c) $w \in S_2(x)$. Then $|S_1(w) \cap S_1(x)| = 2$. Hence there is a $v \in S_1(x) \setminus \{y\}$ with $d(v, w) = 1$ and we are in case 1 again for $c := 1v$.

The next lemma differs slightly from [2, Lemma 17].

Lemma 1: Let C be a binary code of length $n > 1$ with covering radius R . Suppose that for some coordinate i $C_0^{(i)} \neq \emptyset$, $C_1^{(i)} \neq \emptyset$ and that both $CR(C_0^{(i)}) \leq R + 1$ and $CR(C_1^{(i)}) \leq R + 1$. Then C is normal and coordinate i is acceptable.

Fig. 1(a). $B_1(x) \cap C_0^{(i)}$.Fig. 1(b). $B_1(x) \cap C_1^{(i)}$.Fig. 2(a). $B_1(x) \cap D_0^{(i)}$.Fig. 2(b). $B_1(x) \cap D_1^{(i)}$.

Proof: Let $y \in \mathbb{F}_2^{n-1}$ be arbitrary. Put $d_0 := d(y, C_0^{(i)})$ and $d_1 := d(y, C_1^{(i)})$. If $d_0 \leq R-1$, then $d_0 + d_1 \leq R-1 + R+1 = 2R$. If $d_0 = R$, then $d_1 \leq R$ (otherwise $d(1y, C) > R$) and $d_0 + d_1 \leq 2R$. Finally, if $d_0 = R+1$, then $d_1 \leq R-1$ (otherwise $d(0y, C) > R$) and again $d_0 + d_1 \leq 2R$. We conclude that in all cases $d_0 + d_1 \leq 2R$ and hence that for all $x \in \mathbb{F}_2^n$, $d(x, C_0^{(i)}) + d(x, C_1^{(i)}) \leq 2R+1$. $\square$

Corollary 1: Let C be an $(n, M)R = 1$ code with $M = K(n, 1)$. Then C is normal, with every coordinate acceptable.

Proof: In general, any binary code C with covering-radius R and length $n \leq R + 1$ is normal, with all coordinates acceptable. This is because of the convention that $d(x, \phi) = n$: for all $x \in \mathbb{F}_2^n$, and all i , $d(x, C_0^{(i)}) + d(x, C_1^{(i)}) \leq n + R \leq 2R + 1$ (compare [12, Lemma 4]). This covers the cases $n = 1$ and $n = 2$. For $n \geq 3$ the result follows from Theorem 1 and Lemma 1. $\square$

Here a proof of Conjecture c) has been given for the case $R = 1$. At first sight, one might think that the idea of this proof can easily be generalized for $R > 1$, unlike the proof of Conjecture a) in [1], for instance. Unfortunately, this does not seem to be true. Some very detailed and technical problems arise when one tries to generalize for $R > 1$.

On the other hand, the idea has proved its value in a different way, lately; repeating the trick in the proof of Theorem 1 leads to lower bounds on the cardinality and length of abnormal binary codes with covering radius one, as shown in more recent work by Honkala and Hämäläinen [7].

III. CODES WITH VERY HIGH MINIMAL DISTANCE ARE NORMAL

It is known that every perfect binary code, that is a code C with $|C| = 1$ or $d_{\min}(C) = 2 \cdot CR(C) + 1$ otherwise, is normal (see [3, Cor. 8] and [12, Cor. 1]). A stronger result is given in Theorem 2.

Theorem 2: Let C be a binary code with $|C| > 1$, and suppose that $d_{\min}(C) \geq 2 \cdot CR(C)$. Then C is normal, with all coordinates acceptable.

Proof: Put $R := CR(C)$ and let n be the length of C . From the assumptions it follows that $n \geq 2R$. Consider the first coordinate, without loss of generality. For an arbitrary $x \in \mathbb{F}_2^n$ there is a $c \in C$ with $r := d(x, c) \leq R$. A word y exists such that $d(c, y) = R + 1$, $y_1 \neq c_1$ and $d(x, y) = R + 1 - r$. Namely, put $w := x + c$. This is a vector of weight r . Choose any vector w' of

weight $R+1-r$ such that $\text{supp}(w') \cap \text{supp}(w) = \emptyset$, and such that $w'_i = 1$ if $w_i = 0$. This is possible, since $n > R$. Put $y := x + w'$. To continue, there is a $b \in C$ such that $d(y, b) \leq R$. Since $d(c, b) \geq 2R$, we have $b_1 = y_1$ ($\neq c_1$). Hence $d(x, C_0^{(1)}) + (x, C_1^{(1)}) \leq d(x, c) + d(x, b) \leq r + d(x, y) + d(y, b) \leq r + R + 1 - r + R = 2R + 1$. Since x was arbitrary, the first coordinate is acceptable. $\square$

In the rest of this section, we restrict our attention to linear codes. It is already known that every binary linear code of length ≤ 14 or of dimension ≤ 5 , or with minimal distance ≤ 4 or with covering radius ≤ 2 is normal (see [2], [3], [8], [10], [14], and [16]).

Remark 2: Let $s \geq 1$. In general, if it is true that all binary linear codes with minimal distance at most s are normal, then the normality of all binary linear codes with covering radius at most $\lceil s/2 \rceil$ follows.

Indeed, let C be a binary linear code with $R := CR(C) \leq \lceil s/2 \rceil$. If $|C| = 1$, then C is normal. If $|C| > 1$, then C has a minimal distance d . If $d \leq 2R - 1$, then $d \leq 2\lceil s/2 \rceil - 1 \leq s$, hence C is normal. If $d \geq 2R$, then C is normal by Theorem 2.

In [10, Th. 16] it is claimed that all binary linear codes with minimal distance ≤ 5 are normal. From this and Remark 2 we could deduce the following theorem.

Theorem 3: Every binary linear code C with $CR(C) \leq 3$ is normal.

Unfortunately, this proof is not valid, at least not yet. After reading a preprint of this correspondence, in which the previous proof was included, X. Hou pointed out to me that there is a mistake in the proof of [10, Th. 16]. It is still unknown whether all binary linear codes with minimal distance 5 are normal. He also showed that Theorem 3 is true nevertheless. In fact, he proved the general bound $N(C) \leq 3R - 2$, for any binary linear code C with covering radius R , ($R \geq 3$). In the proof of this bound, Theorem 2 is (still) used. For all this, see [8].

Until now, the maximal s that satisfies the condition in Remark 2 is 4. Perhaps this will be improved in the future, yielding a short proof of Theorem 3 (at least).

IV. A GENERALIZATION OF FRANKL'S CONSTRUCTION OF ABNORMAL BINARY CODES

Frankl's construction, published in [10, Th. 19], gives examples of abnormal binary codes only with covering radius $R = 1$. In view of the problems and conjectures that have come up concerning the normality of codes (such as Conjecture c) in the previous section), it would be nice to know whether abnormal binary codes with covering radius R exist for all $R \geq 1$. We generalize Frankl's construction here and show that the answer to this question is affirmative.

Let $R \geq 1$ and $n \geq 4R + 2$. Suppose that we have n words $x^{(i)} \in \mathbb{F}_2^n$ ($i = 1, \dots, n$) and n subsets A_i of $\{1, 2, \dots, n\}$ of size R ($i = 1, \dots, n$), such that for $i = 1, \dots, n$, $i \in A_i$ and $\text{supp}(x^{(i)}) \cap A_i = \emptyset$, and such that $d(x^{(r)}, x^{(s)}) \geq 4R + 2$ if $r \neq s$ (see Remark 3). Then put $T_i := \{y \in \mathbb{F}_2^n \mid d(y, x^{(i)}) \leq 2R \text{ and } y_j = 0 \text{ for some } j \in A_i\}$ ($i = 1, 2, \dots, n$), and define $C := \mathbb{F}_2^n \setminus (T_1 \cup T_2 \cup \dots \cup T_n)$. Note that the T_i are disjoint.

Theorem 4:

- a) For $i = 1, 2, \dots, n$, $N^{(i)}(C) \geq 3R + 1$,
- b) $CR(C) = R$.

In particular, C is abnormal.

Proof:

a) For $i = 1, 2, \dots, n$, $d(x^{(i)}, C_0^{(i)}) = 2R + 1$ and $d(x^{(i)}, C_1^{(i)}) = d(x^{(i)}, c^{(i)}) = R$, where $c^{(i)} = x^{(i)} + v^{(i)}$, with $\text{supp}(v^{(i)}) = A_i$. Hence $N^{(i)}(C) \geq 2R + 1 + R = 3R + 1$.

b) In the proof of a) we already saw that for any i , $d(x^{(i)}, C) = R$. Hence $CR(C) \geq R$. Let $x \in \mathbb{F}_2^n$ be arbitrary. If $x \notin C$, then there is a unique i such that $x \in T_i$. Put $d := d(x, x^{(i)})$. If $d \leq R$, then let c be the word obtained from x by changing x in those coordinates j belonging to A_i where $x_j = 0$. Then $d(x, c) \leq R$. Moreover, $d(x^{(i)}, c) \leq d + R \leq 2R$ and $c_j = 1$ for all $j \in A_i$, hence $c \in C$. If $d \geq R + 1$, then there is a $c \in C$ with $d(x^{(i)}, c) = 2R + 1$ and $d(x, c) = 2R + 1 - d \leq R$. $\square$

Remark 3:¹ For every $R \geq 1$ there is an n_0 such that for all $n \geq n_0$, n words $x^{(i)}$ and n sets A_i exist that meet the conditions in the description of the construction. One way to prove this is the following. Let $A(n, d)$ denote the maximal number of code-words in any binary code of length n and minimal distance d (cf. [11, p. 54] and [13, p. 523]). Fix $R \geq 1$. By the Gilbert–Varshamov bound,

$$A(n, 4R+3) \geq 2^n / \sum_{i=0}^{4R+2} \binom{n}{i}$$

(cf. [11, p. 56]). Hence an n_0 exists such that for all $n \geq n_0$, $A(n+1, 4R+3) \geq n+1$. Now suppose that $y^{(1)}, y^{(2)}, \dots, y^{(n+1)} \in \mathbb{F}_2^{n+1}$ are $n+1$ words with mutual distances $\geq 4R+3$. After complementing coordinates, if necessary, in addition we have $y_i^{(i)} = 0$, for all i . At least n of them, $y^{(1)}, \dots, y^{(j-1)}, y^{(j+1)}, \dots, y^{(n+1)}$ say, have weight $< n+1 - (2R+1)$. Delete the j th coordinate of all these n words, and call the resulting words of length n $x^{(1)}, x^{(2)}, \dots, x^{(n)}$, respectively. We have that $d(x^{(r)}, x^{(s)}) \geq 4R+3-1 = 4R+2$ if $r \neq s$. Moreover, since $x_i^{(i)} = 0$ and $wt(x^{(i)}) < n+1 - (2R+1) < n-R$, for $i=1, \dots, n$, the n sets A_i of size R can be chosen such that $i \in A_i$ and $\text{supp}(x^{(i)}) \cap A_i = \emptyset$.

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¹Of course, more advanced lower bounds on $A(n, d)$, like the Plotkin–Levenshtein bound (see [13, p. 50]), for example, can be used to get a more precise result, i.e., an upper bound on the minimal n_0 (for given R).

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BOUNDS ON PACKINGS AND COVERINGS BY SPHERES IN q -ARY AND MIXED HAMMING SPACES

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ABSTRACT

In a recent paper by the same author general improvements on the sphere covering bound for binary covering codes were obtained. In the present work it is shown how the main idea can be exploited to obtain "improved sphere bounds" also for nonbinary codes. We concentrate on general q -ary codes and on binary/ternary mixed codes. Special attention is paid to the football pool problem; a few new lower bounds are established. Also, it is shown how a similar method yields upper bounds on packing codes.

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I. INTRODUCTION

Consider the Cartesian product $H := Q_1 \times Q_2 \times \cdots \times Q_n$, with $n \in \mathbb{N}$ and the Q_i alphabets of size $q_i \geq 2$. H is a metric space with (Hamming) distance d : for every $\mathbf{x} = (x_1, x_2, \dots, x_n)$, $\mathbf{y} = (y_1, y_2, \dots, y_n)$ in H , $d(\mathbf{x}, \mathbf{y}) = |\{i \mid x_i \neq y_i\}|$. The sphere with radius r and center $\mathbf{x}$, i.e. the set $\{\mathbf{y} \in H \mid d(\mathbf{x}, \mathbf{y}) \leq r\}$, is denoted by $B_r(\mathbf{x})$. The elements of H are called words.

A nonempty subset C of H is called a code. Its elements are called code-words. More specifically, in our general setting, C is usually called a mixed code, and H a mixed (Hamming) space. When all q_i are equal to q , one also speaks of a q -ary code and a q -ary (Hamming) space, respectively. We call n the length of C .

Now let C be a code. By the packing radius $PR(C)$ of C we mean the maximal number $e \in \{0, 1, \dots, n\}$ such that for all $\mathbf{b}, \mathbf{c} \in C$ with $\mathbf{b} \neq \mathbf{c}$, $B_e(\mathbf{b}) \cap B_e(\mathbf{c}) = \emptyset$. The covering radius $CR(C)$ of C is the minimal number $R \in \{0, 1, \dots, n\}$ such that $\bigcup_{\mathbf{c} \in C} B_R(\mathbf{c}) = H$. Clearly $PR(C) \leq CR(C)$. A code with packing radius $\geq e$ is often called an e -packing of H . Similarly, a code with covering radius $\leq R$ is often called an R -covering of H .

For given e, R and H , the numbers

$$N := \max \{ |C| \mid \emptyset \neq C \subseteq H, PR(C) = e \} \text{ and}$$

$$K := \min \{ |C| \mid \emptyset \neq C \subseteq H, CR(C) = R \}$$

are of great interest. Of course, N only depends on n , the q_i and e , and K only depends on n , the q_i and R .

The general problems of determining the values of the functions N and K are known as the (sphere) packing problem, the (sphere) covering problem, respectively. The covering problem for $q_i = 3$ (for all i) and $R = 1$ is also well-known as the football pool problem (for n matches). Packing and covering problems have been widely studied during the last four decades; see, e.g. [1-18], [20-35]. These problems seem hard to handle. Only few values of N and K have been exactly determined. Mostly, only lower and upper bounds are known, with large gaps in between.

This paper is only concerned with upper bounds on N and lower bounds on K . For a given Hamming space H let $V(r)$ denote the number of words in a sphere with radius r . We shall use this notation in Section II too. Then

$$V(r) = 1 + \sum_{j=1}^r \sum_{1 \leq i_1 < \dots < i_j \leq n} (q_{i_1} - 1) \cdots (q_{i_j} - 1) .$$

Now the expression

$$\frac{1}{V(r)} \prod_{i=1}^n q_i \tag{1}$$

is a lower bound ("*sphere covering bound*") on the cardinality of any r -covering of H , and an upper bound ("*sphere packing bound*") on the cardinality of any r -packing of H .

In [32] a new method was introduced to obtain lower bounds on binary covering codes better than (1) (with all $q_i = 2$). It was stated in the introduction of that paper, that the main idea also applies to nonbinary codes. This (kind of) generalization is one of the goals of this paper (and of [19]). In addition, we show that a similar idea can be used to obtain upper bounds for the packing problem. We shall rely heavily on [32].

In Section II we first generalize some notation, definitions and lemmas from [32]. In Section III we treat packings and coverings in q -ary Hamming spaces. In Section IV we pay special attention to the football pool problem and obtain some new lower bounds. In Section V we consider mixed Hamming spaces, but, for simplicity, only in the case that all $q_i \in \{2, 3\}$, which is probably the most interesting. To illustrate how these spaces H can be dealt with, we derive a lower bound on 1-coverings of $\mathbb{F}_3^t \mathbb{F}_2^b$. For general bounds on R -coverings and e -packings of $\mathbb{F}_3^t \mathbb{F}_2^b$, see [19], a work that is completely devoted to this subject.

From [8] and [20], for example, it appears that at present there is a growing interest in bounds such as presented in the Sections III, IV and V, and in [19].

Many of the ideas and results in this paper and [19] were presented earlier by us in [18]. Results similar to those in Section IIIA are derived in [3], a paper which was written independently of this paper and [19], but using [18].

II. SOME PRELIMINARIES

We recall from [32] some definitions and lemmas and generalize for arbitrary Hamming spaces $H = Q_1 \times Q_2 \times \dots \times Q_n$. We also introduce some notation for packings. We omit the (straightforward) proofs in this section.

We have already defined $B_r(\mathbf{x})$. In addition, we define $S_r(\mathbf{x}) := \{y \in H \mid d(\mathbf{x}, y) = r\}$.

DEFINITION 1. Let $r \in \{0, 1, \dots, n\}$ and $W, C \subseteq H$. the nonnegative number $E_C^r(W)$ is defined by

$$E_C^r(W) := \sum_{c \in C} |B_r(c) \cap W| - \left| \bigcup_{c \in C} B_r(c) \cap W \right|. \quad (2)$$

If $C \neq \emptyset$, we define $E_C(W) := E_C^{CR(C)}(W)$, and we call this number the *excess* on W by C .

LEMMA 2. Let C be a code in H with $CR(C) = R$

- a) $E_C(H) = |C| V(R) - q_1 q_2 \cdots q_n$.
- b) For any finite collection $\{W_i\}$ of pairwise disjoint subsets of H we have

$$E_C(\bigcup_i W_i) = \sum_i E_C(W_i).$$

- c) If $W_1 \subseteq W_2 \subseteq H$, then $E_C(W_1) \leq E_C(W_2)$.

LEMMA 3. Let $r \in \{0, 1, \dots, n\}$ and $W, C, C' \subseteq H$.

If $C' \subseteq C$, then $E_{C'}^r(W) \leq E_C^r(W)$.

DEFINITION 4. For a code C in H we define

- a) $Z(C) := \{z \in H \mid E_C(\{z\}) > 0\}$, and
- b) $G(C) := H \setminus \bigcup_{c \in C} B_{PR(C)}(c)$.

If it is obvious which code is meant, we simply write Z, G , respectively.

From (2) and Definition 4a we have

$$Z(C) = \{z \in H \mid |B_{CR(C)}(z) \cap C| > 1\}, \quad (3)$$

and by Lemma 2a and b,

$$|Z| \leq E_C(Z) = E_C(H) = |C| V(CR(C)) - q_1 q_2 \cdots q_n. \quad (4)$$

Clearly,

$$|G| = q_1 q_2 \cdots q_n - |C| V(PR(C)). \quad (5)$$

Also, if $W \subseteq H$ then

$$|G \cap W| = |W| - \sum_{c \in C} |B_{PR(C)}(c) \cap W|. \quad (6)$$

On the analogy of Definition 1, we could call this number the *deficit* on W by C .

III. THE Q-ARY CASE

In this and the next section we confine ourselves to the case that for some $q \geq 2$ $q_1 = q_2 = \dots = q_n = q$. One usually takes $H = \mathbb{Z}_q^n$ (or $\mathbb{F}_q^n$, if possible). The number of words in a sphere of radius r is now denoted by $V_q(n, r)$ (cf. [32, Section VII]). This number equals

$$\sum_{i=1}^r \binom{n}{i} (q-1)^i.$$

A. Coverings

The notation $K_q(n, R) := \min \{ |C| \mid \emptyset \neq C \subseteq H, CR(C) = R \}$ has more or less become standard. In this Part A let C be a code in H with $CR(C) = R$. Furthermore, put

$$A := \{x \in H \mid d(x, C) = R\}, \text{ and}$$

$$\varepsilon := \left\lceil \frac{(n-R)(q-1)}{R+1} \right\rceil (R+1) - (n-R)(q-1).$$

LEMMA 5.

- a) $E_C(B_1(x)) \geq 1$ if $x \in Z$,
- b) $E_C(B_1(x)) \geq \varepsilon$ if $x \in A - Z$.

PROOF. a) Obvious. b) There is a unique $c \in C$ with $d(x, c) = R$. For every $b \in C \setminus \{c\}$ we have $d(b, x) \geq R+1$, and hence $|B_R(b) \cap B_1(x)| \in \{0, R+1\}$. Consequently, by (2),

$$\begin{aligned}
E_C(B_1(\mathbf{x})) &\equiv |B_R(\mathbf{c}) \cap B_1(\mathbf{x})| - |B_1(\mathbf{x})| = \\
&= 1 + R(q-1) - (1 + n(q-1)) = (n-R)(q-1) \pmod{R+1},
\end{aligned}$$

from which b) follows

THEOREM 6. If $\varepsilon > 0$, then

$$K_q(n, R) \geq \frac{(n(q-1)-R-1+2\varepsilon)q^n}{(n(q-1)-R-1+\varepsilon)V_q(n, R)+\varepsilon V_q(n, R-1)}. \quad (7)$$

PROOF. For every $\mathbf{z} \in Z$ we have $|A \cap B_1(\mathbf{z})| \leq n(q-1) - R$ (compare with [32, Eq. (14)]). We shall not prove this here, but refer the reader to [19, Lemma 10], where a more general situation is considered. We now have

$$\begin{aligned}
\varepsilon(q^n - |C| V_q(n, R-1)) - (\varepsilon-1)(|C| V_q(n, R) - q^n) \\
&\leq \varepsilon|A| - (\varepsilon-1)|Z| \quad (\text{by (4)}) \\
&\leq \varepsilon|A| - (\varepsilon-1)|A \cap Z| \\
&= \varepsilon|A - Z| + 1 \cdot |A \cap Z| \\
&\leq \sum_{\mathbf{a} \in A - Z} E_C(B_1(\mathbf{a})) + \sum_{\mathbf{a} \in A \cap Z} E_C(B_1(\mathbf{a})) \quad (\text{by Lemma 5}) \\
&= \sum_{\mathbf{a} \in A} E_C(B_1(\mathbf{a})) \\
&= \sum_{\mathbf{a} \in A} \sum_{\mathbf{x} \in B_1(\mathbf{a})} E_C(\{\mathbf{x}\}) \quad (\text{by Lemma 2b}) \\
&= \sum_{\mathbf{a} \in A} \sum_{\mathbf{x} \in B_1(\mathbf{a}) \cap Z} E_C(\{\mathbf{x}\}) \quad (\text{by Definition 4a}) \\
&= \sum_{\mathbf{x} \in Z} \sum_{\mathbf{a} \in A \cap B_1(\mathbf{x})} E_C(\{\mathbf{x}\}) \\
&= \sum_{\mathbf{x} \in Z} |A \cap B_1(\mathbf{x})| E_C(\{\mathbf{x}\}) \\
&\leq \sum_{\mathbf{x} \in Z} (n(q-1) - R) E_C(\{\mathbf{x}\}) \\
&= (n(q-1) - R) E_C(Z) \quad (\text{by Lemma 2b}) \\
&= (n(q-1) - R)(|C| V_q(n, R) - q^n) \quad (\text{by (4)}),
\end{aligned}$$

and (7) follows.

COROLLARY 7. If q and n are even, then

$$K_q(n, 1) \geq \frac{q^n}{n(q-1)}.$$

Here the sphere covering bound (1) reads

$$K_q(n, 1) \geq \frac{q^n}{1+n(q-1)}.$$

Corollary 7 is a generalization of [32, Corollary 1a].

Remarks:

- 1) If we substitute $\varepsilon = 0$ in (7), we get the sphere bound (provided the denominator is positive). The condition $\varepsilon > 0$ is equivalent to $(n - R)(q - 1) \not\equiv 0 \pmod{R + 1}$, and then (7) is better than (1).
- 2) For $R > 1$, Theorem 6 is not really a generalization of [32, Th. 9]. This is because of the estimation $E_C(B_1(x)) \geq 1$ for $x \in A \cap Z$ that we used. For $q = 2$, Theorem 9 in [32] is always at least as good as (7). A slight improvement on (7) could be achieved by considering the case $x \in A \cap Z$ more carefully.
- 3) The case $q = 3$ is of special interest. Tables with lower and upper bounds on $K_3(n, R)$, for $n \leq 13$ and $R = 1, 2, 3$, recently appeared in [8] and [20]. The new bound (7) improves on the lower bounds in these tables at several places. In [19] Theorem 6 for the cases $q = 2$ and $q = 3$ is generalized to a bound (Th. 5 in [19]) which applies to mixed covering codes with all $q_i \in \{2, 3\}$. For the computed values of (7) for $q = 3$, $n \leq 13$, $R = 1, 2, 3$, the reader is referred to Table I in [19] (the cases with $b = 0$). We list the improvements by Theorem 6 on [20, Table I] in Corollary 8 (the old values of [20] given in brackets). Theorem 6 also gives alternative proofs of $K_3(6, 2) \geq 11$ and $K_3(8, 3) \geq 13$, results which were already found in [20] by linear programming.

COROLLARY 8.

$$K_3(7, 2) \geq 25 \quad (24)$$

$$K_3(9, 2) \geq 128 \quad (121)$$

$$K_3(10, 2) \geq 322 \quad (294)$$

$$K_3(10, 3) \geq 56 \quad (51)$$

$$K_3(12, 2) \geq 1915 \quad (1839)$$

$$K_3(12, 3) \geq 280 \quad (260)$$

$$K_3(13, 2) \geq 5048 \quad (4704) .$$

B. Packings

In the case of packings we can use similar arguments as with coverings to obtain a general bound (Theorem 11) better than the sphere packing bound. Throughout this Part B we assume that C is a code in H with $PR(C) = e$. We put

$$A := \{x \in H \mid d(x, C) = e\} ,$$

$$\delta := (n - e)(q - 1) - \left\lfloor \frac{(n - e)(q - 1)}{e + 1} \right\rfloor (e + 1) , \text{ and}$$

$$\delta' := \left\lfloor \frac{n(q - 1)}{e + 1} \right\rfloor (e + 1) .$$

To have a positive denominator in Theorem 11, we make the (nonsignificant) assumption that $e < n$.

LEMMA 9.

$$|G \cap S_1(x)| \geq \delta \quad \text{for all } x \in A .$$

PROOF. There is a unique $c \in C$ with $d(x, c) = e$. For every $b \in C \setminus \{c\}$ we have $d(b, x) \geq e + 1$ and hence $|B_e(b) \cap S_1(x)| \in \{0, e + 1\}$. By (6) we have

$$\begin{aligned} |G \cap S_1(x)| &= |S_1(x)| - |B_e(c) \cap S_1(x)| = \\ &= n(q - 1) - e(q - 1) = (n - e)(q - 1) \pmod{e + 1} , \end{aligned}$$

and the lemma follows.

LEMMA 10.

$$|A \cap S_1(z)| \leq \delta' \quad \text{for all } z \in G.$$

PROOF. Let $z \in G$. Every $c \in C$ has $d(c, z) \geq e + 1$. If $a \in A \cap S_1(z)$, then a $c \in C$ exists with $d(c, a) = e$. For this c we have $d(c, z) = e + 1$, and $|S_e(c) \cap S_1(z)| = e + 1$. Since $PR(C) = e$, there are at most $\lfloor |S_1(z)| / (e + 1) \rfloor = \lfloor n(q-1)/(e+1) \rfloor$ such codewords c . Therefore $|A \cap S_1(z)| \leq \delta'$.

THEOREM 11.

$$|C| \leq \frac{\delta' q^n}{\delta' V_q(n, e) + \delta \binom{n}{e} (q-1)^e} \quad (8)$$

PROOF.

$$\begin{aligned} \delta |C| \binom{n}{e} (q-1)^e &= \delta |A| \\ &\leq \sum_{a \in A} |G \cap S_1(a)| \quad (\text{by Lemma 9}) \\ &= \sum_{z \in G} |A \cap S_1(z)| \\ &\leq \delta' |G| \quad (\text{by Lemma 10}) \\ &= \delta' (q^n - |C| V_q(n, e)) \quad (\text{by (5)}) , \end{aligned}$$

and (8) follows.

Remarks:

- 1) The bound (8) improves on the sphere packing bound (1) whenever $\delta > 0$, which is the case if and only if $(n - e)(q - 1) \not\equiv 0 \pmod{e + 1}$.

- 2) If $q = 2$ and $\delta > 0$, then $\delta' = 1 + n - \delta$.
- 3) To get an impression of the performance of (8), the reader is referred to [19, Table I]. As was the case with (7), the bound (8) for the special cases $q = 2$ and $q = 3$ is generalized in [19] to a bound (Theorem 9 in [19]) for packing codes in $\mathbb{F}_3^t \mathbb{F}_2^b$. The cases with $b = 0$ in [19, Table I] give the computed values of (8) for $q = 3, n \leq 13, R = 1, 2, 3$.

COROLLARY 12. If $e = 1$ and q and n are even, then

$$|C| \leq \frac{q^n}{2+n(q-1)}.$$

It is interesting to compare this with Corollary 7 and the remark that followed it.

IV. LOWER BOUNDS FOR THE FOOTBALL POOL PROBLEM

We now consider the q -ary covering problem in the particular case that $q = 3$ and $R = 1$. A list of the best lower and upper bounds on $K_3(n, 1)$ known for $n \leq 13$ was given in [20, first column of Table I]. the bound of Theorem 6 does not give any new results in this case, because for all n we have $\varepsilon = 0$ if $q = 3$ and $R = 1$.

Since the football pool problem is of such great interest, we try harder and do something that was called "counting excess on spheres with radius 2" in [32]. We assume throughout that C is a code in H with $CR(C) = 1$.

LEMMA 13.

- a) $E_C(B_2(x)) \geq 2$ if $n \equiv 2 \pmod{3}$ and $x \in C$,
- b) $E_C(B_2(x)) \geq 1$ if $n \not\equiv 1 \pmod{3}$ and $x \notin C$.

PROOF. a) If there is a $c \in C \setminus \{x\}$ with $d(x, c) \leq 2$, then $|B_1(c) \cap B_1(x)| \in \{2, 3\}$; hence, by Lemmas 2c and 3, $E_C(B_2(x)) \geq E_{\{c, x\}}(B_1(x)) \geq 2$. If there is no such c , then for all $b \in C \setminus \{x\}$ $|B_1(b) \cap B_2(x)| \in \{0, 3\}$. Hence, by (2),

$$\begin{aligned}
 E_C(B_2(x)) &\equiv |B_1(x)| - |B_2(x)| \\
 &= - \binom{n}{2} \cdot 4 = -2n(n-1) \equiv 2 \pmod{3},
 \end{aligned}$$

and a) follows.

b) There is a $c \in C$ such that $d(x, c) = 1$. We have $|B_1(c) \cap B_1(x)| = 3$. If $E_C(B_1(x)) \geq 1$, then also $E_C(B_2(x)) \geq 1$ (Lemma 2c). Suppose that $E_C(B_1(x)) = 0$. Then there are no other codewords than c at distance 1 from x . For every $b \in C$ with $d(x, b) = 2$ we have $|B_1(b) \cap B_1(x)| = 2$. By (2), we have

$$\begin{aligned}
 0 &= E_C(B_1(x)) \\
 &= |B_1(c) \cap B_1(x)| + \sum_{b \in C \setminus \{c\}} |B_1(b) \cap B_1(x)| - |B_1(x)| \\
 &= 3 + \sum_{b \in C: d(x, b)=2} 2 - (1 + 2n).
 \end{aligned}$$

Hence the number of codewords b with $d(x, b) = 2$ is $\frac{1}{2}(1 + 2n - 3) = n - 1$. For all such b we have $|B_1(b) \cap B_2(x)| = 5$. For all $d \in C$ with $d(x, d) \geq 3$ we have $|B_1(d) \cap B_2(x)| \in \{0, 3\}$. Substitution in (2) gives

$$\begin{aligned}
 E_C(B_2(x)) &\equiv |B_1(x)| + \sum_{b \in C: d(x, b)=2} |B_1(b) \cap B_2(x)| - |B_2(x)| \\
 &= (n-1) \cdot 5 - \binom{n}{2} \cdot 4 = (5-2n)(n-1) \\
 &\equiv (n-1)^2 \equiv 1 \pmod{3},
 \end{aligned}$$

which implies b).

THEOREM 14.

- a) $K_3(n, 1) \geq \frac{(V_3(n, 2)+1)3^n}{V_3(n, 2)(1+2n)-1}$ if $n \equiv 2 \pmod{3}$,
- b) $K_3(n, 1) \geq \frac{(V_3(n, 2)-1)3^n}{(V_3(n, 2)-2)(1+2n)+1}$ if $n \equiv 0 \pmod{3}$.

PROOF. a)

$$\begin{aligned}
& 2|C| + 1 \cdot (3^n - |C|) \\
& \leq \sum_{x \in C} E_C(B_2(x)) + \sum_{x \in H \setminus C} E_C(B_2(x)) \quad (\text{by Lemma 13}) \\
& = \sum_{x \in H} E_C(B_2(x)) \\
& = \sum_{x \in H} \sum_{y \in B_2(x)} E_C(\{y\}) \quad (\text{by Lemma 2b}) \\
& = \sum_{y \in H} \sum_{x \in B_2(y)} E_C(\{y\}) \\
& = \sum_{y \in H} V_3(n, 2) E_C(\{y\}) \\
& = V_3(n, 2) E_C(H) \quad (\text{by Lemma 2b}) \\
& = V_3(n, 2) (|C| + (1 + 2n) - 3^n) \quad (\text{by Lemma 2a}) .
\end{aligned}$$

b) Put $A := H \setminus C$. If $z \in Z$, then $|A \cap B_2(z)| \leq V_3(n, 2) - 2$ (see (3)). We have

$$\begin{aligned}
1 \cdot (3^n - |C|) & \leq \sum_{a \in A} E_C(B_2(a)) \quad (\text{by Lemma 13b}) \\
& = \sum_{a \in A} \sum_{x \in B_2(a) \cap Z} E_C(\{x\}) \quad (\text{by Lemma 2b and} \\
& \quad \text{Definition 4a}) \\
& = \sum_{x \in Z} \sum_{a \in A \cap B_2(x)} E_C(\{x\}) \\
& \leq \sum_{x \in Z} (V_3(n, 2) - 2) E_C(\{x\}) \\
& = (V_3(n, 2) - 2) E_C(Z) \quad (\text{by Lemma 2b}) \\
& = (V_3(n, 2) - 2) (|C| + (1 + 2n) - 3^n) \quad (\text{by (4)}) .
\end{aligned}$$

Remarks:

- 1) Compare Theorem 14 with [32, Corollary 2].
- 2) According to Lemma 13b the proof of Theorem 14b also holds for $n \equiv 2 \pmod{3}$. It is, however, easily verified that the bound in Theorem 14a is better then.

- 3) We list the improvements by Theorem 14 on Table I of [20] in the following corollary. The old lower bounds, all established by the sphere covering bound, are given in brackets.

COROLLARY 15.

$$K_3(8, 1) \geq 390 \quad (386)$$

$$K_3(9, 1) \geq 1043 \quad (1036)$$

$$K_3(11, 1) \geq 7736 \quad (7703)$$

$$K_3(12, 1) \geq 21329 \quad (21258) .$$

V. BINARY/TERNARY MIXED CODES

In this final section we assume that $H = F_3^t F_2^b$. Coverings of these spaces H have an obvious practical importance for football pools. Through the last decades many interesting covering codes were found by Finnish football pool fans, and often published in Finnish magazines. Also, a recent work of Hämäläinen and Rankinen [8] is concerned with this matter. They included a table with lower and upper bounds on covering codes in H .

It is not immediately clear how methods like those in Section III can be used in the case $H = F_3^t F_2^b$. Here, we treat only the special case of 1-coverings in detail (Theorem 16). By this example we hope to illustrate in a clear and comprehensive way how the idea of counting excess on spheres can be applied. For $R > 1$ things become much more complicated. In a joint work with J.H. van Lint, Jr., Theorem 16 is generalized for all covering radii R (see [19, Theorem 5]).

The situation for packings is quite different. We leave it as an easy exercise to derive Theorem 17 by altering the proof of Theorem 16. In contrast with the situation for coverings, here no extra difficulties arise when one tries to generalize for all packing radii e . We refer the reader for the general bound (which includes Theorem 17) to [19, Theorem 9].

Theorem 16 and its generalization [19, Theorem 5] give various improvements on the lower bounds in [8, Table I], which were all computed using the sphere covering bound (1). In [19] we present a new table with lower bounds on

coverings (and upper bounds on packings). We do not give the computed values of Theorems 16 and 17 here, but simply refer to that table (the column $e = 1, R = 1$).

THEOREM 16. Let C be a code in H with $CR(C) = 1$. Then

- a) $|C| \geq \frac{(2t+b)3^t 2^b}{(2t+b)(1+2t+b)-b}$ if b is even,
 b) $|C| \geq \frac{(2t+b)3^t 2^b}{(2t+b)(1+2t+b)-2t}$ if b is odd.

PROOF. a) Let A' denote the set of words in H that are equal to a codeword or can be obtained from a codeword by changing one ternary coordinate. Put $A := H \setminus A'$. Let $\mathbf{a} \in A$. Then $|B_1(\mathbf{c}) \cap B_1(\mathbf{a})| \in \{0, 2\}$ for all $\mathbf{c} \in C$. Since $|B_1(\mathbf{a})| = 1 + 2t + b$ is odd, $B_1(\mathbf{a}) \cap Z \neq \emptyset$. Therefore

$$A \subseteq \bigcup_{\mathbf{z} \in Z} B_1(\mathbf{z}) .$$

But then also

$$A \subseteq \bigcup_{\mathbf{z} \in Z} (B_1(\mathbf{z}) - C) .$$

We get

$$\begin{aligned} 3^t 2^b - |C| (1 + 2t) &\leq |A| \leq |Z| (1 + 2t + b - 2) \leq \\ &\leq (|C| (1 + 2t + b) - 3^t 2^b) (2t + b - 1) , \end{aligned}$$

and a) follows.

b) In this case let A' denote the set of words in H that are equal to a codeword or can be obtained from a codeword by changing one binary coordinate. Again we put $A := H \setminus A'$. Suppose $\mathbf{a} \in A$. Then there is a $\mathbf{c} \in C$ which differs from $\mathbf{a}$ in exactly one coordinate, which is ternary. We have $|B_1(\mathbf{c}) \cap B_1(\mathbf{a})| = 3$. If $\mathbf{a} \notin Z$, then $|B_1(\mathbf{b}) \cap B_1(\mathbf{a})| \in \{0, 2\}$ for all $\mathbf{b} \in C \setminus \{\mathbf{c}\}$. Since $|B_1(\mathbf{a})| = 1 + 2t + b$ is even, $B_1(\mathbf{a}) \cap Z \neq \emptyset$. If $\mathbf{a} \in Z$, then again $B_1(\mathbf{a}) \cap Z \neq \emptyset$. The rest of the proof is similar to the corresponding part of the proof of a) (use $|A| \geq 3^t 2^b - |C| (1 + b)$ here).

Theorem 16 generalizes Corollary 1a of [32]. It is always at least as good as the sphere covering bound.

This last statement also holds for Theorem 17, compared to the sphere packing bound.

THEOREM 17. Let C be a code in H with $PR(C) = 1$. Then

- a) $|C| \leq \frac{(2t+b)3^t 2^b}{(2t+b)(1+2t+b)+b}$ if b is even,
 b) $|C| \leq \frac{(2t+b)3^t 2^b}{(2t+b)(1+2t+b)+2t}$ if b is odd.

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GENERALIZED BOUNDS ON BINARY/TERNARY MIXED PACKING- AND COVERING CODES

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ABSTRACT

We derive lower bounds on R -coverings and upper bounds on e -packings of $\mathbb{F}_3^t \mathbb{F}_2^b$. These bounds are generalizations of the bounds for the cases $R = 1$, $e = 1$ or $t = 0$ or $b = 0$, which were already derived in a paper by the second-named author. The present work is a supplement to that paper. The computed results for $t + b \leq 13$ and $R, e \leq 3$ are collected in a table.

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I. INTRODUCTION

One of the goals of [5] was to illustrate that the ideas of [6] can also be successfully applied to mixed codes. In Section V of [5] a lower bound on 1-coverings of $\mathbb{F}_3^t \mathbb{F}_2^b$ was derived. This paper may be seen as a supplement to that section of [5]. We generalize Theorem 16 of [5] to a bound which holds for all covering radii R (Theorem 5). The proof of [5, Theorem 16] could be kept short and easy. For $R > 1$, however, one meets several complications. Although Section IIA might look rather technical, we hope to have put things in a readable way.

So far about coverings. As was mentioned in [5, Section V], the situation for packings is quite different. Theorem 17 of [5] already dealt with 1-packings. Without any real difficulty, this bound can be generalized to a bound for all packing radii e . We fulfil this task in Section IIB (Theorem 9).

We assume that the reader is familiar with [5, especially Sections I, II and V], and we shall use the same notation and terminology. In addition, we shall denote the number of words in a sphere with radius r in $H := \mathbb{F}_3^t \mathbb{F}_2^b$ by $V(t, b, r)$ (as in [1]). Hence

$$V(t, b, r) = \sum_{i=0}^r \sum_{j=0}^i \binom{t}{j} 2^j \binom{b}{i-j},$$

(with the convention that $\binom{n}{k} = 0$ if $k > n$ or $k < 0$).

Also, in [1] the notation

$$K(t, b, R) := \min \{ |C| \mid \emptyset \neq C \subseteq H, CR(C) = R \}$$

was used.

The computed values for $n = t + b \leq 13$ of the bounds that we derive in Section II, are assembled in Table I in Section III. This table contains many improvements on the tables in [1] and [3].

Many of our results were already presented in [2].

II. THE BOUNDS

In this paper (except in the Appendix) we are dealing with the case $H = F_3^t F_2^b$. First we define the "ternary" and "binary" distance between two words in H .

DEFINITION 1. Let $x, y \in H$. We define

$$d^t(x, y) := |\{i \in \{1, 2, \dots, t\} \mid x_i \neq y_i\}|, \text{ and}$$

$$d^b(x, y) := |\{i \in \{t+1, t+2, \dots, t+b\} \mid x_i \neq y_i\}|.$$

Hence we have $d^t(x, y) + d^b(x, y) = d(x, y)$. The following observation is fundamental to our investigations.

LEMMA 2. Let $x, y \in H, r \in \{0, 1, \dots, n\}$ and $d(x, y) \geq r$.

Then

$$|B_r(x) \cap B_1(y)| = \begin{cases} 0 & \text{if } d(x, y) > r+1 \\ r+1 & \text{if } d(x, y) = r+1 \\ r+1+\tau & \text{if } d(x, y) = r \text{ and } d^t(x, y) = \tau. \end{cases}$$

A. Coverings

Throughout Part A of this section we assume that C is a code in H with $CR(C) = R$ and $|C| = M$.

Put

$$A := \{x \in H \mid d(x, C) = R\}. \quad (1)$$

For $j = 0, 1, \dots, R$, let $\tau_j \in \{0, 1, \dots, R\}$ be such that

$$\tau_j \equiv 1 + 2t + b + j \pmod{R+1}, \quad (2)$$

and put

$$A_j := \{x \in A \mid \forall c \in C (d(x, c) = R \Rightarrow d^t(x, c) = \tau_j)\}, \quad (3)$$

$$Z_j := A_j \cap Z, \quad (4)$$

$$A'_j := A_j \setminus Z_j . \quad (5)$$

Note that $(A \cap Z) \cup A'_0 \cup A'_1 \cup \dots \cup A'_R$ is a partition of A , and that some of the defined sets may be empty.

LEMMA 3. Let $a \in A$. Then

$$E_C(B_1(a)) \geq \begin{cases} 1 & \text{if } a \in A \cap Z \\ j & \text{if } a \in A'_j \text{ for some } j . \end{cases}$$

PROOF. The case $a \in A \cap Z$ is trivial. If $a \in A'_j$, then there is a unique $c \in C$ with $d(c, a) = R$. By (3) and Lemma 2 $|B_R(c) \cap B_1(a)| = R + 1 + \tau_j$. For all $b \in C \setminus \{c\}$ we have $d(b, a) \geq R + 1$, hence (by Lemma 2) $|B_R(b) \cap B_1(a)| \in \{0, R + 1\}$. Using [5, Definition 1] and (2), we conclude that

$$\begin{aligned} E_C(B_1(a)) &\equiv |B_R(c) \cap B_1(a)| - |B_1(a)| \\ &= R + 1 + \tau_j - (1 + 2t + b) \equiv j \pmod{R + 1} . \end{aligned}$$

Since $0 \leq j \leq R$, the result follows.

LEMMA 4. If $z \in Z$, then $|A \cap B_1(z)| \leq 2t + b - R$.

This lemma is a special case of a more general result, namely Lemma 10, that is proved in the Appendix. Lemma 10 is also used in the proof of [5, Theorem 6].

For $j = 0, 1, \dots, R$ we define

$$T_j := \begin{bmatrix} t \\ \tau_j \end{bmatrix} 2^{\tau_j} \begin{bmatrix} b \\ R - \tau_j \end{bmatrix} \quad (6)$$

(with the usual convention that $\begin{bmatrix} n \\ k \end{bmatrix} = 0$ if $k > n$ or $k < 0$). Note that if $x \in H$, then $T_j = |\{y \in H \mid d(x, y) = R \wedge d'(x, y) = \tau_j\}|$. We also define

$$\begin{aligned} L_j &:= \max(0, 3^t 2^b - M(V(t, b, R) - T_j)) \quad (j = 0, 1, \dots, R), \text{ and} \\ L &:= \max(0, 3^t 2^b - M(V(t, b, R - 1) + T_0)) . \end{aligned}$$

Hence

$$|A_j| \geq L_j \quad (j = 0, 1, \dots, R), \text{ and} \quad (7)$$

$$|A \setminus A_0| \geq L. \quad (8)$$

Furthermore, let $j^* \in \{1, 2, \dots, R\}$ be taken such that $j^* = 1$ if

$$MV(t, b, R) - 3'2^b > \sum_{j=1}^R L_j, \text{ and such that}$$

$$\sum_{j=j^*+1}^R L_j \leq MV(t, b, R) - 3'2^b \leq \sum_{j=j^*}^R L_j \quad (9)$$

otherwise.

THEOREM 5.

$$\begin{aligned} (2t + b - R + j^* - 1) (MV(t, b, R) - 3'2^b) \\ \geq L + \sum_{j=1}^{j^*-1} (j-1) L_j + (j^* - 1) \sum_{j=j^*}^R L_j. \end{aligned} \quad (10)$$

PROOF.

$$\begin{aligned} (2t + b - R) (MV(t, b, R) - 3'2^b) \\ &= \sum_{x \in Z} (2t + b - R) E_C(\{x\}) \quad (\text{by [5, Eq. (4) and Lemma 2b]}) \\ &\geq \sum_{x \in Z} \sum_{a \in A \cap B_1(x)} E_C(\{x\}) \quad (\text{by Lemma 4}) \\ &= \sum_{a \in A} \sum_{x \in B_1(a) \cap Z} E_C(\{x\}) \\ &= \sum_{a \in A} E_C(B_1(a)) \quad (\text{by [5, Definition 4a} \\ &\quad \text{and Lemma 2b]}) \\ &= \sum_{a \in A \cap Z} E_C(B_1(a)) + \sum_{j=0}^R \sum_{a \in A'_j} E_C(B_1(a)) \\ &\geq 1 \cdot |A \cap Z| + \sum_{j=0}^R j |A'_j| \quad (\text{by Lemma 3}) \\ &= |A \setminus \bigcup_{j=0}^R A'_j| + \sum_{j=1}^R j |A'_j| \end{aligned}$$

$$= |A \setminus A_0| + \sum_{j=1}^R (j-1) (|A_j| - |Z_j|) =: (*) .$$

Let $j_0 \in \{1, 2, \dots, R\}$ be such that

$$\sum_{j=j_0+1}^R |A_j| \leq \sum_{j=1}^R |Z_j| \leq \sum_{j=j_0}^R |A_j| .$$

We claim that $j^* \leq j_0$. If $j^* = 1$, this is obviously true. If $j^* > 1$, then (using [5, Eq. (4)], (9) and (7)),

$$\begin{aligned} \sum_{j=1}^R |Z_j| &\leq |Z| \leq MV(t, b, R) - 3^t 2^b \\ &\leq \sum_{j=j^*}^R L_j \leq \sum_{j=j^*}^R |A_j| , \end{aligned} \quad (11)$$

and consequently $j^* \leq j_0$. This completes the proof of our claim.

We get

$$\begin{aligned} (*) &\geq |A \setminus A_0| + \sum_{j=1}^{j_0-1} (j-1) |A_j| + (j_0-1) \left(\sum_{j=j_0}^R |A_j| - \sum_{j=1}^R |Z_j| \right) \\ &\quad + \sum_{j=j_0+1}^R (j-1) (|A_j| - |A_j|) \\ &\geq |A \setminus A_0| + \sum_{j=1}^{j^*-1} (j-1) |A_j| + \sum_{j=j^*}^{j_0-1} (j-1) |A_j| \\ &\quad + (j^*-1) \left(\sum_{j=j_0}^R |A_j| - \sum_{j=1}^R |Z_j| \right) + 0 \quad (\text{since } j^* \leq j_0) \\ &\geq |A \setminus A_0| + \sum_{j=1}^{j^*-1} (j-1) |A_j| + (j^*-1) \left(\sum_{j=j^*}^R |A_j| - \sum_{j=1}^R |Z_j| \right) \\ &\geq L + \sum_{j=1}^{j^*-1} (j-1) L_j + (j^*-1) \left(\sum_{j=j^*}^R L_j - |Z| \right) \quad (\text{by (7) and (8)}). \end{aligned}$$

Substitution of $|Z| \leq MV(t, b, R) - 3^t 2^b$ (see (11)) gives the desired result.

Theorem 5 gives a lower bound on $K(t, b, R)$. If, for given t, b and R , M_0 is the minimal M that satisfies (10), then it follows that $K(t, b, R) \geq M_0$. The only thing that we need now is an easy way to determine this M_0 . Because the L_j, L and j^* depend on M , it is not immediately clear that $M+1$ satisfies (10) if M does. That this is true nevertheless is shown by Lemma 6. By this lemma, M_0

can be determined very quickly, for instance by a binary search on the interval $[Lb, Ub]$, where Lb is taken the best lower bound on $K(t, b, R)$ known (or simply the sphere covering bound), and

$$Ub = \min \{2^k \cdot Lb \mid k \in \mathbb{N} \cup \{0\} \text{ and } M = 2^k \cdot Lb \text{ satisfies (10)}\}.$$

The above method is the one that we used to compute Table I.

LEMMA 6. If for given t, b and R, M satisfies (10), then so does $M + 1$.

PROOF. For convenience, we put

$$\begin{aligned} P(M) &:= MV(t, b, R) - 3^t 2^b, \text{ and} \\ Q(M) &:= L + \sum_{j=1}^{j^*-1} (j-1) L_j + (j^* - 1) \left(\sum_{j=j^*}^R L_j - P(M) \right). \end{aligned} \quad (12)$$

In (12) the L_j, L and j^* are calculated using M . As with $P(M)$ and $Q(M)$, in the rest of the proof we shall write $L_j(M), L(M)$ and $j^*(M)$ to indicate that M is used. Now " M satisfies (10)" is equivalent to " $(2t + b - R)P(M) \geq Q(M)$ " (see the proof of Th. 5). One immediately verifies that $P(M+1) \geq P(M)$, $L_j(M+1) \leq L_j(M)$ for all j , and $L(M+1) \leq L(M)$. It follows that $j^*(M+1) \leq j^*(M)$.

If $j^*(M+1) = j^*(M)$, then

$$\begin{aligned} Q(M+1) &= L(M+1) + \sum_{j=1}^{j^*(M)-1} (j-1) L_j(M+1) \\ &\quad + (j^*(M) - 1) \left(\sum_{j=j^*(M)}^R L_j(M+1) - P(M+1) \right) \\ &\leq Q(M). \end{aligned}$$

If $j^*(M+1) < j^*(M)$, then by the first inequality of (9) (which is always true),

$$\sum_{j=j^*(M+1)}^R L_j(M+1) - P(M+1) \leq L_{j^*(M+1)}(M+1). \quad (13)$$

We get

$$Q(M+1) = L(M+1) + \sum_{j=1}^{j^*(M+1)-1} (j-1) L_j(M+1) +$$

$$\begin{aligned}
& (j^*(M+1) - 1) \left(\sum_{j=j^*(M+1)}^R L_j(M+1) - P(M+1) \right) \\
& \leq L(M+1) + \sum_{j=1}^{j^*(M+1)} (j-1) L_j(M+1) \quad (\text{by (13)}) \\
& \leq L(M) + \sum_{j=1}^{j^*(M)-1} (j-1) L_j(M) \leq Q(M)
\end{aligned}$$

(note that all three terms of (12) are nonnegative). We conclude that in both cases $Q(M+1) \leq Q(M)$. The lemma now follows from

$$(2t+b-R)P(M+1) \geq (2t+b-R)P(M) \geq Q(M) \geq Q(M+1) .$$

Remarks:

- 1) As already said, Theorem 5 generalizes [5, Theorem 16]. It also generalizes the cases $q=2$ and $q=3$ of [5, Theorem 6].
- 2) Remark 2 of [5, Section IIIA] almost literally applies; i.e., Theorem 5 is not a generalization of [6, Theorem 9].
- 3) Theorem 5 is always at least as good as the sphere covering bound.

B. Packings

Here we assume that C is a code in H with $PR(C) = e$ and $|C| = M$. For $j=0, 1, \dots, e$ let $\theta_j \in \{0, 1, \dots, e\}$ be such that

$$\theta_j \equiv 1 + 2t + b - j \pmod{e+1} . \quad (14)$$

We define A , the A_j and T_j ($j=0, 1, \dots, e$) as in (1), (3) and (6), with R replaced by e and τ_j by θ_j , and we put

$$\delta' := \left\lfloor \frac{2t+b}{e+1} \right\rfloor (e+1) .$$

In order to have a positive denominator in Theorem 9, we make the (nonsignificant) assumption that $e < t+b$.

LEMMA 7.

$$|G \cap S_1(a)| \geq j \text{ if } a \in A_j .$$

PROOF. Let $a \in A_j$. Then there is a unique $c \in C$ with $d(c, a) = e$. By the definition of A_j and Lemma 2, $|B_e(c) \cap S_1(a)| = e + 1 + \theta_j - 1 = e + \theta_j$. For every $b \in C \setminus \{c\}$ we have $d(b, a) \geq e + 1$ and hence $|B_e(b) \cap S_1(a)| = |B_e(b) \cap B_1(a)| \in \{0, e + 1\}$. By [5, Eq. (6)] and (14) we have

$$\begin{aligned} |G \cap S_1(a)| &\equiv |S_1(a)| - |B_e(c) \cap S_1(a)| \\ &= 2t + b - (e + \theta_j) \equiv j \pmod{e + 1}, \end{aligned}$$

from which the lemma follows.

LEMMA 8. For all $z \in G$

$$|A \cap S_1(z)| \leq \delta'.$$

PROOF. The proof of [5, Lemma 10] is valid here if we use $|S_1(z)| = 2t + b$ instead of $n(q - 1)$.

THEOREM 9.

$$|C| \leq \frac{\delta' 3^t 2^b}{\delta' V(t, b, e) + \sum_{j=0}^e j T_j} \quad (15)$$

PROOF.

$$\begin{aligned} \sum_{j=0}^e j |C| T_j &= \sum_{j=0}^e j |A_j| \\ &\leq \sum_{j=0}^e \sum_{a \in A_j} |G \cap S_1(a)| \quad (\text{by Lemma 7}) \\ &= \sum_{a \in A} |G \cap S_1(a)| \\ &= \sum_{z \in G} |A \cap S_1(z)| \\ &\leq \delta' |G| \quad (\text{by Lemma 8}) \\ &= \delta' (3^t 2^b - |C| V(t, b, e)) \quad (\text{by [5, Eq. (5)]}) \end{aligned}$$

Remarks:

- 1) Theorem 9 generalizes [5, Theorem 17] and it also generalizes the cases $q = 2$ and $q = 3$ of [5, Theorem 11].
- 2) The bound (15) is always at least as good as the sphere packing bound.

III. THE TABLE

Here we present the table that was announced in this paper and in [5] many times already. We have computed the bounds in Theorems 5 and 9 for $n = t + b \leq 13$. We did not include the cases with $t = 0$. The reason for this is that for binary coverings a better table already appeared in [6] (see Remark 2 in Section IIA), and that for binary packings Theorem 9 does not improve on the table in [4, p. 674].

The results for the cases with $e = 1$, $R = 1$ or $b = 0$ (or $t = 0$) were already established in [5, Theorems 6, 11, 16 and 17]. All other results in the table are due to Theorems 5 and 9 of the present work. For comparison, in the table we have given the sphere packing bound and the sphere covering bound in brackets below the bounds obtained from Theorem 9 and Theorem 5, respectively.

APPENDIX

Here we prove a lemma that has been used in the proof of [5, Theorem 6] as well as in this paper (Lemma 4). We state it in a general form which applies to both situations. Lemma 10 could also be used instead of [6, Lemma 7b] in the proof of [6, Theorem 9].

Consider a general Hamming space H (as in [5, Sections I and II]). Let C be a code in H with $CR(C) = R$ and put $A := \{x \in H \mid d(x, C) = R\}$.

LEMMA 10. If $z \in Z(C)$, then

$$|A \cap B_1(z)| \leq \sum_{i=1}^n (q_i - 1) - R. \quad (16)$$

PROOF. Let $z \in Z$. By [5, Eq. (3)] there are different codewords b and c such that $d_1 := d(z, b) \leq R$ and $d_2 := d(z, c) \leq R$. W.l.o.g. $d_1 \leq d_2$. If $d_1 \leq R - 2$, then $B_1(z) \subseteq B_{R-1}(b)$, hence $|A \cap B_1(z)| = 0$, and (16) holds. Now suppose that $d_1 \geq R - 1$. Let i be a coordinate such that $c_i \neq z_i$ and $b_i \neq c_i$. Put

$$x := (z_1 z_2 \cdots z_{i-1} c_i z_{i+1} \cdots z_n),$$

$$F_1 := B_1(z) \cap B_{R-1}(b)$$

and

$$F_2 := B_1(z) \cap B_{R-1}(c) - B_{R-1}(b).$$

We have

$$|A \cap B_1(z)| \leq |B_1(z) \cap (B_{R-1}(b) \cup B_{R-1}(c))| = |B_1(z)| - |F_1| - |F_2|.$$

We claim that

$$|F_1| + |F_2| \geq R + 1.$$

To prove this, we distinguish between the cases i) $d_1 = R - 1$ and $b_i \neq z_i$, and ii) $d_1 = R$ or $b_i = z_i$. In case i) F_1 contains the $R - 1$ words

$$(z_1 z_2 \cdots z_{j-1} b_j z_{j+1} \cdots z_n),$$

where j runs through the coordinates where b and z differ, as well as z and x . Hence

$$|F_1| + |F_2| \geq R + 1 + 0 = R + 1.$$

In case ii) we have $d(x, z) = 1$, $d(x, b) \geq R$ and $d(x, c) \leq R - 1$, hence $x \in F_2$. Since $|F_1| \geq R$, we conclude that also in this case

$$|F_1| + |F_2| \geq R + 1.$$

This completes the proof of our claim, and (16) follows.

TABLE I
Upper Bounds on Packings and Lower Bounds on Coverings of $F_3^t F_2^b$

t	b	e=1	R=1	e=2	R=2	e=3	R=3
		Th. 9	Th. 5	Th. 9	Th. 5	Th. 9	Th. 5
1	0	- (1)	- (1)				
1	1	1 (1)	2 (2)	- (1)	- (1)		
1	2	2 (2)	3 (3)	1 (1)	2 (2)	- (1)	- (1)
1	3	3 (4)	5 (4)	1 (1)	2 (2)	1 (1)	2 (2)
1	4	6 (6)	8 (7)	2 (2)	3 (3)	1 (1)	2 (2)
1	5	11 (12)	13 (12)	2 (3)	4 (4)	1 (1)	2 (2)
1	6	19 (21)	24 (22)	4 (5)	6 (6)	2 (2)	3 (3)
1	7	37 (38)	40 (39)	8 (8)	10 (9)	2 (3)	4 (4)
1	8	65 (69)	76 (70)	12 (13)	16 (14)	3 (4)	6 (5)
1	9	125 (128)	130 (128)	21 (23)	24 (24)	6 (6)	8 (7)
1	10	222 (236)	253 (237)	37 (39)	44 (40)	10 (10)	12 (11)
1	11	433 (438)	444 (439)	60 (67)	74 (68)	15 (16)	19 (17)
1	12	774 (819)	869 (820)	112 (117)	120 (118)	23 (26)	30 (27)
2	0	1 (1)	2 (2)	- (1)	- (1)		
2	1	2 (3)	4 (3)	1 (1)	2 (2)	- (1)	- (1)
2	2	4 (5)	6 (6)	1 (1)	2 (2)	1 (1)	2 (2)
2	3	8 (9)	10 (9)	2 (2)	4 (3)	1 (1)	2 (2)
2	4	15 (16)	17 (16)	3 (4)	5 (5)	1 (1)	2 (2)
2	5	27 (28)	31 (29)	6 (6)	7 (7)	2 (2)	3 (3)

Table continued

TABLE I - continued

t	b	e=1	R=1	e=2	R=2	e=3	R=3
		Th. 9	Th. 5	Th. 9	Th. 5	Th. 9	Th. 5
2	6	49 (52)	56 (53)	9 (10)	12 (11)	3 (3)	5 (4)
2	7	92 (96)	99 (96)	16 (17)	19 (18)	4 (5)	6 (6)
2	8	168 (177)	187 (178)	28 (29)	32 (30)	7 (8)	9 (9)
2	9	321 (329)	337 (330)	46 (51)	56 (52)	12 (13)	15 (14)
2	10	586 (614)	646 (615)	83 (88)	92 (89)	18 (20)	23 (21)
2	11	1131 (1152)	1172 (1152)	149 (154)	165 (155)	30 (33)	35 (34)
3	0	3 (3)	4 (4)	1 (1)	2 (2)	- (1)	- (1)
3	1	6 (6)	8 (7)	2 (2)	3 (3)	1 (1)	2 (2)
3	2	11 (12)	13 (12)	2 (3)	4 (4)	1 (1)	2 (2)
3	3	20 (21)	24 (22)	4 (5)	6 (6)	1 (2)	3 (3)
3	4	37 (39)	41 (40)	7 (8)	9 (9)	2 (2)	4 (3)
3	5	68 (72)	76 (72)	12 (13)	15 (14)	3 (4)	5 (5)
3	6	128 (132)	139 (133)	21 (22)	24 (23)	6 (6)	7 (7)
3	7	238 (246)	256 (247)	36 (38)	43 (39)	9 (10)	12 (11)
3	8	443 (460)	480 (461)	61 (67)	71 (68)	14 (16)	18 (17)
3	9	841 (864)	887 (864)	113 (117)	123 (118)	22 (25)	28 (26)
3	10	1568 (1626)	1689 (1627)	192 (206)	223 (207)	40 (42)	45 (43)
4	0	9 (9)	9 (9)	2 (2)	3 (3)	1 (1)	2 (2)

Table continued

TABLE I - *continued*

t	b	e=1	R=1	e=2	R=2	e=3	R=3
		Th. 9	Th. 5	Th. 9	Th. 5	Th. 9	Th. 5
4	1	14 (16)	18 (17)	3 (3)	5 (4)	1 (1)	2 (2)
4	2	28 (29)	30 (30)	6 (6)	7 (7)	2 (2)	3 (3)
4	3	50 (54)	58 (54)	9 (10)	12 (11)	2 (3)	4 (4)
4	4	97 (99)	103 (100)	16 (17)	18 (18)	4 (5)	6 (6)
4	5	176 (185)	194 (186)	27 (29)	32 (30)	7 (7)	9 (8)
4	6	336 (345)	356 (346)	46 (50)	55 (51)	11 (12)	14 (13)
4	7	625 (648)	671 (648)	85 (88)	92 (89)	17 (19)	22 (20)
4	8	1184 (1219)	1257 (1220)	147 (155)	168 (156)	30 (32)	34 (33)
4	9	2241 (2304)	2366 (2304)	257 (276)	290 (277)	51 (53)	58 (54)
5	0	22 (22)	23 (23)	4 (4)	5 (5)	1 (1)	2 (2)
5	1	37 (40)	44 (41)	7 (7)	9 (8)	2 (2)	4 (3)
5	2	73 (74)	76 (75)	12 (13)	14 (14)	3 (3)	5 (4)
5	3	131 (138)	147 (139)	21 (22)	24 (23)	5 (6)	7 (7)
5	4	254 (259)	265 (260)	35 (38)	42 (39)	9 (9)	11 (10)
5	5	465 (486)	508 (486)	63 (67)	70 (68)	13 (15)	17 (16)
5	6	895 (914)	936 (915)	112 (117)	126 (118)	22 (25)	27 (26)
5	7	1670 (1728)	1787 (1728)	193 (208)	222 (209)	39 (41)	44 (42)
5	8	3199 (3274)	3353 (3275)	359 (372)	385 (373)	64 (68)	76 (69)

Table continued

TABLE I - continued

t	b	e=1	R=1	e=2	R=2	e=3	R=3
		Th. 9	Th. 5	Th. 9	Th. 5	Th. 9	Th. 5
6	0	56 (56)	57 (57)	8 (9)	11 (10)	2 (3)	4 (4)
6	1	97 (104)	112 (105)	16 (16)	18 (17)	4 (4)	6 (5)
6	2	192 (194)	197 (195)	26 (29)	33 (30)	7 (7)	8 (8)
6	3	345 (364)	384 (365)	47 (50)	53 (51)	10 (11)	14 (12)
6	4	676 (686)	697 (687)	86 (89)	94 (90)	17 (19)	21 (20)
6	5	1244 (1296)	1349 (1296)	145 (157)	170 (158)	30 (31)	33 (32)
6	6	2413 (2455)	2500 (2456)	268 (281)	290 (282)	49 (52)	58 (53)
6	7	4515 (4665)	4818 (4666)	486 (504)	532 (505)	80 (88)	96 (89)
7	0	145 (145)	146 (146)	20 (22)	25 (23)	5 (5)	6 (6)
7	1	257 (273)	291 (274)	34 (38)	41 (39)	8 (9)	11 (10)
7	2	510 (514)	519 (515)	66 (67)	70 (68)	13 (14)	17 (15)
7	3	926 (972)	1019 (972)	110 (119)	130 (120)	22 (24)	25 (25)
7	4	1820 (1841)	1864 (1842)	200 (212)	220 (213)	38 (40)	44 (41)
7	5	3368 (3499)	3634 (3500)	370 (380)	397 (381)	61 (67)	74 (68)
7	6	6571 (6665)	6762 (6666)	642 (686)	729 (687)	105 (114)	121 (115)
8	0	385 (385)	386 (386)	50 (50)	51 (51)	10 (11)	13 (12)
8	1	690 (729)	770 (729)	84 (89)	99 (90)	16 (18)	20 (19)
8	2	1373 (1381)	1390 (1382)	148 (160)	167 (161)	30 (30)	33 (31)

Table continued

TABLE I — *continued*

t	b	e=1	R=1	e=2	R=2	e=3	R=3
		Th. 9	Th. 5	Th. 9	Th. 5	Th. 9	Th. 5
8	3	2521 (2624)	2740 (2625)	282 (286)	295 (287)	47 (51)	57 (52)
8	4	4951 (4998)	5047 (4999)	485 (517)	555 (518)	80 (87)	93 (88)
8	5	9208 (9543)	9886 (9544)	892 (937)	964 (938)	142 (149)	155 (150)
9	0	1035 (1035)	1036 (1036)	109 (120)	128 (121)	23 (23)	24 (24)
9	1	1874 (1968)	2067 (1969)	215 (216)	219 (217)	37 (39)	45 (40)
9	2	3731 (3749)	3768 (3750)	367 (389)	422 (390)	61 (66)	72 (67)
9	3	6876 (7157)	7448 (7158)	664 (706)	731 (707)	105 (113)	118 (114)
9	4	13585 (13692)	13802 (13693)	1264 (1285)	1321 (1286)	191 (196)	207 (197)
10	0	2811 (2811)	2812 (2812)	279 (293)	322 (294)	46 (50)	56 (51)
10	1	5134 (5368)	5611 (5369)	493 (531)	555 (532)	78 (86)	91 (87)
10	2	10228 (10269)	10311 (10270)	960 (968)	983 (969)	147 (149)	154 (150)
10	3	18964 (19683)	20423 (19683)	1679 (1769)	1894 (1770)	243 (258)	283 (259)
11	0	7702 (7702)	7703 (7703)	729 (729)	729 (729)	113 (113)	114 (114)
11	1	14171 (14762)	15376 (14763)	1271 (1331)	1436 (1332)	187 (196)	217 (197)
11	2	28249 (28343)	28439 (28344)	2297 (2443)	2528 (2444)	316 (341)	365 (342)
12	0	21257 (21257)	21258 (21258)	1708 (1838)	1915 (1839)	242 (259)	280 (260)
12	1	39366 (40880)	42448 (40881)	3374 (3384)	3405 (3385)	415 (454)	470 (455)
13	0	59049 (59049)	59049 (59049)	4529 (4703)	5048 (4704)	606 (606)	607 (607)

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A Note on Perfect Multiple Coverings of the Hamming Space

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Abstract — Let Q be an alphabet of size $q \geq 2$. The *Hamming* space Q^n that consists of all n -tuples of elements of Q , is a metric space, provided with the *Hamming* distance function. A *perfect multiple covering (PMC)* is a code C in Q^n such that there exist fixed numbers r and μ with the property that every word in Q^n is within distance r from exactly μ codewords of C . We give a few constructions of PMC's, and investigate in detail the problem of determining all possible parameters of PMC's with $r=1$.

Index Terms: multiple coverings, covering radius, perfect codes, football pool problems.

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I. INTRODUCTION

Let q and n be integers, with $q \geq 2$ and $n \geq 1$. Let Q be an alphabet of size q . In this paper we usually take Q equal to $\mathbb{Z}_q$, the group of integers modulo q . Sometimes, if q is a prime power, we take $Q = \mathbb{F}_q$, the finite field with q elements. The set of n -tuples of elements of Q is a metric space, with the (*Hamming*-) distance d defined by $d(x, y) = |\{i | x_i \neq y_i\}|$, where $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n) \in Q^n$. Sometimes it is more convenient to write $x_1 x_2 \dots x_n$ instead of $(x_1, x_2, \dots, x_n)$. The elements of Q^n are usually called *words*. We also define $wt(x) = d(0, x)$, the *weight* of x , where the word with all coordinates equal to zero is denoted by 0 .

Let K be a nonempty set of possible messages, and suppose that

$$M := |K| \leq q^n.$$

Encoding the set of messages K means that one chooses an injective map

$$\phi: K \rightarrow Q^n.$$

The subset

$$C := \phi(K)$$

of Q^n is called a *code* (hence any nonempty subset of Q^n is a code). If q is a prime power, and C is a linear subspace of $\mathbb{F}_q^n$, C is called a *linear code* over $\mathbb{F}_q$. When we want to send a message $k \in K$ through a channel, the q -ary symmetric memoryless channel (see [8, p. 11]), say, the associated word

$$x := \phi(k) \in Q^n$$

is transmitted over this channel. The receiver receives a word y , which might differ from x , because of noise on the channel. The word y need not even be in the code C . In general we have

$$y = x + e,$$

where $e \in Q^n$ is the error vector. The weight of e is the number of errors that occurred during the transmission of x .

Nearest neighbour decoding (cf. [8, p. 9]) means that the receiver determines an $x' \in C$, such that

$$d(y, x') = d(y, C).$$

The outcome of this decoding procedure is this x' , or, in fact, the unique message k' such that $\phi(k') = x'$. We will not make this distinction between codewords and messages anymore. Correct decoding in this sense means that $x = x'$.

Under certain circumstances a different decoding procedure, called *list decoding*, can be useful. We shall not go into details about the practical purpose of list decoding. For this, see [1] and [3]. We only describe the procedure here. Let μ be a positive integer. When y is received, in the list decoding procedure the μ codewords in C closest to y are determined. More precisely, a set (list) L of μ codewords in C is determined, such that

$$d(x', y) \leq d(c, y), \text{ for all } x' \in L \text{ and } c \in C \setminus L.$$

The list L is the output of this decoding procedure. Correct decoding means that the originally transmitted word x is in the list L . The nearest neighbour decoding procedure described earlier is a special instance of list decoding, namely when we choose the parameter μ equal to 1.

For $x \in Q^n$, we denote

$$B_r(x) := \{y \in Q^n \mid d(x, y) \leq r\},$$

and this set is called the *sphere* with *radius* r and *center* x . We put

$$V_q(n, r) := |B_r(x)| = \sum_{i=0}^r \binom{n}{i} (q-1)^i.$$

Now let μ be fixed. Suppose we want to know how many errors our list decoder can handle. That is, we ask for the maximal number r , with $r \leq n$, such that the following holds: assuming that we always have $wt(e) \leq r$, the list decoder decodes correctly, for any $x \in C$ that is transmitted. One easily verifies that this number equals

$$\max\{r \in \{0, 1, \dots, n\} \mid \forall x \in Q^n: |B_r(x) \cap C| \leq \mu\}.$$

This brings us closer to the actual subject of this paper. A code $C \subseteq Q^n$ with

$$|B_r(x) \cap C| \leq \mu, \quad \text{for all } x \in Q^n \quad (1)$$

is called a μ -fold r -packing of Q^n . We have

$$|C| V_q(n, r) \leq \mu q^n.$$

A code C with

$$|B_r(x) \cap C| \geq \mu, \quad \text{for all } x \in Q^n \quad (2)$$

is called a μ -fold r -covering of Q^n . In this case,

$$|C| V_q(n, r) \geq \mu q^n.$$

If a code C satisfies both (1) and (2), it is called *perfect*. We then have

$$|C| V_q(n, r) = \mu q^n \quad (3)$$

(the *sphere packing* or *-covering equation*). One often speaks of (perfect) *multiple packings* or *-coverings*.

By a (q, n, M, r, μ) perfect multiple covering (PMC, cf. [2]) we mean a μ -fold r -covering of Q^n , with $|Q| = q$, that has M code-words. For example, the code C that equals Q^n is a

$$(q, n, q^n, r, V_q(n, r))$$

PMC, for any $r \in \{0, 1, \dots, n\}$. Other trivial examples are the codes with $r = n$: every code in Q^n is a (q, n, M, n, M) PMC, where M is the cardinality of the code.

In this paper we investigate for which parameters PMC's exist. We shall pay most attention to the case $r=1$. In Section II we give a few simple constructions of PMC's. In Section III we determine all possible parameters of PMC's in the case $r=1$ (Theorems 2 and 3), provided q is a prime. This restriction on q can be dropped if we restrict ourselves to *linear* PMC's. We also give a result that applies in the case that none of these two conditions is fulfilled (Theorem 4 and Corollary 1). After reading this paper, it will be clear that in this area there is still a lot of work left to be done. One of the goals of this paper is to stimulate further research.

When $\mu=1$ (r arbitrary), we are simply dealing with perfect codes in the ordinary sense (see [8, p. 19]). The determination of all parameters q , n , and r for which such a code exists is a well-known and widely studied problem. Extensive research has been done on this. For a survey and many references, see [5].

Multiple packings and coverings, perfect or not, are the topic of Clayton's thesis [2]. This thesis contains many interesting results. In [2], the definition of a PMC is somewhat different from the definition given above: Clayton allows a 'code' to be a multiset, i.e., a code can have repeated codewords. Every PMC by our definition is a PMC by Clayton's definition, but not the other way around. In this context, [2, Corollary 4.7] is most interesting. It says that if the parameters q , n , and r admit a PMC (repeated codewords allowed), then there also exists a subgroup of $\mathbb{Z}_q^n$ that is a

$$(q, n, q^{n-1}, r, V_q(n, r)/q)$$

PMC (without repeated codewords). Therefore, the results in [2] concerning the (non-)existence of PMC's also apply if we use our definition. For the sake of clearness: we only use our own definition in this paper.

Related to his different definition of a PMC, is the fact that Clayton always poses the question as follows: given q, n and r , does a (q, n, M, r, μ) PMC for *some* μ and *some* M with $1 < M < q^n$ exist? In other words, the parameter μ is not fixed (M follows from the parameters q, n, r , and μ , by (3)). In this paper we investigate a different problem: For which q, n, r , and μ does a (q, n, r, μ) PMC exist?

From the numerous results in [2], we only mention two (see [2, Theorems 4.8 and 4.9]):

For given r and q , there always exists a (q, n, M, r, μ) PMC for some n, μ and M with $r < n$ and $M < q^n$. More precisely, for given r and q , there exists a PMC for $n=rq+1$, $M=q^{rq}=q^{n-1}$, and $\mu=V_q(n, r)/q$. (4)

For $r=1$, a PMC exists for some μ and M with $M < q^n$ if and only if $n \equiv 1 \pmod{q}$. (5)

In this introduction we mentioned list decoding in connection to (perfect) multiple packings and coverings. Multiple coverings also come up in a different natural way. Suppose we have a football pool with 7 matches, each with $q=3$ possible outcomes: win, lose or draw. A forecast wins a prize if for at most one of the matches the predicted outcome is wrong. After the matches have been played, there is usually more than one winning forecast, and in this case the money has to be split. Now if a participant has μ winning forecasts by himself, and the total number of winning forecasts in the pool is much greater than μ , then he earns about μ times as much money as when he had only one winning forecast.

Suppose a participant wants to make sure that he wins a prize, whatever the outcomes of the matches are. Then he needs a 1-fold 1-covering of $\mathbb{F}_3^7$. The smallest such covering known at

present has 186 codewords, and is given in [4]. If this participant wants to win $\mu=5$ prizes, he could of course send in five times these 186 forecasts. This would take $5 \cdot 186 = 930$ forecasts. This strategy is not profitable, since for five times the costs he earns (less than) five times as much money. However, he can reach his goal in a cheaper way, namely by using a

$$(3, 7, 3^6 = 729, 1, 5)$$

PMC, which exists by Theorem 2(a). In this way the costs are considerably less than five times as high! By [7, Table I], any 1-fold 1-covering of $\mathbb{F}_3^7$ has at least 146 codewords, so even when the number 186 of [4] is improved, the first method still needs at least $5 \cdot 146 = 730 > 729$ forecasts.

II. SOME CONSTRUCTIONS

In this section we mention a few simple ways to construct PMC's. The 'constructions' (a), (b), and (c) were already mentioned in the Introduction.

- (a) Every ordinary perfect code is a PMC with $\mu=1$. Examples: binary repetition codes of odd length, Hamming codes, binary Golay code, ternary Golay code (see [8, p. 179, 180]).
- (b) The PMC's with $r=n$: every code C in Q^n is a (q, n, M, n, M) PMC, where $|C|=M$.
- (c) $C=Q^n$; this is a

$$(q, n, q^n, r, V_q(n, r))$$

PMC for any $r \in \{0, 1, \dots, n\}$.

- (d) "Complementing": This only works for binary codes. From any $(2, n, M, r, \mu)$ PMC C with $r < n$, we obtain a $(2, n, M, n-r-1, M-\mu)$ PMC, by replacing every codeword of C by its complement (that is, replace all zeroes by ones and all ones by zeroes).
- (e) "Shifting" of a group code: Suppose a (q, n, M, r, μ) PMC C is a subgroup of Q^n . For any $\mu' \in \{1, 2, \dots, q^n/M\}$, we get a $(q, n, \mu'M, r, \mu'\mu)$ PMC C' if we take C' equal to the union of C and $\mu'-1$ of its cosets. In particular, this works if q is a prime power, $Q = \mathbb{F}_q$ and C is a linear PMC.
- (f) The "Vasil'ev construction" (see [6, p. 90] and [8, p. 77]): Suppose q is a prime power and let $\mathbb{F}_q = \{0 = \alpha_0, \alpha_1, \dots, \alpha_{q-1}\}$. Let C be a code in $\mathbb{F}_q^n$. Define the maps π and T by

$$\pi: \mathbb{F}_q^{(q-1)n} \rightarrow \mathbb{F}_q, \quad \pi(v) = \sum_{i=1}^{q-1} \alpha_i \sum_{j=1}^n v_{(i-1)n+j},$$

and

$$T: \mathbb{F}_q^{(q-1)n} \rightarrow \mathbb{F}_q^n, \quad T(v) = (T_1(v), T_2(v), \dots, T_n(v)),$$

where

$$T_j(v) = \sum_{i=1}^{q-1} v_{(i-1)n+j} \quad (1 \leq j \leq n),$$

and

$$v = (v_1, v_2, \dots, v_{(q-1)n}) \in \mathbb{F}_q^{(q-1)n}.$$

Define the code $C' \subseteq \mathbb{F}_q^{qn+1}$ by

$$C' = \{(v|c + T(v) | \pi(v)) \in \mathbb{F}_q^{qn+1} | v \in \mathbb{F}_q^{(q-1)n} \text{ and } c \in C\}.$$

We have

$$|C'| = q^{(q-1)n} |C|.$$

In Theorem 1 we summarize a few important properties of this construction.

Theorem 1: Construction (f) has the following properties:

(a) If C is a μ -fold 1-packing, then so is C' .
 (b) Let $r \in \{1, 2, \dots, n\}$. If C is a μ -fold r -covering, then so is C' .

(c) If C is a $(q, n, M, 1, \mu)$ PMC, then C' is a

$$(q, qn+1, q^{(q-1)n}M, 1, \mu)$$

PMC.

(d) If C is linear, then so is C' .

Remark 1: Theorem 1(a) does not generalize to μ -fold r -packings with $r > 1$. For instance, if C is a 1-fold 2-packing with $q=2$, $n=3$, and $|C|=1$, then C' is not a 1-fold 2-packing by the sphere-packing bound.

Proof of Theorem 1: (a): Suppose an $x \in \mathbb{F}_q^{qn+1}$ and $\mu+1$ different codewords $b^{(1)}, b^{(2)}, \dots, b^{(\mu+1)} \in C'$ exist such that

$$d(x, b^{(k)}) \leq 1, \text{ for } k=1, 2, \dots, \mu+1.$$

Write

$$x = (y|z|\alpha),$$

with $y \in \mathbb{F}_q^{(q-1)n}$, $z \in \mathbb{F}_q^n$ and $\alpha \in \mathbb{F}_q$. For $k=1, 2, \dots, \mu+1$, write

$$b^{(k)} := (v^{(k)}|c^{(k)} + T(v^{(k)})|\pi(v^{(k)})),$$

where $v^{(k)} \in \mathbb{F}_q^{(q-1)n}$ and $c^{(k)} \in C$. We claim that

$$i) d(z, c^{(k)} + T(y)) \leq 1, \text{ for } k=1, 2, \dots, \mu+1,$$

and that

$$ii) \text{ the } c^{(k)} \text{ are all different.}$$

This gives a contradiction with

$$|B_1(z - T(y)) \cap C| \leq \mu.$$

Hence

$$|B_1(x) \cap C'| \leq \mu, \text{ for all } x \in \mathbb{F}_q^{qn+1},$$

i.e., C' is a μ -fold 1-packing.

First we prove *i*). Let $k \in \{1, 2, \dots, \mu+1\}$. Put

$$d_1 := d(y, v^{(k)}),$$

$$d_2 := d(z, c^{(k)} + T(v^{(k)})), \text{ and}$$

$$d'_2 := d(z, c^{(k)} + T(y)).$$

From

$$d(x, b^{(k)}) \leq 1,$$

it follows that $d_1 \leq 1$, $d_2 \leq 1$, and $d_1 + d_2 \leq 1$. If $d_1 = 0$, then

$$T(y) = T(v^{(k)}),$$

hence $d'_2 = d_2 \leq 1$. If $d_1 = 1$, then $d_2 = 0$ and

$$d(T(y), T(v^{(k)})) = 1.$$

Hence, by the triangle inequality, $d'_2 \leq d_2 + 1 = 0 + 1 = 1$. So in both cases $d'_2 \leq 1$, which proves *i*).

To prove *ii*), suppose that we had $i, j \in \{1, 2, \dots, \mu+1\}$ with $i \neq j$ and

$$c^{(i)} = c^{(j)}.$$

Put

$$d := d(b^{(i)}, b^{(j)}),$$

$$d_1 := d(v^{(i)}, v^{(j)}),$$

$$d_2 := d(c^{(i)} + T(v^{(i)}), c^{(j)} + T(v^{(j)})) = d(T(v^{(i)}), T(v^{(j)})), \text{ and}$$

$$d_3 := d(\pi(v^{(i)}), \pi(v^{(j)})).$$

We have

$$0 < d_1 + d_2 + d_3 = d \leq d(x, b^{(i)}) + d(x, b^{(j)}) \leq 1 + 1 = 2.$$

If $d_1 = 0$, then

$$T(v^{(i)}) = T(v^{(j)}), \text{ and}$$

$$\pi(v^{(i)}) = \pi(v^{(j)}),$$

hence also $d_2=d_3=0$, contradicting $d>0$. If $d_1=1$, then

$$d(T(v^{(i)}), T(v^{(j)}))=1, \text{ and}$$

$$\pi(v^{(i)}) \neq \pi(v^{(j)}),$$

hence also $d_2=d_3=1$, contradicting $d \leq 2$. Finally, if $d_1=2$, it follows from $d \leq 2$ that $d_2=d_3=0$. Let $(e-1)n+g$ and $(f-1)n+h$ be the two coordinate positions at which $v^{(i)}$ differs from $v^{(j)}$, where $e, f \in \{1, 2, \dots, q-1\}$ and $g, h \in \{1, 2, \dots, n\}$. Define $\beta, \gamma \in \mathbb{F}_q \setminus \{0\}$ by

$$v^{(i)}_{(e-1)n+g} - v^{(j)}_{(e-1)n+g} = \beta, \text{ and}$$

$$v^{(i)}_{(f-1)n+h} - v^{(j)}_{(f-1)n+h} = \gamma.$$

From $d_2=0$ it follows that $g=h$ and $\gamma=-\beta$. From $d_3=0$ and $\gamma=-\beta$ it follows that

$$\alpha_e \beta = -\alpha_f \gamma = \alpha_f \beta,$$

which implies $e=f$. Consequently,

$$(e-1)n+g = (f-1)n+h,$$

a contradiction. This completes the proof of *ii*), and thereby the proof of part (a) of the theorem.

(b): Let $x \in \mathbb{F}_q^{qn+1}$. Write

$$x = (y|z|\alpha),$$

with $y \in \mathbb{F}_q^{(q-1)n}$, $z \in \mathbb{F}_q^n$ and $\alpha \in \mathbb{F}_q$. There are μ distinct codewords

$$c^{(1)}, c^{(2)}, \dots, c^{(\mu)} \in C,$$

such that

$$d(z - T(y), c^{(k)}) \leq r \quad (k=1, 2, \dots, \mu).$$

We shall now define μ distinct (!) words $b^{(k)} \in \mathbb{F}_q^{qn+1}$ which are all in the code C' , and which satisfy

$$d(x, b^{(k)}) \leq r \quad (k=1, 2, \dots, \mu).$$

Since x was arbitrary, this proves that C' is a μ -fold r -

covering. We distinguish between two cases: *i*) and *ii*).

Case *i*): $\alpha = \pi(y)$. Then, for $k=1,2,\dots,\mu$, define

$$b^{(k)} := (y|c^{(k)} + T(y)|\pi(y)).$$

Case *ii*): $\alpha \neq \pi(y)$. For every k such that

$$d(z - T(y), c^{(k)}) \leq r-1,$$

define $b^{(k)}$ as in case *i*). For the other k , we do the following. We have

$$d(z - T(y), c^{(k)}) = r \geq 1.$$

Let $p_k \in \{1, 2, \dots, n\}$ be a coordinate position at which $z - T(y)$ and $c^{(k)}$ differ. Let $s(k) \in \{1, 2, \dots, q-1\}$ be such, that the p_k -th coordinate of

$$z - T(y) - c^{(k)}$$

is $\alpha_{s(k)}$. Define $t(k) \in \{1, 2, \dots, q-1\}$ by

$$\alpha_{t(k)} \alpha_{s(k)} = \alpha - \pi(y).$$

Let w be the word in $\mathbb{F}_q^{(q-1)n}$ with the $((t(k)-1)n + p_k)$ -th coordinate equal to $\alpha_{s(k)}$ and all other coordinates zero. We have

$$d(z, c^{(k)} + T(y+w)) = r-1, \text{ and}$$

$$\pi(y+w) = \alpha.$$

Now define

$$b^{(k)} := (y+w|c^{(k)} + T(y+w)|\alpha).$$

(*c*): This follows immediately from (*a*) and (*b*). A different way to prove it is by combining (*a*) (or (*b*)) and the fact that C' satisfies the sphere packing equation (3), because C does:

$$\begin{aligned} |C'| V_q(qn+1, 1) &= q^{(q-1)n} M(1 + (qn+1)(q-1)) \\ &= q^{(q-1)n+1} M(1 + n(q-1)) \\ &= q^{(q-1)n+1} \mu q^n \\ &= \mu q^{qn+1}. \end{aligned}$$

(d): Trivial.

Example: The code $C = \{00, 01, 10, 11\} = \mathbb{F}_2^2$ is a linear $(2, 2, 4, 1, 3)$ PMC, (cf. Construction (c)). For this C , Construction (f) gives

$$C' = \{00000, 00010, 00100, 00110, \\ 01011, 01001, 01111, 01101, \\ 10101, 10111, 10001, 10011, \\ 11110, 11100, 11010, 11000\},$$

a linear $(2, 5, 16, 1, 3)$ PMC.

III. PERFECT MULTIPLE 1-COVERINGS

In this section we discuss the case $r=1$. We try to determine all possible parameters q , n , M , and μ for which a $(q, n, M, 1, \mu)$ PMC exists. Of course, it is sufficient to specify only q , n , and μ , since M follows from the other parameters, by (3).

Let us first return to the ordinary case, $\mu=1$. Let q be a prime power and let C_1 be the code $\{0\}$ in $\mathbb{F}_q^1$. This is a $(q, 1, 1, 1, 1)$ PMC. We apply Construction (f) of the previous section to C_1 , and obtain a linear

$$(q, q+1, q^{(q-1)}, 1, 1)$$

PMC C_2 in $\mathbb{F}_q^{q+1}$. If we apply Construction (f) to C_2 , we get a linear

$$(q, q^2+q+1, q^{q^2+q-2}, 1, 1)$$

PMC C_3 . This can be continued indefinitely. We get an infinite sequence of linear PMC's

$$\{C_i\}_{i=1}^{\infty},$$

where C has the parameters

$$(q, (q^i-1)/(q-1), q^{(q^i-1)/(q-1)-i}, 1, 1).$$

This sequence of codes is exactly the well-known sequence of Hamming codes over $\mathbb{F}_q$ (see [8, p. 193]). In Theorem 2 we generalize this, and construct sequences of perfect multiple 1-coverings for *any* μ .

It is also well-known (see [5] or [8, p. 180], for example), that, for q a prime power, the parameters of the Hamming codes are the only parameters for which there exists a (linear or nonlinear) perfect 1-error-correcting code over $\mathbb{F}_q$, that is, a $(q, n, M, 1, 1)$ PMC. We shall show that a similar statement is true in the case of perfect *multiple* 1-coverings, i.e., when also $\mu > 1$ is allowed (Theorem 3).

Theorem 2: Let q be a prime power and $n, \mu \in \mathbb{N}$.

(a): If

$$n = (\mu q^i - 1)/(q - 1) \text{ for some } i \in \mathbb{N} \cup \{0\}, \quad (6)$$

then a *linear* $(q, n, q^{n-i}, 1, \mu)$ PMC exists.

(b): If

$$\left. \begin{aligned} n &= (\mu_0 q^i - 1)/(q - 1) \text{ for some } i \in \mathbb{N} \cup \{0\} \text{ and } \mu_0 \in \mathbb{N}, \\ &\text{with } \mu_0 \mid \mu \text{ and } \mu \leq q^i \mu_0, \end{aligned} \right\} \quad (7)$$

then a $(q, n, \frac{\mu}{\mu_0} q^{n-i}, 1, \mu)$ PMC exists.

Proof: (a): If $\mu = 1$, then $i \geq 1$ (since $n \geq 1$). The sequence $\{C_i\}_{i=1}^{\infty}$ defined in the second paragraph of this section does the job then. If $\mu > 1$, then we define a similar sequence $\{C_i\}_{i=1}^{\infty}$ of PMC's, but we add a C_0 in front. By (6),

$$\mu - 1 \equiv \mu q^i - 1 = (q - 1)n \equiv 0 \pmod{q - 1}.$$

Hence there is an integer n_0 such that $\mu = n_0(q - 1) + 1$. Since $\mu > 1$, we have $n_0 \geq 1$. Define

$$C_0 := \mathbb{F}_q^{n_0},$$

a linear

$$(q, n_0, q^{n_0}, 1, \mu)$$

PMC. Now apply Construction (f) of Section II to C_0 , call the resulting code C_1 , apply (f) to C_1 , call the resulting code C_2 , and so on (just as in the case $\mu=1$). We get an infinite sequence

$$\{C_i\}_{i=1}^{\infty}$$

of linear $(q, n_i, M_i, 1, \mu)$ PMC's over $\mathbb{F}_q$, where

$$n_i = (\mu q^i - 1)/(q-1), \text{ and}$$

$$M_i = q^{n_i - i}.$$

Indeed,

$$n_0 = (\mu q^0 - 1)/(q-1),$$

and if

$$n_i = (\mu q^i - 1)/(q-1),$$

then

$$n_{i+1} = q n_i + 1 = q(\mu q^i - 1)/(q-1) + 1 = (\mu q^{i+1} - 1)/(q-1).$$

Also,

$$M_0 = q^{n_0 - 0},$$

and if

$$M_i = q^{n_i - i},$$

then

$$M_{i+1} = q^{(q-1)n_i + 1 - i} = q^{n_i - i} = q^{n_{i+1} - (i+1)}.$$

(b): By part (a) of this theorem (with μ_0 substituted for μ), a linear

$$(q, n = (\mu_0 q^i - 1)/(q-1), M = q^{n-i}, 1, \mu_0)$$

PMC C over $\mathbb{F}_q$ exists. By (7),

$$\mu/\mu_0 \in \{1, 2, \dots, q^n/M\}.$$

Apply Construction (e) of Section II ("shifting") with $\mu' := \mu/\mu_0$. This yields a PMC C' with the desired parameters.

Theorem 3:

- (a): Let q be a prime power and $n, \mu \in \mathbb{N}$. A linear $(q, n, 1, \mu)$ PMC exists if and only if (6) holds.
- (b): Let q be a prime and $n, \mu \in \mathbb{N}$. A $(q, n, 1, \mu)$ PMC exists if and only if (7) holds.

Proof: (a): Suppose a linear $(q, n, M, 1, \mu)$ PMC exists. Then

$$M = q^j,$$

for some $j \in \{0, 1, \dots, n\}$. By (3),

$$q^j(1 + (q-1)n) = \mu q^n.$$

Consequently,

$$n = (\mu q^i - 1)/(q-1),$$

where $i = n - j \in \{0, 1, \dots, n\}$. Hence (6) holds. The "if" part is Theorem 2(a).

(b): Suppose a $(q, n, M, 1, \mu)$ PMC exists. By (3),

$$M(1 + (q-1)n) = \mu q^n,$$

hence

$$1 + (q-1)n = \mu_0 q^i,$$

for some $\mu_0 \in \mathbb{N}$ and $i \in \{0, 1, \dots, n\}$ with $\mu_0 \mid \mu$. Also,

$$\mu_0 q^i = \mu q^n / M \geq \mu.$$

The "if" part is a consequence of Theorem 2(b).

Theorem 3 and the proof of Theorem 2 show that Constructions (b), (c), (e), and (f) of Section II are the only constructions needed to obtain all possible parameters of PMC's with $\mu=1$, provided q is a prime, or when we restrict ourselves

to linear PMC's. Theorem 3(b) leaves open the cases when q is *not* a prime. For any q , prime power or not, the cases with $M=q^n$ are trivial: by (3) and Construction (c) of Section II, the only possible parameters of PMC's with $M=q^n$ are

$$(q, n, q^n, 1, 1+(q-1)n), \text{ for } n=1, 2, 3, \dots$$

These parameters satisfy (7).

From now on, let q be a prime power. In view of Theorem 2(b), one could conjecture that the answer to the following question is affirmative.

Open problem 1: Is it allowed to replace "prime" by "prime power" in Theorem 3(b)?

While studying this problem, we can restrict ourselves to PMC's with $M < q^n$, as previously pointed out. To prove the "only if" part of Theorem 3(b), we only used the sphere packing equation (3). To this condition, we can add the condition that

$$n \equiv 1 \pmod{q},$$

by (5), which was proved in [2] using properties of Lloyd polynomials. These two necessary conditions are still not sufficient to rule out all parameters n and μ which do not satisfy (7): the parameters

$$(q=4, n=13, M=2^{23}, r=1, \mu=5)$$

satisfy (3) and $n \equiv 1 \pmod{q}$, but they do not satisfy (7). In fact, this is the 'smallest' instance (according to the values of q , n , and μ , in this order) for which we left the question whether a PMC with $r=1$ exists unanswered.

We conclude with a condition (Theorem 4(c)) on the parameters $(q, n, 1, \mu)$ of a PMC, in the case that they do *not* satisfy (7). As a consequence, we get a lower bound on the parameter μ

(Corollary 1). This could be a first step in the investigation of the open problem mentioned above.

Theorem 4: Let p be a prime, $m, n, \mu, M \in \mathbb{N}$ and $q = p^m$. Suppose that a $(q, n, M, 1, \mu)$ PMC exists. Define $\lambda \in \mathbb{N}, k \in \mathbb{N} \cup \{0\}$ with $p \nmid \lambda$ by

$$1 + (q-1)n = \mu q^n / M = \lambda p^k. \quad (8)$$

We have the following:

- (a): If $m \mid k$, then (7) holds.
- (b): If $\lambda = 1$, then $m \mid k$, and (7) holds.
- (c): If $\lambda > 1$, write $k = sm + t$, where $s, t \in \mathbb{N} \cup \{0\}$ and $0 \leq t < m$. Then

$$\lambda = p^{m-t} + (q-1)a,$$

for some $a \in \mathbb{N}$. In particular,

$$\lambda \geq p + q - 1.$$

Proof: Equation (8) follows immediately from (3).

- (a): Put $i := k/m$ and $\mu_0 := \lambda$. Then

$$p^k = q^i,$$

and by (8),

$$n = (\mu_0 q^i - 1) / (q - 1).$$

Moreover, $i \in \mathbb{N} \cup \{0\}$, $\mu_0 \in \mathbb{N}$, $\mu_0 \mid \mu$ and

$$\mu = \lambda p^k M / q^n \leq \lambda p^k = \lambda q^i = \mu_0 q^i.$$

Hence (7) holds.

- (b): See also [6, Theorem (5.1.1)] or [8, Lemma 34, p.181].

By (8), we have

$$1 + (p^m - 1)n = p^k.$$

It follows that

$$p^m - 1 \mid p^k - 1,$$

hence

$$m \mid k$$

(see [8, Lemma 9, p. 102]). By (a), (7) holds.

(c): Put $n_0 := n$. For $i=1, \dots, s$ define

$$n_i := (n_{i-1} - 1)/q.$$

We have $n_0 \in \mathbb{N}$ and

$$1 + (q-1)n_0 = \lambda p^{sm+t}.$$

Now if for some $i \in \{0, 1, \dots, s-1\}$, n_i is an integer and

$$1 + (q-1)n_i = \lambda p^{(s-i)m+t},$$

then

$$n_i \equiv 1 \pmod{q},$$

hence $n_{i+1} \in \mathbb{Z}$. Furthermore,

$$1 + (q-1)n_{i+1} = 1 + (q-1)(n_i - 1)/q = (1 + (q-1)n_i)/q = \lambda p^{(s-(i+1))m+t}.$$

This proves that for $i=0, 1, \dots, s$, $n_i \in \mathbb{Z}$ and

$$1 + (q-1)n_i = \lambda p^{(s-i)m+t}.$$

Moreover, $n_i > 1$, since $\lambda > 1$ and $p \nmid \lambda$. In particular,

$$1 + (q-1)n_s = \lambda p^t.$$

Hence

$$n_s \equiv 1 \pmod{p^t}.$$

Write

$$n_s = ap^t + 1,$$

with $a \in \mathbb{Z}$. Since $n_s > 1$, $a \geq 1$. We get

$$\lambda p^t = 1 + (q-1)n_s = 1 + (q-1)(ap^t + 1) = q + (q-1)ap^t,$$

hence

$$\lambda = p^{m-t} + (q-1)a.$$

Remark 2: In Theorem 4, when $\lambda > 1$ and $m \nmid k$, it is possible that (7) holds, nevertheless. Example: By Theorem 2(a), with $i=1$, a (linear)

$$(q=4, n=13, M=4^{12}, r=1, \mu=10)$$

PMC exists. Certainly (7) holds; namely, choose $i=1$ and $\mu_0 = \mu = 10$. But also (8) is satisfied with $\lambda > 1$ and $m \nmid k$, if we choose $\lambda = \mu/2 = 5$

and $k=3$:

$$1+(4-1)13=10\cdot 4^{13}/4^{12}=5\cdot 2^3.$$

Corollary 1: Let p be a prime, $m, n, \mu \in \mathbb{N}$ and $q=p^m$. If a $(q, n, \cdot, 1, \mu)$ PMC exists, and these parameters do not satisfy (7), then

$$\mu \geq p+q-1.$$

Remark 3: We have been informed by one of the (anonymous) referees that a $(4, 13, 2^{23}, 1, 5)$ PMC exists. Therefore, the answer to open problem 1 is negative. This 5-fold PMC, while obviously not linear over $\mathbb{F}_4$, is linear over $\mathbb{F}_2$.

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Which Linear Codes Are Algebraic-Geometric ?

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Abstract – An infinite series of curves is constructed in order to show that all linear codes can be obtained from curves using Goppa's construction. If one imposes conditions on the degree of the divisor used, then we derive criteria for linear codes to be algebraic-geometric. In particular, we investigate the family of q -ary Hamming codes, and prove that only those with redundancy one or two, and the binary $[7, 4, 3]$ code are algebraic-geometric in this sense. For these codes we explicitly give a curve, rational points and a divisor. We prove that this triple is in a certain sense *unique* in the case of the $[7, 4, 3]$ code.

Index Terms: algebraic-geometric codes, algebraic curves, divisors, generalized Goppa codes, geometric Goppa codes.

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I. Introduction

Since the early papers by Goppa [5], [6], [7], algebraic-geometric codes have been in the spotlight of coding theoretic research for about a decade. As is well-known, numerous exciting results have been achieved using Goppa's construction of linear codes from algebraic curves over finite fields, both by algebraic geometers and coding theorists. Because of the difficulty of the subject, several explanatory papers and text books have appeared, see for instance [9] or [16]. In this paper we

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investigate which linear codes can be constructed by Goppa's method. It turns out that it makes sense to distinguish between three types of codes, according to the degree of the divisor used in the construction. For more details, see Section II (Definition 2).

Although this paper is quite self-contained, a certain knowledge of algebraic geometry is taken for granted. For this, we refer to [2], [4], [11], [16] or [22]. For coding theory, see [15], [16], or [17].

Outline of the paper

In Section II we define *weakly algebraic-geometric* (WAG), *algebraic-geometric* (AG), and *strongly algebraic-geometric* (SAG) codes (Definition 2). The class of SAG codes is a proper subset of the class of AG codes, and the class of AG codes is a proper subset of the class of WAG codes. Furthermore, we also explain what we mean by a WAG, AG or SAG representation of a code. Some basic properties are mentioned. Section II actually serves as an introduction to the rest of the paper. At the end of Section II we introduce the notion of a *minimal* representation. We prove that every WAG, AG or SAG code of dimension at least two has a minimal WAG, AG or SAG representation, respectively. This is useful in Sections IV and V.

The WAG codes are the codes which can be obtained by Goppa's construction when no restrictions are imposed on the degree of the divisor used. Inspired by the notion of a covering curve of Goppa [9] and a paper by Hansen and Stichtenoth [10], we prove in Section III that *every* linear code is WAG. In this way we solve problem (3.1.19) of [22]. The curves are given explicitly. Goppa [7, p.78] claimed that every linear code is WAG, but his proof is not sufficient, see Remark 5. Lachaud [14, (5.10)] made a weaker claim, namely that every linear code is a subcode of a WAG code.

In Section IV we derive several conditions on linear codes to be AG. As proved at the end of that section, all binary SAG codes have length ≤ 8 . By the results of Section III, the class of AG codes therefore seems to be the most interesting. Special attention is paid to Reed-Muller codes, Hamming codes and the binary Golay code and its extension. For example, the conditions on AG codes imply that a q -ary Hamming code of redundancy r is not AG if $r > 2$ and $(r, q) \neq (3, 2)$.

In Section V we are interested in explicit WAG, AG or SAG representations of codes, and in the question whether something can be

said about the uniqueness of these representations. As an example, we investigate the family of q -ary Hamming codes in close detail (Section V-A). We prove that these codes are SAG in the cases left open in Section IV. In the case $(r, q) = (3, 2)$, that is, for the binary $[7, 4, 3]$ code, we obtain the nice result that this code has a *unique* minimal representation as an AG code. In Section V-B we discuss another example, namely a code which was mentioned in [13], and prove that it is SAG.

Notation

We use $\mathbf{F}_q$ to denote the finite field of q elements. We use $\mathbf{P}^l$ to denote the l -dimensional projective space; it will be clear from the context over which field (usually $\mathbf{F}_q$ or the algebraic closure $\overline{\mathbf{F}}_q$). If any confusion is possible, we use $\mathbf{P}^l(\mathbf{F}_q)$ to denote the finite set of $(q^{l+1} - 1)/(q - 1)$ points over $\mathbf{F}_q$ in $\mathbf{P}^l$, for instance. Similarly, $\mathbf{A}^l$ denotes the l -dimensional affine space. By a curve over a field k we mean a projective, reduced scheme over k of dimension one. As with $\mathbf{P}^l$ and $\mathbf{A}^l$, we sometimes write $\mathcal{X}(\mathbf{F}_q)$ to indicate the finite set of $\mathbf{F}_q$ -rational points on $\mathcal{X}$. The function field of $\mathcal{X}$ over k is denoted by $k(\mathcal{X})$. The group of divisors on $\mathcal{X}$ is denoted by $\text{Div}(\mathcal{X})$. If

$$\varphi: \mathcal{X} \rightarrow \mathcal{X}'$$

is a morphism of curves, then we denote by φ^* both the induced homomorphism

$$k(\mathcal{X}') \rightarrow k(\mathcal{X})$$

and the induced homomorphism

$$\text{Div}(\mathcal{X}') \rightarrow \text{Div}(\mathcal{X}),$$

see [11, p.137]. If

$$f \in k(\mathcal{X}) \setminus \{0\},$$

we denote by (f) its divisor, a so-called *principal* divisor. The notation $\text{div}(f)$ is also used in the literature. Similarly, if ω is a nonzero rational differential form on $\mathcal{X}$, then we denote its divisor by (ω) , a so-called *canonical* divisor. If P is a place of $k(\mathcal{X})$ over k , that is, P is a discrete valuation ring of $k(\mathcal{X})$ over k , then we denote by v_P the discrete valuation function at P . In the literature the notation ord_P

is also customary. If D is a divisor on $\mathcal{X}$, then $\text{supp}(D)$ denotes the *support* of D , that is, the set of places with nonzero coefficient in D . If D_1 and D_2 are divisors on a curve $\mathcal{X}$, then we denote by

$$D_1 \sim D_2$$

that D_1 and D_2 are linearly equivalent. By $[D]$ we denote the linear equivalence class of D , that is, the set consisting of all the divisors on $\mathcal{X}$ linearly equivalent with D . The *complete linear system* associated to D is denoted by $|D|$. This is the set of all effective divisors in $[D]$. We denote by $\text{Pic}(\mathcal{X})$ the group of divisors on $\mathcal{X}$ modulo principal divisors, the so-called *divisor class group*. By $\text{Pic}_0(\mathcal{X})$ we denote the subgroup of $\text{Pic}(\mathcal{X})$ consisting of the divisors of degree 0 modulo principal divisors. By $\text{Pic}_m(\mathcal{X})$ we denote the coset of $\text{Pic}_0(\mathcal{X})$ in $\text{Pic}(\mathcal{X})$ consisting of the divisors of degree m modulo principal divisors. By D_m we denote the set of effective divisors on $\mathcal{X}$ of degree m . We define

$$h := \#\text{Pic}_0(\mathcal{X}).$$

In fact, we have $h = \#\text{Pic}_m(\mathcal{X})$ for every m . For all this, see [16]. If C is a linear code, we denote by $d(C)$ its minimum distance.

II. Algebraic-Geometric Codes and Representations

Definition 1 Let $\mathcal{X}$ be a projective, nonsingular, absolutely irreducible curve defined over $\mathbf{F}_q$. The genus of $\mathcal{X}$ is denoted by $g(\mathcal{X})$, or simply by g , if it is clear which curve is meant. Let $P_1, \dots, P_n$ be n distinct $\mathbf{F}_q$ -rational points of $\mathcal{X}$. We denote both the n -tuple

$$(P_1, \dots, P_n)$$

and the divisor

$$P_1 + \dots + P_n$$

by D (the order of the P_i is fixed). Let G be a divisor on $\mathcal{X}$ of degree m with support disjoint from the support of D . Let $\mathbf{F}_q(\mathcal{X})$ be the function field of $\mathcal{X}$ over $\mathbf{F}_q$ and

$$L(G) = \{f \in \mathbf{F}_q(\mathcal{X})^* \mid (f) \geq -G\} \cup \{0\}.$$

Let $\Omega_{\mathcal{X}}$ be the vector space of rational differential forms on $\mathcal{X}$ and

$$\Omega(G) = \{\omega \in \Omega_{\mathcal{X}} \setminus \{0\} \mid (\omega) \geq G\} \cup \{0\}.$$

We define the map

$$\alpha_L : L(G) \longrightarrow \mathbf{F}_q^n,$$

by

$$\alpha_L(f) = (f(P_1), \dots, f(P_n)),$$

and the map

$$\alpha_{\Omega} : \Omega(G - D) \longrightarrow \mathbf{F}_q^n,$$

by

$$\alpha_{\Omega}(\omega) = (\text{res}_{P_1}(\omega), \dots, \text{res}_{P_n}(\omega)).$$

We define

$$C_L(\mathcal{X}, D, G) = \text{Image}(\alpha_L),$$

and

$$C_{\Omega}(\mathcal{X}, D, G) = \text{Image}(\alpha_{\Omega}).$$

We abbreviate $C_L(\mathcal{X}, D, G)$ and $C_{\Omega}(\mathcal{X}, D, G)$ by

$$C_L(D, G),$$

and

$$C_{\Omega}(D, G),$$

respectively, if it is clear which curve is meant. See Goppa [5], [6], [7], [8], [9], or [16], or [22].

Theorem 1 .

a) If $m < n$ then $C_L(D, G)$ is a linear $[n, k, d]$ code with

$$k \geq m + 1 - g \quad \text{and} \quad d \geq n - m.$$

If moreover $2g - 2 < m$ then $k = m + 1 - g$.

b) If $2g - 2 < m$ then $C_{\Omega}(D, G)$ is a linear $[n, k, d]$ code with

$$k \geq n - m - 1 + g \quad \text{and} \quad d \geq m + 2 - 2g.$$

If moreover $m < n$ then $k = n - m - 1 + g$.

Proof: See [5], [16], or [22].

Proposition 1 *The linear code $C_{\mathbf{n}}(D, G)$ is the dual of $C_L(D, G)$.*

Proof: See [8], [16], or [22].

Definition 2 We call a q -ary linear code C *weakly algebraic-geometric* (WAG) if there exists a projective, nonsingular, absolutely irreducible curve $\mathcal{X}$ defined over $\mathbb{F}_q$ of genus g , and n distinct rational points $P_1, \dots, P_n$ on $\mathcal{X}$ and a divisor G with support disjoint from the support of D , where

$$D = P_1 + \dots + P_n,$$

such that

$$C = C_L(\mathcal{X}, D, G).$$

We call the triple

$$(\mathcal{X}, D, G)$$

a *weakly algebraic-geometric representation* (WAG representation), or shortly, a *representation* of C . An *algebraic-geometric representation* (AG representation) is a representation $(\mathcal{X}, D, G)$ with

$$\deg(G) < n.$$

We call a code *algebraic-geometric* (AG) if it has an AG representation. A *strongly algebraic-geometric representation* (SAG representation) is a representation $(\mathcal{X}, D, G)$ with

$$2g - 2 < \deg(G) < n.$$

A code is called *strongly algebraic-geometric* (SAG) if it has a SAG representation.

Remark 1 There exists a differential form ω with a simple pole at each P_i and such that

$$\text{res}_{P_i}(\omega) = 1 \quad \text{for } i = 1, \dots, n.$$

We have

$$C_{\mathbf{n}}(\mathcal{X}, D, G) = C_L(\mathcal{X}, D, (\omega) - G + D),$$

see [21, Corollary 2.6] or [16, Lemma 3.5]. As a consequence we have that C is WAG if and only if

$$C = C_{\mathbf{n}}(\mathcal{X}, D, G)$$

for some curve $\mathcal{X}$ and divisors D and G (without the constraints on the degree of G). The code C is AG if moreover

$$2g - 2 < \deg(G).$$

The code C is SAG if moreover

$$2g - 2 < \deg(G) < n.$$

In view of Proposition 1 we therefore have the following corollary.

Corollary 1 *If C is WAG or SAG, then $C^\perp$ is WAG, SAG, respectively.*

Remark 2 There exist codes which are AG while the dual is not. For an example, see Remark 13.

Definition 3 Let $n > 1$. Let

$$\pi_i : \mathbf{F}_q^n \rightarrow \mathbf{F}_q^{n-1}$$

be the projection defined by deleting the i^{th} coordinate. If C is a code in $\mathbf{F}_q^n$ then define C_i by

$$C_i = \pi_i(C).$$

We say that C_i is obtained from C by *puncturing* at the i^{th} coordinate.

Lemma 1 *If C is WAG then C_i is WAG.*

Proof: Suppose that $C = C_L(\mathcal{X}, D, G)$, where $D = (P_1, \dots, P_n)$. Let

$$D_i = (P_1, \dots, P_{i-1}, P_{i+1}, \dots, P_n).$$

Then $C_i = C_L(\mathcal{X}, D_i, G)$.

Remark 3 If C is AG or SAG, then C_i need not be AG, SAG, respectively, see Remark 19.

Definition 4 Let C be a linear code in $\mathbf{F}_q^n$ and σ a permutation of $\{1, \dots, n\}$. Define

$$\sigma C = \{(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \mid (x_1, \dots, x_n) \in C\}.$$

Two linear codes C_1 and C_2 in $\mathbf{F}_q^n$ are called *equivalent* if $C_2 = \sigma C_1$ for some permutation σ of $\{1, \dots, n\}$. Let

$$\lambda = (\lambda_1, \dots, \lambda_n)$$

be an n -tuple of nonzero elements in $\mathbf{F}_q$. Define

$$\lambda C = \{(\lambda_1 x_1, \dots, \lambda_n x_n) | (x_1, \dots, x_n) \in C\}.$$

The codes C_1 and C_2 are called *generalized equivalent* or *isometric* if there is an n -tuple $\lambda = (\lambda_1, \dots, \lambda_n)$ of nonzero elements in $\mathbf{F}_q$ and a permutation σ such that

$$C_2 = \lambda \sigma C_1.$$

Lemma 2 *If C_1 and C_2 are isometric codes and C_1 is WAG, AG or SAG, then C_2 is WAG, AG, SAG, respectively.*

Proof: Suppose $C_1 = C_L(\mathcal{X}, D, G)$ and $C_2 = \lambda \sigma C_1$ for some non zero elements $\lambda_1, \dots, \lambda_n$ in $\mathbf{F}_q$ and a permutation σ . There exists a rational function f such that

$$f(P_{\sigma(i)}) = \lambda_i \quad \text{for all } i,$$

by the independence of valuations, see [2, p.11]. Let

$$\sigma D = (P_{\sigma(1)}, \dots, P_{\sigma(n)}).$$

Then the divisor $G - (f)$ has disjoint support with σD , since all the λ_i are nonzero. We have

$$C_2 = C_L(\mathcal{X}, \sigma D, G - (f)),$$

and C_2 is WAG. The degrees of G and $G - (f)$ are equal. So, if C_1 is AG or SAG, then C_2 is AG, SAG, respectively.

Definition 5 We call a q -ary linear $[n, k]$ code *projective* if every two columns of a generator matrix of C are linearly independent. Thus if we view the columns of a generator matrix as points in the $(k-1)$ -dimensional projective space $\mathbf{P}^{k-1}$, expressed in homogeneous coordinates, then we get n distinct points. This definition is obviously independent from the generator matrix chosen. By $S(r, q)$ we denote

any q -ary projective code of dimension r and length $(q^r - 1)/(q - 1)$. Such a code is called a *Simplex* code. By $H(r, q)$ we denote the dual of $S(r, q)$. This is a q -ary *Hamming* code of redundancy r . If all the n points of a projective code lie in the complement of a hyperplane then we call the code *affine*.

Remark 4 If $n \geq 3$, then a code C is projective if and only if $d(C^\perp) \geq 3$. The code C is affine if and only if C is projective and there exists a codeword with weight equal to the word length. The maximal word length of a projective code of dimension r is $(q^r - 1)/(q - 1)$. For fixed r and q all q -ary Simplex codes of dimension r are isometric. The same holds for Hamming codes. The maximal possible word length of an affine code of dimension r is q^{r-1} . For fixed q and r all affine q -ary codes of dimension r and word length q^{r-1} are isometric and are called *q -ary first order Reed-Muller codes*.

Remark 5 Suppose C is an affine code and we want to show that it is WAG. By Lemma 2, we may assume after an isometry, that the all one vector is a codeword and it is the first row of a generator matrix of C . Let the n points $Q_1, \dots, Q_n$ in $\mathbf{P}^{k-1}$ correspond to the n columns of the generator matrix. Suppose there exists an absolutely irreducible, projective curve $\mathcal{X}$ over $\mathbf{F}_q$ in $\mathbf{P}^{k-1}$, which goes through $Q_1, \dots, Q_n$. The curve may be singular, but suppose there exists a rational point P_i in $n^{-1}(Q_i)$, for every i , where

$$n: \tilde{\mathcal{X}} \rightarrow \mathcal{X}$$

is the normalization. Let $x_0, \dots, x_{k-1}$ be homogeneous coordinates of $\mathbf{P}^{k-1}$ corresponding to the first up to the k^{th} row of the generator matrix. Then none of the points $Q_1, \dots, Q_n$ lies in the hyperplane H , given by $x_0 = 0$. Let

$$G = n^*(\mathcal{X} \cdot H)$$

be the pull back of the intersection divisor $\mathcal{X} \cdot H$ to the normalization. Let

$$f_i = (x_i/x_0) \circ n.$$

Then

$$f_0, \dots, f_{k-1} \in L(G),$$

and they are linearly independent, since the rank of the generator matrix of C is k . So $l(G) \geq k$. If $l(G) = k$ then

$$C = C_L(\tilde{\mathcal{X}}, D, G),$$

where

$$D = (P_1, \dots, P_n),$$

that is to say C is WAG. In other words, we are looking for a curve $\mathcal{X}$ in $\mathbf{P}^{k-1}$ going through $Q_1, \dots, Q_n$ such that the linear system of hyperplane sections of $\mathcal{X}$ is complete, and such that for every i there is a rational point in $n^{-1}(Q_i)$. In the next section we show that indeed there exists such a curve, going through all the q^{k-1} rational points of $\mathbf{P}^{k-1}$ outside a hyperplane. Such curves were called *covering* curves by Goppa [9, Ch.4, Sect.10]. Goppa [7, p.78] claimed that every linear code is WAG. In the proof he only mentioned that if $Q_1, \dots, Q_n$ are n distinct points in $\mathbf{P}^{k-1}$, then there exists a curve passing through $Q_1, \dots, Q_n$. First of all this reasoning only applies to *projective* codes, and secondly, the linear system of hyperplane sections of this curve does not need to be *complete*. This would only prove that every projective code is a subcode of a WAG code, see Lachaud [14, (5.10)].

Remark 6 Let C be a q -ary projective code of dimension at least 2. Suppose

$$C = C_L(\mathcal{X}, D, G)$$

for some curve $\mathcal{X}$ and divisors D and G . If

$$L(G) = L(G - P)$$

for some point P of $\mathcal{X}$, then P is not in the support of D . Otherwise $P = P_i$ for some $i \in \{1, \dots, n\}$, so all the codewords have a zero at place i , contradicting the assumption that C is projective. Thus $G - P$ has disjoint support with D and

$$C = C_L(\mathcal{X}, D, G - P).$$

Repeating this procedure we may assume without loss of generality that G is a divisor such that

$$L(G) \neq L(G - P)$$

for all points P , that is to say G has no base points. Let $l(G) = l$ and let $f_0, \dots, f_{l-1}$ be a basis of $L(G)$. Consider the morphism

$$\varphi_G : \mathcal{X} \rightarrow \mathbf{P}^{l-1},$$

given by the collection of morphisms

$$\{\varphi_j : \mathcal{X} \setminus \text{supp}(G_j) \rightarrow \mathbf{P}^{l-1}\}_{j=0}^{l-1},$$

where

$$G_j = G + (f_j),$$

and where φ_j is defined by

$$\varphi_j(P) = \left(\frac{f_0}{f_j}(P) : \dots : \frac{f_{l-1}}{f_j}(P) \right), \quad \text{for } P \in \mathcal{X} \setminus \text{supp}(G_j),$$

see [12, p.128]. Then

$$\varphi_G(P) = (f_0(P) : \dots : f_{l-1}(P)), \quad \text{for } P \in \mathcal{X} \setminus \text{supp}(G).$$

This holds in particular for the P_i . The morphism φ_G depends only on the linear equivalence class of G , and on the choice of the basis $f_0, \dots, f_{l-1}$ of $L(G)$. A different choice of a basis of $L(G)$ gives a morphism which differs by an automorphism of $\mathbf{P}^{l-1}$ (see [11, p.158]). Let $\mathcal{X}_0$ be the reduced image of $\mathcal{X}$ under the morphism φ_G . Then $\mathcal{X}_0$ is not a single point. Even stronger, $\mathcal{X}_0$ is not contained in any hyperplane. This follows from the fact that $f_0, f_1, \dots, f_{l-1}$ are linearly independent. Hence

$$\varphi_G : \mathcal{X} \rightarrow \mathcal{X}_0$$

is a finite dominant morphism of curves. Since $\mathcal{X}$ is absolutely irreducible, $\mathcal{X}_0$ is absolutely irreducible too. Finally, we have

$$\deg(G) = \deg(\varphi_G) \cdot \deg(\mathcal{X}_0),$$

since G has no base points, see [12, p.213].

Definition 6 Let C be a projective code of dimension at least 2. If $(\mathcal{X}, D, G)$ is a (WAG, AG or SAG) representation of C and G is a divisor without base points and $\deg(\varphi_G) = 1$, then we call $(\mathcal{X}, D, G)$ a *minimal* (WAG, AG or SAG) representation of C (respectively).

Proposition 2 *Suppose C is a projective WAG code of dimension at least two. If $(\mathcal{X}, D, G)$ is a representation of C , with G base point free, then there exists a minimal representation $(\tilde{\mathcal{X}}_0, \tilde{D}, \tilde{G})$ of C and a finite morphism $\varphi: \mathcal{X} \rightarrow \tilde{\mathcal{X}}_0$ with the following properties:*

- i) $\tilde{D} = (\varphi(P_1), \dots, \varphi(P_n))$.
- ii) $\varphi^*(\tilde{G}) \sim G$, where $\varphi^*(\tilde{G})$ is the pull back of $\tilde{G}$ under φ .
- iii) $\deg(\varphi) = \deg(\varphi_G)$.
- iv) $\deg(\tilde{G}) = \deg(G)/\deg(\varphi) \leq \deg(G)$.
- v) $g(\tilde{\mathcal{X}}_0) \leq g(\mathcal{X})$, with equality if and only if $\deg(\varphi) = 1$.
- vi) If $(\mathcal{X}, D, G)$ is an AG representation, then so is $(\tilde{\mathcal{X}}_0, \tilde{D}, \tilde{G})$.
- vii) If $(\mathcal{X}, D, G)$ is a SAG representation, then so is $(\tilde{\mathcal{X}}_0, \tilde{D}, \tilde{G})$.

Proof: Let $l(G) = l$. The kernel of the linear map α_L is $L(G - D)$, see Definition 1. We have

$$k = \dim(C) = l(G) - l(G - D).$$

Let $f_0, \dots, f_{l-1}$ be a basis of $L(G)$ such that $f_k, \dots, f_{l-1}$ is a basis of $L(G - D)$. Let A be the $(l \times n)$ -matrix

$$(f_j(P_i))_{j=0, \dots, l-1; i=1, \dots, n}.$$

The first k rows of A form a generator matrix of C . The remaining $l - k$ rows have only zero entries. Let the morphism φ_G be defined by this basis of $L(G)$. The reduced image $\mathcal{X}_0$ of $\mathcal{X}$ under φ_G is possibly singular. Let

$$n: \tilde{\mathcal{X}}_0 \rightarrow \mathcal{X}_0$$

be the normalization of $\mathcal{X}_0$. Then n is a birational morphism. Hence we have a rational map

$$\tilde{\varphi}_G: \mathcal{X} \rightarrow \tilde{\mathcal{X}}_0$$

such that

$$n \circ \tilde{\varphi}_G = \varphi_G.$$

The curve $\mathcal{X}$ is nonsingular, hence $\tilde{\varphi}_G$ is a morphism. The n points

$$\tilde{\varphi}_G(P_i) \quad (i = 1, \dots, n)$$

are rational and we claim that they are all distinct. Indeed, if

$$\tilde{\varphi}_G(P_s) = \tilde{\varphi}_G(P_t),$$

then

$$\varphi_G(P_s) = \varphi_G(P_t).$$

But $\varphi_G(P_s)$ corresponds to the s^{th} column of the matrix A , and C is projective, hence $s = t$. Put

$$\tilde{P}_i = \tilde{\varphi}_G(P_i),$$

and

$$\tilde{D} = (\tilde{P}_1, \dots, \tilde{P}_n).$$

For $j = 0, \dots, l-1$, we denote by g_j the function x_j/x_0 , which is a rational function on $\mathcal{X}_0$ such that

$$f_j/f_0 = g_j \circ \varphi_G.$$

We denote $g_j \circ n$ by $\tilde{g}_j$. Let H be the hyperplane in $\mathbf{P}^{l-1}$ with equation $x_0 = 0$ and let $H \cdot \mathcal{X}_0$ be the intersection divisor of H with $\mathcal{X}_0$. Define

$$G_0 := G + (f_0).$$

The pull back $\varphi_G^*(H \cdot \mathcal{X}_0)$ is equal to G_0 . Let

$$\tilde{G}_0 = n^*(H \cdot \mathcal{X}_0).$$

Then $\tilde{\varphi}_G$ induces an injective map $\tilde{\varphi}_G^*$ from the function field of $\tilde{\mathcal{X}}_0$ into the function field of $\mathcal{X}$, and maps $L(\tilde{G}_0)$ injectively into $L(G_0)$. This map is also surjective since

$$\tilde{\varphi}_G^*(\tilde{g}_j) = f_j/f_0,$$

for $j = 0, \dots, l-1$, and since

$$1, f_1/f_0, \dots, f_{l-1}/f_0$$

is a basis of $L(G_0)$. Let $\varphi_{\tilde{G}_0}$ be defined by the basis $\tilde{g}_0, \dots, \tilde{g}_{l-1}$ of $L(\tilde{G}_0)$. Note that $\varphi_{\tilde{G}_0}$ is equal to the normalization map n . There exists a divisor $\tilde{G}$ which is linearly equivalent with $\tilde{G}_0$ and has disjoint support with $\tilde{D}$, by the theorem of independence of valuations, see [2, p.11]. We have $\varphi_{\tilde{G}} = \varphi_{\tilde{G}_0}$, where $\varphi_{\tilde{G}}$ is defined by a suitable choice of a basis of $L(\tilde{G})$. Hence

$$\varphi_{\tilde{G}}(\tilde{P}_i) = n \circ \tilde{\varphi}_G(P_i) = \varphi_G(P_i),$$

for $i = 1, \dots, n$. All these points have their last $l - k$ coordinates equal to zero. Thus there is an n -tuple

$$\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbf{F}_q^n,$$

with all $\lambda_i \neq 0$, such that

$$C_L(\tilde{\mathcal{X}}_0, \tilde{D}, \tilde{G}) = \lambda C_L(\mathcal{X}, D, G).$$

As we see from the proof of Lemma 2, we may assume without loss of generality that

$$C_L(\tilde{\mathcal{X}}_0, \tilde{D}, \tilde{G}) = C_L(\mathcal{X}, D, G).$$

In the proposition choose $\varphi = \tilde{\varphi}_G$. We have

$$\deg(\varphi_{\tilde{G}}) = \deg(n) = 1,$$

$$\deg(\varphi) = \deg(\varphi_G),$$

and

$$\deg(\tilde{G}) = \deg(\tilde{G}_0) = \frac{\deg(G_0)}{\deg(\varphi)} = \frac{\deg(G)}{\deg(\varphi_G)} \leq \deg(G),$$

see the end of Remark 6. Since G is base point free and φ^* restricts to an isomorphism

$$L(\tilde{G}_0) \rightarrow L(G_0),$$

$\tilde{G}$ is base point free too. Since φ^* preserves linear equivalence and $\tilde{G} \sim \tilde{G}_0$, we have

$$\varphi^*(\tilde{G}) \sim \varphi^*(\tilde{G}_0) = G_0 \sim G.$$

This proves everything in the proposition, except *v*), *vi*), and *vii*). Note that *vi*) is an immediate consequence of *iv*). Part *v*) and part *vii*) will follow by the genus formula of Zeuthen-Hurwitz, see [16, p.52] or [11, p.301]. First we prove that φ is separable. The morphism $\varphi : \mathcal{X} \rightarrow \tilde{\mathcal{X}}_0$ factorizes into $\varphi = \varphi_* \circ \varphi_i$, where

$$\varphi_i : \mathcal{X} \rightarrow \mathcal{X}_i$$

is purely inseparable and

$$\varphi_* : \mathcal{X}_i \rightarrow \tilde{\mathcal{X}}_0$$

is separable, see [11, p.303, Example 2.5.4]. The morphism φ_i induces an inclusion

$$\varphi_i^* : \mathbf{F}_q(\mathcal{X}_i) \hookrightarrow \mathbf{F}_q(\mathcal{X}),$$

and the image is equal to

$$\{f^{p^r} | f \in \mathbf{F}_q(\mathcal{X})\} = \mathbf{F}_q(\mathcal{X})^{p^r},$$

where

$$p^r = \deg(\varphi_i),$$

p is the characteristic of $\mathbf{F}_q$, and r is some nonnegative integer. The curve $\mathcal{X}_i$ is isomorphic with $\mathcal{X}$, see [11, p.302, Prop.2.5]. Let

$$\psi : \mathcal{X}_i \rightarrow \mathcal{X}$$

be the isomorphism (of curves) which induces the isomorphism (of function fields)

$$\psi^* : \mathbf{F}_q(\mathcal{X}) \rightarrow \mathbf{F}_q(\mathcal{X}_i), f \mapsto f^{p^r}.$$

Put

$$G_i := \varphi_i^*(\tilde{G}_0).$$

Define the divisor G'_i on $\mathcal{X}$ by

$$\psi^*(G'_i) = G_i.$$

Then

$$G_0 = \varphi^*(\tilde{G}_0) = \varphi_i^*(\varphi_s^*(\tilde{G}_0)) = \varphi_i^*(G_i) = p^r G'_i.$$

The map φ_i^* maps $L(G_i)$ injectively into $L(G_0)$. The morphism φ_s induces an inclusion

$$\varphi_s^* : \mathbf{F}_q(\tilde{\mathcal{X}}_0) \hookrightarrow \mathbf{F}_q(\mathcal{X}_i),$$

and φ_s^* maps $L(\tilde{G}_0)$ injectively into $L(G_i)$. Thus

$$l(\tilde{G}_0) \leq l(G_i) \leq l(G_0).$$

But, as we saw earlier in the proof of this proposition,

$$l(G_0) = l(\tilde{G}_0),$$

hence

$$l(G_i) = l(G_0). \quad (1)$$

Now suppose that $\deg(\varphi_i) > 1$, that is to say $r > 0$. Let

$$P \in \text{supp}(G'_i).$$

Then

$$G'_i \leq (p^r - 1)G'_i \leq p^r G'_i - P = G_0 - P \leq G_0.$$

Hence

$$L(G'_i) \subseteq L(G_0 - P) \subseteq L(G_0). \quad (2)$$

On the other hand, ψ^* restricts to an isomorphism

$$L(G_i) \rightarrow L(G'_i),$$

hence

$$l(G_i) = l(G'_i),$$

and by (1),

$$l(G_0) = l(G'_i).$$

This implies that the inclusions in (2) are equalities, and hence that P is a basepoint of G_0 , a contradiction. Thus $\deg(\varphi_i) = 1$, and φ is separable. So we can apply the genus formula of Zeuthen-Hurwitz to φ :

$$2g(\mathcal{X}) - 2 = (2g(\tilde{\mathcal{X}}_0) - 2)\deg(\varphi) + \deg(R),$$

where R is the ramification divisor of φ , which is effective. As shown in [11, p.303, Example 2.5.4], it follows that

$$g(\mathcal{X}) \geq g(\tilde{\mathcal{X}}_0).$$

Note that if $\deg(\varphi) = 1$, then $R = 0$. One easily verifies that

$$g(\mathcal{X}) = g(\tilde{\mathcal{X}}_0)$$

if and only if

$$\deg(\varphi) = 1,$$

or

$$g(\mathcal{X}) = 0,$$

or

$$g(\mathcal{X}) = 1 \text{ with } \varphi \text{ unramified.}$$

However, in our situation, the second and the third case are included in the first. Namely, suppose that

$$g(\mathcal{X}) = g(\tilde{\mathcal{X}}_0) =: g \leq 1.$$

We have

$$2 \leq k \leq l(G) = l(\tilde{G}_0),$$

hence

$$\deg(G) > 0 \geq 2g - 2,$$

and

$$\deg(\tilde{G}_0) > 0 \geq 2g - 2.$$

By Riemann-Roch,

$$l(G_0) = \deg(G) + 1 - g,$$

$$l(\tilde{G}_0) = \deg(\tilde{G}_0) + 1 - g.$$

Since $l(G_0) = l(\tilde{G}_0)$, we get

$$\deg(G_0) = \deg(\tilde{G}_0) = \deg(G_0)/\deg(\varphi),$$

hence $\deg(\varphi) = 1$. This proves *v*). Finally, if

$$\deg(G) > 2g(\mathcal{X}) - 2,$$

then

$$\deg(\tilde{G}) = \frac{\deg(G)}{\deg(\varphi)} > \frac{2g(\mathcal{X}) - 2}{\deg(\varphi)} \geq 2g(\tilde{\mathcal{X}}_0) - 2.$$

This proves *vii*) and completes the proof of the proposition.

Corollary 2 *Suppose that $(\mathcal{X}, D, G)$ is a WAG representation of a projective code C of dimension at least two, with G base point free, and such that $g(\mathcal{X})$ is minimal, that is to say, for all WAG representations $(\mathcal{X}', D', G')$ of C we have $g(\mathcal{X}) \leq g(\mathcal{X}')$. Then $(\mathcal{X}, D, G)$ is a minimal WAG representation of C . This corollary is also true if 'WAG' is replaced by 'AG' everywhere, or by 'SAG'.*

Proof: Let $(\tilde{\mathcal{X}}_0, \tilde{D}, \tilde{G})$ be a minimal WAG representation of C with the properties as in Proposition 2. By the assumption on $g(\mathcal{X})$, and by Prop. 2(*v*), we have

$$g(\tilde{\mathcal{X}}_0) = g(\mathcal{X}),$$

and hence

$$\deg(\varphi_G) = \deg(\varphi) = 1,$$

by Prop. 2(iii). Since G is base point free, moreover, $(\mathcal{X}, D, G)$ is minimal. The two assertions in the second part of the corollary are proved similarly, using Prop. 2(vi) and 2(vii), respectively.

III. All Linear Codes are Weakly Algebraic-Geometric

Remark 7 Hansen and Stichtenoth [10] considered the curve $\mathcal{X}$ in $\mathbf{P}^2$ defined by the homogeneous equation

$$x^{q_0}(x^q + xz^{q-1}) = z^{q_0}(y^q + yz^{q-1}),$$

where $q_0 = 2^n$ and $q = 2^{2n+1}$. This curve is absolutely irreducible, has exactly one (singular) point P_∞ at the line $z = 0$, and goes through all the rational points outside the line $z = 0$. The linear system of hyperplane sections of this curve is complete. Inspired by their result we consider the following series of curves.

Definition 7 Let p be a prime number and q a power of p . Let $\mathcal{X}(l, q)$ be the closed subscheme over $\mathbf{F}_p$ in $\mathbf{P}^l$ defined by the homogeneous ideal

$$I(l, q) = (x_i^{q+1} - x_i^2 x_0^{q-1} + x_{i+1} x_0^q - x_{i+1}^q x_0, i = 1, \dots, l-1)$$

in $\mathbf{F}_p[x_0, \dots, x_l]$.

Proposition 3 *The scheme $\mathcal{X}(l, q)$ is a projective, absolutely irreducible, reduced curve over $\mathbf{F}_p$. It has exactly one point P_∞ at the hyperplane H with equation $x_0 = 0$, the curve is nonsingular outside P_∞ and goes through all the q^l rational points of $\mathbf{P}^l$ outside the hyperplane H .*

Proof: The scheme $\mathcal{X}(l, q)$ is defined by $l-1$ equations, hence all the irreducible components are at least one dimensional.

$$P_\infty = (0 : \dots : 0 : 1)$$

is the only point in the intersection with H , which follows directly from the equations. Let

$$f_i = y_i^{q+1} - y_i^2 + y_{i+1} - y_{i+1}^q \quad \text{for } i = 1, \dots, l-1.$$

Then

$$f_1 = \dots = f_{l-1} = 0$$

are the equations of $\mathcal{X}(l, q)$ on the complement of H , which is isomorphic with affine l -space with coordinates $y_1, \dots, y_l$, where $y_i = x_i/x_0$. For every fixed y_1 there are exactly q^{l-1} solutions $\mathbf{y} = (y_1, \dots, y_l)$ in $\mathbf{F}_q^l$ as well as in $\overline{\mathbf{F}}_q^l$ of the equations $f_1, \dots, f_{l-1} = 0$. Hence $\mathcal{X}(l, q)$ has dimension one and is a complete intersection. Let

$$f = (f_1, \dots, f_{l-1}).$$

Computing the derivative of f gives

$$df = \begin{pmatrix} y_1^q - 2y_1 & 1 & & 0 \\ & \ddots & \ddots & \\ 0 & & y_{l-1}^q - 2y_{l-1} & 1 \end{pmatrix}$$

Hence df has maximal rank at all points not equal to P_∞ of $\mathcal{X}(l, q)$ over $\overline{\mathbf{F}}_q$. Thus the curve is nonsingular outside P_∞ . Thus $\mathcal{X}(l, q)$ is reduced outside P_∞ and a complete intersection, and therefore it is reduced. Let

$$n: \tilde{\mathcal{X}}(l, q) \longrightarrow \mathcal{X}(l, q)$$

be the normalization of $\mathcal{X}(l, q)$ and $\tilde{P}_\infty$ any point in $n^{-1}(P_\infty)$. Let v_∞ be the discrete valuation at $\tilde{P}_\infty$. Let $z_i = y_i \circ n$. Then $z_1, \dots, z_l$ are rational functions on $\tilde{\mathcal{X}}(l, q)$ and have no poles outside $n^{-1}(P_\infty)$. Furthermore

$$z_i^{q+1} - z_i^2 = z_{i+1}^q - z_{i+1}$$

Thus

$$(q+1)v_\infty(z_i) = qv_\infty(z_{i+1})$$

Now z_1 has a pole at $\tilde{P}_\infty$, hence $v_\infty(z_1)$ is negative. Hence by induction one shows that there exists a positive integer a such that

$$v_\infty(z_i) = -aq^{l-i}(q+1)^{i-1}.$$

Consider the map

$$\varphi: \mathcal{X}(l, q) \rightarrow \mathcal{Y},$$

which is the projection of the curve in $\mathbf{P}^l$ with center with equations

$$x_0 = x_1 = 0,$$

onto the line $\mathcal{Y}$ defined by the equations

$$x_2 = \dots = x_l = 0.$$

Then $t = x_0/x_1$ is a local parameter of the point

$$Q_\infty = (0 : 1 : 0 : \dots : 0)$$

in $\mathcal{Y}$. Let the map

$$\tilde{\varphi} : \tilde{\mathcal{X}}(l, q) \rightarrow \mathcal{Y}$$

be defined by

$$\tilde{\varphi} = \varphi \circ n.$$

Then $\tilde{P}_\infty$ is a point of $\tilde{\varphi}^{-1}(Q_\infty)$ and

$$v_\infty(t) = aq^{l-1},$$

so the ramification index $e_{\tilde{P}_\infty}$ of $\tilde{\varphi}$ at $\tilde{P}_\infty$ is at least q^{l-1} . For every other point Q of $\mathcal{Y}(\overline{\mathbf{F}}_q)$ not equal to Q_∞ , the inverse image $\tilde{\varphi}^{-1}(Q)$ consists of exactly q^{l-1} points over $\overline{\mathbf{F}}_q$, all with ramification index one, since the map

$$d\tilde{\varphi} : T_Q(\tilde{\mathcal{X}}(l, q)) \longrightarrow T_Q(\mathcal{Y})$$

between the tangent spaces, is surjective, as one sees from the derivative df of f . Thus

$$\deg(\tilde{\varphi}) = q^{l-1} \leq e_{\tilde{P}_\infty}.$$

Therefore $\tilde{\mathcal{X}}(l, q)$ is absolutely irreducible and

$$n^{-1}(P_\infty) = \{\tilde{P}_\infty\},$$

by the following lemma, and thus $\mathcal{X}(l, q)$ is absolutely irreducible. This proves the proposition.

Lemma 3 *Let $\mathcal{X}$ and $\mathcal{Y}$ be projective, nonsingular curves over an algebraically closed field. Suppose $\mathcal{Y}$ is irreducible. Let*

$$\varphi : \mathcal{X} \rightarrow \mathcal{Y}$$

be a finite morphism. Suppose there exist points P_∞ in $\mathcal{X}$ and Q_∞ in $\mathcal{Y}$ such that

$$\varphi(P_\infty) = Q_\infty$$

and such that the ramification index e_{P_∞} of φ at P_∞ is at least $\deg(\varphi)$. Then

$$\deg(\varphi) = e_{P_\infty},$$

$\mathcal{X}$ is irreducible, and

$$\{P_\infty\} = \varphi^{-1}(Q_\infty).$$

Proof: Suppose $\mathcal{X}_1, \dots, \mathcal{X}_s$ are the irreducible components of $\mathcal{X}$. Let φ_i be the restriction of φ to $\mathcal{X}_i$. Then

$$\deg(\varphi) = \sum_{i=1}^s \deg(\varphi_i) = \sum_{P \in \varphi^{-1}(Q)} e_P,$$

for every point Q of $\mathcal{Y}$. Suppose $P_\infty \in \mathcal{X}_1$. Then

$$\deg(\varphi) \geq \deg(\varphi_1) \geq e_{P_\infty} \geq \deg(\varphi).$$

Thus

$$\deg(\varphi) = \deg(\varphi_1) = e_{P_\infty}.$$

So

$$\{P_\infty\} = \varphi^{-1}(Q_\infty),$$

and $\mathcal{X}_1$ is the only irreducible component of $\mathcal{X}$, that is to say $\mathcal{X}$ is irreducible. This proves the lemma.

Proposition 4 *The normalization of $\mathcal{X}(l, q)$ has genus $g(l, q)$, where*

$$g(l, q) = \frac{1}{2} \left\{ \sum_{i=1}^{l-1} q^{l+1-i} (q+1)^{i-1} - (q+1)^{l-1} + 1 \right\}$$

Proof: It follows from the proof of Proposition 3 that

$$v_\infty(z_i) = -q^{l-i} (q+1)^{i-1}$$

and $n^{-1}(P_\infty)$ consists of exactly one point $\tilde{P}_\infty$. Let

$$u = \prod_{i=1}^l z_i \binom{l-1}{i-1} (-1)^{l-1-i}$$

Then u is a local parameter of $\tilde{P}_\infty$, since

$$\begin{aligned} v_\infty(u) &= \sum_{i=1}^l \binom{l-1}{i-1} (-1)^{l-1-i} v_\infty(z_i) \\ &= \sum_{i=0}^{l-1} \binom{l-1}{i} (-q)^{l-1-i} (q+1)^i \\ &= [-q + (q+1)]^{l-1} = 1. \end{aligned}$$

Differentiating the equation

$$z_{i+1} - z_{i+1}^q = z_i^{q+1} - z_i^2,$$

for $1 \leq i \leq l-1$, with respect to z_i , gives

$$\frac{dz_{i+1}}{dz_i} = 2z_i - z_i^q.$$

Hence we get by the chain rule and induction

$$\frac{dz_j}{dz_1} = \prod_{i=1}^{j-1} (2z_i - z_i^q).$$

Let $t = z_1^{-1}$. Then t is a local parameter of Q_∞ in $\mathcal{Y}$. Thus

$$\begin{aligned} \frac{du}{dt} &= -z_1^2 \frac{du}{dz_1} \\ &= -z_1^2 \left\{ \sum_{i=1}^l \prod_{j \neq i} z_j \binom{l-1}{j-1} (-1)^{l-1-j} \binom{l-1}{i-1} (-1)^{l-1-i} \right. \\ &\quad \times \left. z_i \binom{l-1}{i-1} (-1)^{l-1-i-1} \frac{dz_i}{dz_1} \right\} \\ &= -z_1^2 \left\{ \sum_{i=1}^l \binom{l-1}{i-1} (-1)^{l-1-i} u z_i^{-1} \frac{dz_i}{dz_1} \right\}. \end{aligned}$$

Now

$$v_\infty(z_i^{-1} \frac{dz_i}{dz_1}) = q^{l-i} (q+1)^{i-1} - \sum_{j=1}^{i-1} q q^{l-j} (q+1)^{j-1} > v_\infty(z_{i+1}^{-1} \frac{dz_{i+1}}{dz_1}).$$

Therefore

$$v_{\infty}\left(\frac{du}{dt}\right) = v_{\infty}\left(z_1^2 u z_1^{-1} \frac{dz_1}{dz_1}\right).$$

And we conclude

$$v_{\infty}\left(\frac{dt}{du}\right) = \sum_{i=1}^{l-1} q^{l+1-i} (q+1)^{i-1} + 2q^{l-1} - 1 - (q+1)^{l-1}.$$

The map $\tilde{\varphi}$ is separable, has degree q^{l-1} and is only ramified at $\tilde{P}_{\infty}$. Let $g = g(l, q)$. Then

$$2g - 2 = -2\deg(\tilde{\varphi}) + v_{\infty}\left(\frac{dt}{du}\right),$$

by the theorem of Hurwitz-Zeuthen, see [16]. Thus

$$g = \frac{1}{2} \left\{ \sum_{i=1}^{l-1} q^{l+1-i} (q+1)^{i-1} - (q+1)^{l-1} + 1 \right\}.$$

This proves the proposition.

Remark 8 Let P be a point on a nonsingular, absolutely irreducible curve $\mathcal{X}$ of genus g over a field. Let

$$N_n = \dim(L(nP)),$$

for $n \in \mathbb{N}$. Then

$$1 = N_0 \leq N_1 \leq \dots \leq N_{2g-1} = g,$$

so there are exactly g numbers

$$0 < n_1 < \dots < n_g < 2g$$

such that

$$L(n_i P) = L((n_i - 1)P).$$

These n_i are called *Weierstrass gaps* of P . Furthermore, if $m \in \mathbb{N}$ then

$$N_n = \#\{m \in \mathbb{N} | m \leq n \text{ and } m \text{ is not a gap at } P\}.$$

See [4].

Definition 8 Let

$$\mathcal{G}(l, q) = \{n_1, n_2, \dots, n_g\}$$

be the set of all gaps of $\tilde{P}_\infty$ on the curve $\tilde{\mathcal{X}}(l, q)$ of genus $g = g(l, q)$.

Definition 9 Let

$$\mathcal{P}(l, q) = \left\{ \sum_{i=1}^l k_i q^{l-i} (q+1)^{i-1} \mid k_i \in \mathbf{Z} \text{ and } k_i \geq 0 \right\}.$$

Proposition 5

$$\mathcal{G}(l, q) = \mathbf{N} \setminus \mathcal{P}(l, q)$$

To prove this proposition we need the following lemmas.

Lemma 4 For every $m \in \mathbf{Z}$, there are unique $u, v \in \mathbf{Z}$, such that

$$m = uq + v(q+1)^{l-1} \quad \text{and} \quad 0 \leq v < q.$$

Moreover, $m \in \mathcal{P}(l, q)$ if and only if $u \in \mathcal{P}(l-1, q)$.

Proof: Since

$$1 = -\left\{ \sum_{i=1}^{l-1} \binom{l-1}{i} q^{i-1} \right\} q + (q+1)^{l-1},$$

we have that

$$m = -m \left\{ \sum_{i=1}^{l-1} \binom{l-1}{i} q^{i-1} \right\} q + m(q+1)^{l-1},$$

for every $m \in \mathbf{N}$, furthermore there exist $a, b \in \mathbf{N}$ such that $m = aq + b$ and $0 \leq b < q$, so

$$m = \{a(q+1)^{l-1} - m \sum_{i=1}^{l-1} \binom{l-1}{i} q^{i-1}\} q + b(q+1)^{l-1}.$$

Let

$$u = a(q+1)^{l-1} - m \sum_{i=1}^{l-1} \binom{l-1}{i} q^{i-1} \quad \text{and} \quad v = b,$$

then

$$m = uq + v(q+1)^{l-1},$$

and

$$0 \leq v < q.$$

If there is another pair $(u_1, v_1) \in \mathbf{Z}^2$, such that

$$m = u_1 q + v_1 (q + 1)^{l-1}$$

and

$$0 \leq v_1 < q,$$

then we can assume without loss of generality that $u_1 \geq u$, thus

$$(u_1 - u)q + (v_1 - v)(q + 1)^{l-1} = 0,$$

so q divides $(v_1 - v)$, so $v_1 = v$, and $u_1 = u$ as well. Therefore such u and v are unique.

Now suppose

$$m = uq + v(q + 1)^{l-1},$$

and

$$0 \leq v < q.$$

If $m \in \mathcal{P}(l, q)$, then

$$m = \sum_{i=1}^l k_i q^{l-i} (q + 1)^{i-1},$$

where k_i is a non negative integer for $i = 1, \dots, l$. But $k_l = aq + b$, where $a, b \in \mathbf{N}$ and $0 \leq b < q$, so

$$m = \left\{ \sum_{i=1}^{l-1} k_i q^{l-1-i} (q + 1)^{i-1} + a(q + 1)(q + 1)^{l-2} \right\} q + b(q + 1)^{l-1},$$

hence

$$u = \sum_{i=1}^{l-1} j_i q^{l-1-i} (q + 1)^{i-1},$$

by the uniqueness of u , where $j_i = k_i$ for $i = 1, \dots, l-2$ and

$$j_{l-1} = k_{l-1} + a(q + 1).$$

Thus $u \in \mathcal{P}(l-1, q)$.

If $u \in \mathcal{P}(l-1, q)$, then $u = \sum_{i=1}^{l-1} j_i q^{l-1-i} (q + 1)^{i-1}$ for some nonnegative integers $j_1, \dots, j_{l-1}$, so

$$m = \left\{ \sum_{i=1}^{l-1} j_i q^{l-1-i} (q + 1)^{i-1} \right\} q + v(q + 1)^{l-1} \in \mathcal{P}(l, q).$$

This proves the lemma.

Lemma 5

$$\#(\mathbf{N} \setminus \mathcal{P}(l, q)) = g(l, q).$$

Proof: By induction on l .

(i) We have

$$\mathcal{P}(2, q) = \{iq + j(q+1) \mid i, j \in \mathbf{N}\},$$

so

$$\mathbf{N} \setminus \mathcal{P}(2, q) = \bigcup_{k=0}^{q-2} \{kq + (k+1), kq + (k+1) + 1, \dots, (k+1)q - 1\},$$

which is a union of mutually disjoint sets. Hence

$$\#(\mathbf{N} \setminus \mathcal{P}(2, q)) = (q-1) + (q-2) + \dots + 2 + 1 = \frac{1}{2}q(q-1),$$

which satisfies the conclusion.

(ii) Assume the conclusion is true for $l-1$. By Lemma 4 we have

$$\begin{aligned} \mathbf{N} = & \{uq + v(q-1)^{l-1} \mid u < 0, 0 \leq v < q\} \cup \\ & \{uq + v(q+1)^{l-1} \mid u \geq 0, 0 \leq v < q\}, \end{aligned}$$

where the two sets are disjoint. We denote the first set by $\mathbf{N}_1$, and the second by $\mathbf{N}_2$. Then

$$\mathbf{N} \setminus \mathcal{P}(l, q) = (\mathbf{N}_1 \setminus \mathcal{P}(l, q)) \cup (\mathbf{N}_2 \setminus \mathcal{P}(l, q)).$$

1) For each

$$uq + v(q-1)^{l-1} \in \mathbf{N}_2 \setminus \mathcal{P}(l, q)$$

we have

$$u \in \mathbf{N} \setminus \mathcal{P}(l-1, q),$$

by Lemma 4, so

$$\begin{aligned} \#(\mathbf{N}_2 \setminus \mathcal{P}(l, q)) &= q\#(\mathbf{N} \setminus \mathcal{P}(l-1, q)) \\ &= \frac{1}{2}q\left\{\sum_{j=1}^{l-2} q^{l-j}(q+1)^{j-1} - (q+1)^{l-2} + 1\right\}. \end{aligned}$$

2) For each

$$uq + v(q-1)^{l-1} \in \mathbf{N}_1 \setminus \mathcal{P}(l, q)$$

we have $u < 0$ and $0 \leq v < q$. Hence

$$\begin{aligned} uq + v(q+1)^{l-1} \geq 1 &\Leftrightarrow -uq \leq v(q+1)^{l-1} - 1 \\ &\Leftrightarrow -uq \leq v \left\{ \sum_{i=1}^{l-1} \binom{l-1}{i} q^i \right\} + v - 1 \\ &\Leftrightarrow -u \leq v \sum_{i=1}^{l-1} \binom{l-1}{i} q^{i-1}, \end{aligned}$$

since $v - 1 < q - 1$. Hence

$$\begin{aligned} \#(\mathbf{N}_1 \setminus \mathcal{P}(l, q)) &= \sum_{v=1}^{q-1} v \sum_{i=1}^{l-1} \binom{l-1}{i} q^{i-1} \\ &= \frac{1}{2} q(q-1) \sum_{i=1}^{l-1} \binom{l-1}{i} q^{i-1} \\ &= \frac{1}{2} \{q(q+1)^{l-1} - (q+1)^{l-1} - q + 1\}. \end{aligned}$$

Combining 1) and 2) gives

$$\begin{aligned} \#(\mathbf{N} \setminus \mathcal{P}(l, q)) &= \frac{1}{2} \left\{ \sum_{j=1}^{l-2} q^{l+1-j} (q+1)^{j-1} - q(q+1)^{l-2} + q \right. \\ &\quad \left. + q(q+1)^{l-1} - (q+1)^{l-1} - q + 1 \right\} \\ &= \frac{1}{2} \left\{ \sum_{i=1}^{l-1} q^{l+1-i} (q+1)^{i-1} - (q+1)^{l-1} + 1 \right\} \\ &= g(l, q). \end{aligned}$$

This proves the lemma.

Proof of Proposition 5: If $m \in \mathcal{P}(l, q)$ then

$$m = \sum_{i=1}^l k_i q^{l-i} (q+1)^i,$$

where k_i is a non negative integer for $i = 1, \dots, l$. Now

$$v_{\infty}(z_1^{k_1} z_2^{k_2} \dots z_l^{k_l}) = - \sum_{i=1}^l k_i q^{l-i} (q+1)^i = -m,$$

since

$$v_{\infty}(z_i) = -q^{l-i} (q+1)^{i-1},$$

for $i = 1, \dots, l$. So

$$z_1^{k_1} z_2^{k_2} \dots z_l^{k_l}$$

is an element of $L(m\tilde{P}_\infty)$ and not of $L((m-1)\tilde{P}_\infty)$, hence m is not a gap of $\tilde{P}_\infty$, so

$$\mathcal{G}(l, q) \subseteq N \setminus \mathcal{P}(l, q).$$

But by Lemma 5 we have

$$\#\mathcal{G}(l, q) = g(l, q) = \#(N \setminus \mathcal{P}(l, q)).$$

Therefore

$$\mathcal{G}(l, q) = N \setminus \mathcal{P}(l, q).$$

This proves the proposition.

Proposition 6 *The vector space $L(m\tilde{P}_\infty)$ is generated by*

$$\{z_1^{k_1} \dots z_l^{k_l} \mid \sum_{i=1}^l k_i q^{l-i} (q+1)^{i-1} \leq m\}.$$

Proof: This follows from Proposition 5 and Remark 8.

Corollary 3 *If*

$$2q^{l-1} > q^{l-i}(q+1)^{i-1},$$

then $1, z_1, \dots, z_i$ is a basis of

$$L(q^{l-i}(q+1)^{i-1}\tilde{P}_\infty).$$

Proof: It follows from Proposition 6 and the assumption that

$$1, z_1, \dots, z_i$$

generate the vector space we consider. The valuations at P_∞ of these $i+1$ elements are mutually distinct, so they are independent.

Corollary 4 *A q -ary first order Reed-Muller code of dimension 3 is AG.*

Proof: A q -ary first order Reed-Muller code of dimension 3 is represented by

$$(\mathcal{X}(2, q), D, G),$$

by Corollary 3, where $P_1, \dots, P_{q^2}$ are the q^2 rational points of the complement in $\mathbf{P}^2$ of the line with equation $x_0 = 0$,

$$D = \sum_{i=1}^{q^2} P_i,$$

and

$$G = (q + 1)\tilde{P}_\infty.$$

The divisor G has degree $q + 1$ which is smaller than q^2 . This proves the corollary.

Proposition 7 *If C is a q -ary linear code which has a codeword of weight equal to the word length, then C is WAG.*

Proof: Let C have dimension k . We may assume that the all one vector is a codeword, by Lemma 2. Choose a generator matrix of C such that the all one vector is the first row. Let $Q_1, \dots, Q_n$ be the points of $\mathbf{P}^{k-1}$ corresponding to the n columns of the generator matrix. Define

$$s = \max\{t \mid \text{there exist } i_1 < \dots < i_t \text{ such that } Q_{i_1} = \dots = Q_{i_t}\}.$$

Let

$$l = [k + \log_q s].$$

Then

$$s \leq q^{l-k+1},$$

and there are n distinct points $P_1, \dots, P_n$, rational over $\mathbf{F}_q$, in $\mathbf{P}^l$ such that $\pi(P_i) = Q_i$, where

$$\pi: \mathbf{P}^l \setminus H \rightarrow \mathbf{P}^{k-1}$$

is defined by

$$\pi(x_0 : \dots : x_l) = (x_0 : \dots : x_{k-1}),$$

and H is the hyperplane with equation $x_0 = 0$, since the fibres of π are isomorphic with $\mathbf{A}^{l-k+1}$. Choose a power q_0 of q such that

$$2q_0^{l-1} > q_0^{l-k}(q_0 + 1)^{k-1}.$$

Let

$$\mathcal{X} = \tilde{\mathcal{X}}(l, q_0),$$

$$G = q_0^{l-k}(q_0 + 1)^{k-1} \tilde{P}_\infty,$$

and

$$D = P_1 + \dots + P_n.$$

Then $C = C_L(\mathcal{X}, D, G)$ by Corollary 3, and C is WAG. This proves the proposition.

Theorem 2 *Every linear code is WAG.*

Proof: Let C be a linear code. Then the dual of the extended code $\overline{C}$ of C , has word length $n + 1$ and the all one vector is an element of $(\overline{C})^\perp$. Thus $(\overline{C})^\perp$ is WAG by Proposition 7, so $\overline{C}$ is WAG by Corollary 1. But C can be obtained from $\overline{C}$ by puncturing at the last coordinate. Therefore C is WAG, by Lemma 1. This proves the theorem.

IV. Criteria for Linear Codes to be Algebraic-Geometric

We first mention a few well-known theorems (Theorems 3, 4, and 5) and bounds on the genus of a curve.

Definition 10 For any divisor D on a nonsingular, absolutely irreducible curve $\mathcal{X}$ over a field we define

$$l(D) = \dim L(D),$$

and

$$i(D) = \dim \Omega(D).$$

Remark 9 If $\deg(D) < 0$ then $l(D) = 0$. If $\deg(D) > 2g - 2$ then $i(D) = 0$, where g is the genus of the curve. The Riemann-Roch Theorem states that

$$l(D) = \deg(D) + 1 - g + i(D).$$

So it gives a lower bound on $l(D)$ in terms of the degree of D . The following theorem gives an upper bound.

Theorem 3 (Clifford) *If $l(D) > 0$ and $i(D) > 0$, then*

$$l(D) \leq \frac{1}{2} \deg(D) + 1.$$

Proof: See [11].

Remark 10 A *hyperelliptic* curve is an absolutely irreducible, non-singular curve of genus at least two, which has a morphism of degree two to the projective line. The pull back under this morphism of a point of degree one on the projective line is called a *hyperelliptic* divisor. A hyperelliptic curve over $\mathbf{F}_q$ has at most $2q + 2$ rational points. We have equality in Clifford's theorem if and only if D is a principal or a canonical divisor or the curve is hyperelliptic and the divisor D is linearly equivalent with a multiple of a hyperelliptic divisor, see [11, p.343].

Definition 11 Let $N_q(g)$ be the maximal number of rational points on a nonsingular, absolutely irreducible curve, over $\mathbb{F}_q$ of genus g .

Theorem 4 (Serre's bound)

$$N_q(g) \leq q + 1 + g[2\sqrt{q}].$$

Furthermore,

$$N_2(g) \leq 0.83g + 5.35.$$

Proof: See [20].

Remark 11 Table I gives some exactly determined values of $N_q(g)$. See [16, p.34], [19], and [20].

Table I
Some known values of $N_q(g)$

[illegible]

Theorem 5 (Castelnuovo's bound) *Let $l \geq 1$. If $\mathcal{X}$ is an absolutely irreducible curve, over $\mathbb{F}_q$, in $\mathbb{P}^l$ and not contained in any hyperplane, then*

$$g(\mathcal{X}) \leq \pi(m, l).$$

Here m is the degree of $\mathcal{X}$ in $\mathbb{P}^l$, and $\pi(m, l)$ is defined by

$$\pi(m, 1) = 0,$$

$$\pi(m, l) = \frac{t(t-1)}{2}(l-1) + t\varepsilon, \quad \text{if } l > 1,$$

where t is an integer such that

$$m - 1 = t(l - 1) + \varepsilon \quad \text{and} \quad 0 \leq \varepsilon < l - 1.$$

Proof: See [1], where the proof is given for curves over the complex numbers. In [11] the proof is given in arbitrary characteristic for $l = 3$. One can easily make a proof for arbitrary l and in any characteristic, by a combination of [1] and [11].

Remark 12 It is easily verified that $\pi(m, l) \leq \pi(m', l)$, if $m \leq m'$.

The following proposition is hidden in a remark of Katsman and Tsfasman, see [13].

Proposition 8 *Let C be an $[n, k]$ code. If C is AG, then $2k \leq n + d^\perp - 1$, where $d^\perp = d^\perp(C)$ is the minimum distance of $C^\perp$.*

Proof: If $C^\perp$ is MDS then $d^\perp = k + 1$, hence $2k \leq n + d^\perp - 1$. So we may assume that $C^\perp$ is not MDS, that is to say $d^\perp \leq k$. If C is an AG code, then $C = C_L(D, G)$ for some divisor G of degree $m < n$ and $k = l(G)$. Now $C^\perp = C_\Omega(D, G)$, so there exist $d^\perp$ distinct indexes $i_1, \dots, i_{d^\perp}$ and a differential $\omega \in \Omega(G - D)$, such that

$$\text{res}_{P_{i_j}}(\omega) \neq 0 \quad \text{for } j = 1 \dots d^\perp,$$

and

$$\text{res}_{P_i}(\omega) = 0 \quad \text{for } i \notin \{i_1, \dots, i_{d^\perp}\}.$$

Put

$$D_1 = \sum_{j=1}^{d^\perp} P_{i_j}.$$

Then ω is an element of $\Omega(G - D_1)$ and not of $\Omega(G)$. But $\Omega(G - D_1)$ contains $\Omega(G)$, so

$$i(G - D_1) \geq i(G) + 1 > 0.$$

Hence

$$\begin{aligned} l(G - D_1) &= m - d^\perp + 1 - g + i(G - D_1) \geq \\ &\geq m + 1 - g + i(G) - d^\perp + 1 = k - d^\perp + 1 > 0, \end{aligned}$$

using the Riemann-Roch Theorem twice. We have

$$k - d^\perp + 1 \leq l(G - D_1) \leq 1 + \frac{m - d^\perp}{2},$$

by Clifford's Theorem. Therefore

$$2k \leq m + d^\perp \leq n + d^\perp - 1,$$

since $m \leq n - 1$. This proves the proposition.

Remark 13 The q -ary first order Reed-Muller code C of dimension 3 has length q^2 and minimum distance $q(q - 1)$, see [3]. By Corollary 4 this code is AG. If $q \geq 7$, then $C^\perp$ is not AG, by Proposition 8, since

$$2(q^2 - 3) > q^2 + q(q - 1) - 1 \quad \text{if } q \geq 7.$$

Thus we have examples of codes C such that C is AG and $C^\perp$ is not AG (see Remark 2).

Definition 12 let $g_q(n)$ be the minimal genus of a nonsingular, absolutely irreducible curve $\mathcal{X}$ over $\mathbf{F}_q$, with at least n rational points.

Remark 14 Serre's bound implies

$$n \leq q + 1 + g_q(n)[2\sqrt{q}] \quad \text{for all } n.$$

Proposition 9 Suppose $(\mathcal{X}, D, G)$ is an AG representation of a q -ary $[n, k]$ code and let $m = \deg(G)$ ($< n$).

- a) If $m \leq 2g - 2$, then $k \leq [(n + 1)/2]$.
- b) If $m > 2g - 2$, then $g_q(n) \leq g \leq n - k$.

Proof: a) If $k = 0$, then there is nothing to prove. So assume $k > 0$, hence $l(G) = k > 0$.

If $g \leq n - k$, then

$$k = l(G) = m + 1 - g + i(G) \leq g - 1 + i(G) \leq n - k - 1 + i(G),$$

hence

$$2k \leq n - 1 + i(G).$$

It follows that $i(G) > 0$ or $k \leq (n - 1)/2$.

If $g > n - k$, then

$$k = l(G) = m + 1 - g + i(G) < m + 1 - n + k + i(G) \leq k + i(G),$$

hence $i(G) > 0$.

If $i(G) > 0$, then

$$k = l(G) \leq m/2 + 1 \leq (n + 1)/2,$$

by Clifford's Theorem. Thus in every case $k \leq [(n + 1)/2]$.

b) If $m > 2g - 2$, then $i(G) = 0$. Hence

$$k = m + 1 - g \leq n - g.$$

This proves the proposition.

Corollary 5 *There exists a q -ary $[n, k]$ SAG code if and only if*

$$g_q(n) \leq \min\{k, n - k\}.$$

Proof: If a q -ary $[n, k]$ code has a SAG representation, then $g_q(n) \leq n - k$, by Proposition 9b. The dual of this code is a SAG $[n, n - k]$ code, by Corollary 1. Hence, again by Proposition 9b, $g_q(n) \leq k$. Conversely, by definition, there exists a nonsingular, absolutely irreducible curve $\mathcal{X}$ over $\mathbb{F}_q$ of genus $g = g_q(n)$, having (at least) n distinct rational points, $P_1, \dots, P_n$ say. Put $D = P_1 + \dots + P_n$. There exists a divisor G of degree $k + g - 1$ and with disjoint support with D , by the theorem of independence of valuations. Now

$$2g - 2 < \deg(G) = k + g - 1 < n,$$

since $g \leq k$ and $g \leq n - k$. Thus $(\mathcal{X}, D, G)$ represents a SAG $[n, k]$ code. This proves the corollary.

Corollary 6 *If there exists a q -ary $[n, k]$ AG code, then*

$$k \leq \left\lfloor \frac{n+1}{2} \right\rfloor, \quad \text{if } g_q(n) > n - k,$$

and

$$k \leq \left\lfloor \frac{([2\sqrt{q}] - 1)n + q + 1}{[2\sqrt{q}]} \right\rfloor, \quad \text{if } g_q(n) \leq n - k.$$

Proof: Suppose $(\mathcal{X}, D, G)$ is an AG representation of a q -ary $[n, k]$ code. Let $m = \deg(G)$. If $g_q(n) > n - k$, then $m \leq 2g - 2$, by Proposition 9b. Thus $k \leq [(n+1)/2]$, by Proposition 9a. If $g_q(n) \leq n - k$ then

$$n \leq q + 1 + g_q(n)[2\sqrt{q}] \leq q + 1 + (n - k)[2\sqrt{q}],$$

by Serre's bound, so

$$k \leq \left\lfloor \frac{([2\sqrt{q}] - 1)n + q + 1}{[2\sqrt{q}]} \right\rfloor.$$

Remark 15 Here and in Section V we shall investigate which Hamming codes $H(r, q)$ are AG. The code $H(r, q)$ is only determined up to isometries (see Definition 5), but this question makes sense anyway, by Lemma 2.

Corollary 7 *If $r \geq 3$ and the Hamming code $H(r, q)$ is AG then $(r, q) = (3, 2)$.*

Proof: Let $r \geq 3$,

$$n = (q^r - 1)/(q - 1),$$

and $k = n - r$. Then $H(r, q)$ is an $[n, k]$ code, see Definition 5. The minimum distance of its dual is q^{r-1} . If $H(r, q)$ is AG then Proposition 8 implies that

$$2\left(\frac{q^r - 1}{q - 1} - r\right) \leq \frac{q^r - 1}{q - 1} + q^{r-1} - 1,$$

so

$$\frac{q^{r-1} - 1}{q - 1} < 2r.$$

This is only possible in case the pair (r, q) is equal to $(3, 2)$, $(4, 2)$, $(3, 3)$ or $(3, 4)$. To exclude the last three possibilities, observe that $g_2(15) > 4$, $g_3(13) > 3$ and $g_4(21) > 3$, by Table I, hence

$$g_q(n) > r = n - k,$$

in these three cases, and apply Corollary 6. Since in all three cases $k > [(n+1)/2]$, we get a contradiction. This proves the corollary.

Remark 16 In Section V we shall see that $H(1, q)$ and $H(2, q)$ are SAG, for every q , and that $H(3, 2)$ is SAG.

Proposition 10 *Let $k \geq 2$. Let $(\mathcal{X}, D, G)$ be a minimal representation of a projective q -ary $[n, k]$ code. Let $l = l(G)$. Then*

$$g_q(n) \leq g(\mathcal{X}) \leq \pi(\deg(G), l-1).$$

In particular, if $(\mathcal{X}, D, G)$ is AG, moreover, then

$$g_q(n) \leq g(\mathcal{X}) \leq \pi(\deg(G), k-1).$$

Proof: By assumption, the divisor G has no base points and the morphism

$$\varphi_G : \mathcal{X} \rightarrow \mathbf{P}^{l-1}$$

has degree one. Hence $\deg(\mathcal{X}_0) = \deg(G)$, where $\mathcal{X}_0$ is the reduced image of $\mathcal{X}$ under φ_G , see Remark 6. Since $\mathcal{X}$ has (at least) n rational points, we have $g_q(n) \leq g(\mathcal{X})$. Since $\deg(\varphi_G) = 1$, we have $g(\mathcal{X}) = g(\mathcal{X}_0)$. The result now follows from Castelnuovo's bound, applied to the curve $\mathcal{X}_0$, which is absolutely irreducible and does not lie in any hyperplane. The second part of the proposition follows from the fact that $\deg(G) < n$ implies $l = k$.

Corollary 8 *Let $k \geq 2$. If there exists a q -ary projective AG $[n, k]$ code, then*

$$g_q(n) \leq \pi(n-1, k-1).$$

Proof: If a q -ary projective AG $[n, k]$ code exists, then there exists a minimal AG representation

$$(\tilde{\mathcal{X}}_0, \tilde{D}, \tilde{G})$$

of this code, by Proposition 2. The result now follows from Proposition 10, applied to this minimal representation, and Remark 12, using $\deg(\tilde{G}) \leq n-1$.

Proposition 11 *If there exists a binary projective AG $[n, k]$ code, then*

a) *If $n \geq 14$ or $n = 12$ then $k < [n/2]$,*

b) *If $n = 11$ or $n = 13$ then $k < n/2$.*

Proof: If $k < \lfloor n/2 \rfloor$, then there is nothing to prove. Suppose $k \geq \lfloor n/2 \rfloor$. If $g_2(n) \leq n - k$ then

$$n \leq 0.83(n - \lfloor \frac{n}{2} \rfloor) + 5.35,$$

by Serre's bound, which implies $n < 10$. Suppose $n \geq 10$. Then $g_2(n) > n - k$. So by Corollary 6, we have

$$\lfloor \frac{n}{2} \rfloor \leq k \leq \lfloor \frac{n+1}{2} \rfloor.$$

There are the following possibilities, a priori:

- i) $k \geq 5$ and $n = 2k$,
- ii) $k \geq 5$ and $n = 2k + 1$,
- iii) $k \geq 6$ and $n = 2k - 1$.

In the first case $\pi(n-1, k-1) = k+2$. Hence $g_2(n) \leq k+2$, by Corollary 8. So $2k \leq 0.83(k+2) + 5.35$, by Serre's bound. Thus $k \leq 5$, and $n \leq 10$, and there is nothing to prove. Similarly we get $k \leq 6, n \leq 13$ in the second case, but now $k < n/2$. Finally, we get $k \leq 5, n \leq 9$ in the third case, which therefore cannot occur. Combining these we get the desired result.

Corollary 9 *The binary Golay code and its extension are not AG.*

Proof: As we know, the minimal distances of the dual codes of the binary Golay code and its extension are greater than 3, see [17]. The binary Golay code is a $[23, 12]$ code and its extension is a $[24, 12]$ code, so they are not AG, by Proposition 11.

Remark 17 Our results do not yield a similar result concerning the ternary Golay code and its extension. The question whether these codes are AG is still unanswered.

Corollary 10 *For every $t \geq 2$, $r \geq t$, and $\epsilon \in \{0, 1\}$, the r -th order binary Reed-Muller code $RM(r, 2t + \epsilon)$ of length $2^{2t+\epsilon}$ is not AG.*

Proof: Let $r > 1$ and $m = 2t + \epsilon$, where $t \geq 2$ and $\epsilon \in \{0, 1\}$. The code $RM(m - r - 1, m)$ is the dual code of $RM(r, m)$. The length of the codewords of $RM(r, m)$ is $n = 2^m$, the dimension of $RM(r, m)$ is

$$1 + \binom{m}{1} + \dots + \binom{m}{r},$$

and the minimum distance of $RM(r, m)$ is 2^{m-r} , see [3] and [17]. So

$$d^{\perp}(RM(r, m)) = 2^{r+1} > 3.$$

If $RM(r, m)$ is AG then, since $n \geq 16$,

$$\dim RM(r, m) \leq \left\lfloor \frac{n}{2} \right\rfloor - 1 = 2^{m-1} - 1,$$

by Proposition 11. However, if $m = 2t$ or $m = 2t + 1$, and $r \geq t$, then

$$\dim RM(r, m) = 1 + \binom{m}{1} + \dots + \binom{m}{r} \geq 2^{m-1} > 2^{m-1} - 1,$$

which gives a contradiction. This proves the proposition.

Lemma 6 . Suppose $(\mathcal{X}, D, G)$ is an AG representation of an $[n, k]$ code. Then $\deg(G) \leq k + g - 1$. If $k \neq 0$ and $\deg(G) < k + g - 1$, then $\deg(G) \geq 2k - 2$.

Proof: By the Riemann-Roch Theorem,

$$k = l(G) \geq \deg(G) - g + 1,$$

so $\deg(G) \leq k + g - 1$. If $\deg(G) \neq k + g - 1$, then by Clifford's Theorem $k \leq \frac{1}{2}\deg(G) + 1$, since $l(G) = k > 0$ and

$$i(G) = k - \deg(G) + g - 1 > 0.$$

Thus $\deg(G) \geq 2k - 2$.

Corollary 11 . If $(\mathcal{X}, D, G)$ is an AG representation of an $[n, k]$ code, and $g \leq n - k$ and $k > [n/2]$, then $\deg(G) = k + g - 1$.

Proof: If $\deg(G) \neq k + g - 1$ then $2k - 2 \leq \deg(G) < k + g - 1 \leq n - 1$, by Lemma 6, so $k \leq [n/2]$. This contradicts the assumption on k .

Proposition 12 (See Table II). Let C be a binary $[n, k]$ code with $4 \leq n \leq 10$. Let k_0 and k' be given by Table II.

a) If $k > k_0$, then C is not AG.

b) Suppose that C is AG and projective, and that $k = k'$. Let $(\mathcal{X}, D, G)$ be a minimal AG representation of C . Let g be the genus of $\mathcal{X}$ and let $m = \deg(G)$. Then $(g, m) = (g', m')$ for one of the pairs (g', m') given in the last column.

Table II

Restrictions on
binary AG $[n, k]$ codes, see Proposition 12

n	k_0	k'	(g', m')
10	5	5	(6, 9) (7, 9)
9	5	5	(5, 8)
8	4	4	(4, 6) (4, 7) (5, 7) (6, 7)
7	4	4	(3, 6) (4, 6)
6	4	4	(2, 5)
		3	(2, 4) (3, 4) (3, 5) (4, 5) (5, 5) (6, 5)
5	4	4	(1, 4)
		3	(1, 3) (2, 4) (3, 4)
4	3	3	(1, 3)

Proof: a) For every n , the proof goes as follows. If $k > k_0$, then $k > [(n+1)/2]$, while

$$n - k < n - k_0 \leq g_2(n),$$

by Table I. By Corollary 6, C cannot be AG.

b) We shall only give the proof for the case $n = 6$, $k = 3$. The proofs in the other cases are analogous, and sometimes simpler. So let $n = 6$ and $k = 3$. By Table I, $g_2(6) = 2$, hence $g \geq 2$. We have $m < n = 6$. If $m = 5$, then

$$2 \leq g \leq \pi(5, 2) = 6,$$

by Proposition 10. The case $(g, m) = (2, 5)$ is excluded by Lemma 6. If $m = 4$, then

$$2 \leq g \leq \pi(4, 2) = 3,$$

by Proposition 10. Since

$$\pi(3, 2) = 1 < 2 \leq g,$$

it is not possible that $m \leq 3$, by Proposition 10 and Remark 12.

Remark 18 In Proposition 12b we do *not* claim that for every pair (g', m') given in the table a minimal AG representation with $(g, m) = (g', m')$ actually exists. As a matter of fact, in the next section we shall prove that for $n = 7$, $k' = 4$, the case $(g', m') = (4, 6)$ is impossible!

Proposition 13 *There exists a binary SAG code of length n if and only if $n \leq 8$.*

Proof: By Proposition 11, SAG codes of length $n \geq 11$ do not exist, since a SAG code is AG and its dual is too, but they cannot both have dimension smaller than $n/2$. The cases with $n \leq 10$ are dealt with by Corollary 5 and Table I: only for $n \leq 8$ there exists a k such that

$$g_2(n) \leq \min\{k, n - k\}.$$

Remark 19 (See Remark 3) There exists a binary SAG $[5, 4]$ code, by Corollary 5, since $g_2(5) = 1$, by Table I. This code is a fortiori AG. Puncturing this code gives a binary $[4, 4]$ code, which is not AG (and not SAG), by Proposition 12.

V. Explicit Representations

In part A of this section we shall give a complete answer to the question: for which r and q is the Hamming code $H(r, q)$ AG? In the affirmative case we shall give an explicit AG representation, and discuss uniqueness. In part B of this section we shall discuss an example of a code which was mentioned in a different paper, and prove that it is SAG.

A. Hamming Codes

Remark 20 Suppose that C is a linear code and that $(\mathcal{X}, D, G)$ is a representation of C , where $D = (P_1, \dots, P_n)$. Now let G' be a divisor on $\mathcal{X}$ which is linearly equivalent with G , and which has disjoint support with D too. Let $C' = C_L(\mathcal{X}, D, G')$. Let f be a rational function on $\mathcal{X}$ such that $G = G' + (f)$. Then f is defined at P_i and $f(P_i) \neq 0$, for all i . In the special case (which is the only possible case if C is binary), that $f(P_i) = f(P_j)$ for all i and j , we have $C = C'$. This is a sufficient, but, in general, not a necessary condition, by the way. By the theorem of independence of valuations, see [2, p.11], there are infinitely many rational functions f on $\mathcal{X}$ with $f(P_i) = 1$ for all i . Hence C has infinitely many representations $C_L(\mathcal{X}, D, G')$. Now return to the general case, where G' and f are arbitrary. Then $C' = \lambda C$,

where $\lambda = (\lambda_1, \dots, \lambda_n)$ and $\lambda_i = f(P_i) \in \mathbb{F}_q \setminus \{0\}$. If σ is a permutation of $\{1, \dots, n\}$, then $C_L(\mathcal{X}, \sigma D, G) = \sigma C$. For the definition of λC , σC and σD , see Definition 4 and the proof of Lemma 2. We have $C_L(\mathcal{X}, \sigma D, G) = C$ if and only if $\sigma \in \text{Aut}(C)$. Here $\text{Aut}(C)$ is the automorphism group of C , see [17, p.229]. We see that the triple $(\mathcal{X}, \sigma D, G')$ represents a code that is isometric with C . The proof of Lemma 2 shows that every code isometric with C can be represented this way, that is to say, by a triple $(\mathcal{X}, \sigma D, G')$ for a suitable permutation σ and a divisor G' linearly equivalent with G . If $\mathcal{X}'$ is a curve and

$$\varphi: \mathcal{X}' \rightarrow \mathcal{X}$$

is an isomorphism, then

$$C_L(\mathcal{X}', \varphi^*(D), \varphi^*(G))$$

is also a representation of C . Here $\varphi^*(D)$ and $\varphi^*(G)$ denote the pull backs of D and G to $\mathcal{X}'$ under φ , respectively. When discussing uniqueness of representations of codes, one doesn't wish to distinguish between isomorphic curves, nor between linearly equivalent divisors G , nor between divisors D which can be obtained from each other by a permutation of the rational points in their support. By the above reasoning, it is therefore more convenient and more useful to think of a representation as a representation of the whole collection of codes isometric to a particular code, rather than to consider it as a representation of a *single* code. For given $\mathcal{X}$ and D it is actually sufficient only to specify the linear equivalence class of G , since in the linear equivalence class of any divisor there is a divisor which has disjoint support with D , by the independence of valuations.

Therefore, we introduce the following concepts.

Definition 13 a) If two linear codes C and C' are isometric, we denote this by $C \sim C'$. We define the *isometry class* of a linear code C to be the set of all codes which are isometric with C , and we denote this class by $[C]$.

b) Let $(\mathcal{X}, D, G)$ and $(\mathcal{X}', D', G')$ be representations (not necessarily of the same code). Let $D = (P_1, \dots, P_n)$ and $D' = (Q_1, \dots, Q_n)$. We call these representations *isometric*, denoted by

$$(\mathcal{X}, D, G) \sim (\mathcal{X}', D', G'),$$

if there exists an isomorphism $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$ and a permutation σ of $\{1, \dots, n\}$, such that $\varphi(Q_{\sigma(i)}) = P_i$, for all i , and such that the pull back $\varphi^*(G)$ of G is linearly equivalent with G' . We define the *isometry class* of a representation $(\mathcal{X}, D, G)$ to be the set of all representations isometric with this representation, and denote it by $[(\mathcal{X}, D, G)]$.

c) We call an isometry class $[(\mathcal{X}, D, G)]$ of representations a *representation class* of an isometry class $[C]$ of codes if $C_L(\mathcal{X}, D, G)$ is isometric with C .

Remark 21 a) Isometry of codes and isometry of representations are equivalence relations in the sets of codes and representations, respectively. The isometry classes defined in Definition 13a) and b) are the equivalence classes under these equivalence relations.

b) We call a representation class (WAG), AG, SAG or *minimal* according to whether there is a representation in this class which is (WAG), AG, SAG, or minimal, respectively. This definition is obviously independent from the choice of the representation. Besides, $\deg(G)$ and $g(\mathcal{X})$ do not depend on this choice either. Similarly, we can speak about a WAG, AG, or SAG isometry class of codes, by Lemma 2.

c) As pointed out in Remark 20, isometric representations represent isometric codes, and if a code C is represented by $(\mathcal{X}, D, G)$, then for every code in $[C]$ there is a representation of this code in $[(\mathcal{X}, D, G)]$.

d) To specify a representation class it is of course sufficient only to give one of the representations in this class.

e) We shall not be too careful with the language we use to express that a (class of) code(s) is represented by a (class of) representation(s). But by Remark 20 there will never be misunderstandings about the right interpretation.

Remark 22 Recall from Section II (Definition 5 and Remark 4) that $H(r, q)$ denotes any q -ary linear code with parameters

$$[n = (q^r - 1)/(q - 1), n - r, 3]$$

(codes of the same length and dimension, but with minimum distance greater than 3 do not exist). All such codes are isometric. From now on, we shall use the notation $H(r, q)$ also to denote the isometry class consisting of all the q -ary linear codes with these parameters. It is well-known and easily deduced from Definition 5 and Remark 4 that a q -ary

linear code is a Simplex code $S(r, q)$ if and only if it has parameters

$$[n = (q^r - 1)/(q - 1), r, q^{r-1}].$$

Similar to the case of the Hamming codes, we shall also use the notation $S(r, q)$ to denote the isometry class consisting of all the q -ary linear codes with these parameters.

In Section IV (Corollary 7) we already saw that the only Hamming codes that can possibly be AG are those with $r = 1$, $r = 2$, or $(r, q) = (3, 2)$. The cases $r = 1$ and $r = 2$ are dealt with by the following proposition.

Proposition 14 *For every q , $H(1, q)$ and $H(2, q)$ are SAG.*

Proof: Let $\mathcal{X} = \mathbf{P}^1$, the projective line over $\mathbf{F}_q$. Let $P_1, \dots, P_{q+1}$ be the $\mathbf{F}_q$ -rational points on $\mathcal{X}$. We have

$$H(1, q) = \{0\} = C_L(\mathcal{X}, D, G)$$

if we choose $D = P_1$, $G = -P_2$. In this case

$$2g - 2 = -2 < -1 = \deg(G) < 1 = n.$$

Hence $H(1, q)$ is SAG. To prove that $H(2, q)$ is SAG, take the same curve $\mathcal{X}$, but now take $D = P_1 + \dots + P_{q+1}$, and let G be any divisor of degree $q - 2$, with support disjoint from the support of D . Since $g = 0$, $C_L(\mathcal{X}, D, G)$ is an MDS code with parameters

$$[n = q + 1, k = q - 2 + 1 = q - 1, d = q + 1 - (q - 2) = 3],$$

by Theorem 1, i.e., $C_L(\mathcal{X}, D, G)$ is a Hamming code $H(2, q)$. We have

$$2g - 2 = -2 < q - 2 = \deg(G) < q + 1 = n.$$

Hence $H(2, q)$ is SAG.

Remark 23 *a)* The number of SAG representation classes of a given code is always finite, because the genus g is upper bounded by $2g - 2 < m < n$, hence $g \leq n/2$, and because there are only finitely many nonisomorphic curves of a given genus, and the number h of linear equivalence classes of divisors of degree m is finite for each curve.

b) The number of *minimal* AG representation classes of a given projective code of dimension at least two is finite, since the genus g is upper bounded by $g \leq \pi(n-1, k-1)$, by Proposition 10 and Remark 12.

Remark 24 a) The SAG representation class of $H(1, q)$ given in the proof of Proposition 14 is *unique*, that is to say, $H(1, q)$ has no other SAG representation classes. Namely, suppose that $(\mathcal{X}, D, G)$ is a SAG representation of $H(1, q)$ and let $m = \deg(G)$. Then

$$2g - 2 < m < n = 1$$

implies $g = 0$ and $m \in \{-1, 0\}$. If $m = 0$, then the dimension of $C_L(\mathcal{X}, D, G)$ is one. Hence $m = -1$. All divisors of degree -1 on $\mathcal{X}$ are linearly equivalent.

b) There are infinitely many AG representation classes of $H(1, q)$. Namely, choose any curve $\mathcal{X}$ over $\mathbb{F}_q$ having at least one rational point, P_1 say. Put $D = P_1$. Let G be any divisor on $\mathcal{X}$ with $P_1 \notin \text{supp}(G)$ and $\deg(G) < 0$. Then $L(G) = \{0\}$ and $C_L(\mathcal{X}, D, G)$ is an $H(1, q)$. We could also let G be a divisor of degree 0 on $\mathcal{X}$ which is not principal (such a G exists if and only if $h > 1$).

c) For example in the case $q = 2$ we find infinitely many AG representation classes of $H(2, q)$ as follows. Choose any curve $\mathcal{X}$ over $\mathbb{F}_2$ having at least three rational points, P_1, P_2 and P_3 , say, and put $D = P_1 + P_2 + P_3$. Take $G = 0$. Then $L(G) = \{0, 1\}$, hence

$$C_L(\mathcal{X}, D, G) = \{000, 111\},$$

which is an $H(2, 2)$. If there is a fourth rational point on $\mathcal{X}$, P_4 say, then we could also take $G = P_4$ (and the same D). Namely, it follows that $g > 0$, and by Riemann-Roch and Clifford's theorem (see also [11, p.138, Example 6.10.1]), $l(G) = 1$, hence again $L(G) = \{0, 1\}$. If we choose for $\mathcal{X}$ an elliptic curve with at least four rational points, this latter example gives a representation which is not only AG, but even SAG, which shows that the SAG representation class of $H(2, q)$ given in the proof of Proposition 14 is *not* unique (at least for $q = 2$).

Let us now concentrate on $H(3, 2)$. We shall prove that $H(3, 2)$ is indeed AG (we shall even prove that it is SAG), and, moreover, that it has a *unique* minimal AG representation class. The latter statement

is not true if we replace AG by WAG, as we shall see. To do so, let us first try to find a triple $(\mathcal{X}, D, G)$ such that the code $C_L(\mathcal{X}, D, G)$ is a binary code with parameters $[7, 4, 3]$, hence is equal to an $H(3, 2)$. First of all, we need a nonsingular, absolutely irreducible projective curve defined over $\mathbf{F}_2$, having at least seven rational points. Such a curve cannot be hyperelliptic (since then it would have at most six rational points), and it has genus at least three (see Table I). If it has genus equal to three, such a curve, since not hyperelliptic, is isomorphic to a nonsingular and absolutely irreducible plane projective curve of degree four. Let S be the set consisting of all the (not necessarily nonsingular or absolutely irreducible, *a priori*) plane projective curves $\mathcal{X}$ of degree four, which have the following property: $\mathcal{X}$ goes through all the seven $\mathbf{F}_2$ -rational points of $\mathbf{P}^2$, and none of these seven points is a singularity of $\mathcal{X}$. The set S is easily computed. It has 24 elements.

One of the curves in S is the following one, which we call $\mathcal{X}_1$, defined by

$$xy(x+y)(x+z) + xz^2(x+z) + y^2z(y+z) = 0. \quad (3)$$

This curve was mentioned earlier by Serre [20]. We have checked that $\mathcal{X}_1$ is nonsingular. By Bézout's theorem it is also absolutely irreducible. Let L be one of the seven lines defined over $\mathbf{F}_2$ in $\mathbf{P}^2$. By Bézout's theorem, the degree of the intersection divisor $L \cdot \mathcal{X}_1$ is 4. There are three rational points on L , which are also on $\mathcal{X}_1$. It follows that $\mathcal{X}_1$ intersects L with multiplicity 2 at exactly one of them, and that the intersection is transversal at the two remaining points. In other words, the tangents to $\mathcal{X}_1$ at the seven rational points are precisely the seven lines defined over $\mathbf{F}_2$. We have named these points and lines, and computed the intersection divisors with the curve in Table III. We shall denote by L_i the tangent line to $\mathcal{X}_1$ at P_i , and by L_{ij} the line through P_i and P_j .

Table III

The $\mathbf{F}_2$ -rational points P_i on the curve $\mathcal{X}_1$,
the tangents L_i to $\mathcal{X}_1$ at these points,
and the intersection divisors $L_i \cdot \mathcal{X}_1$

P_i	L_i	$L_i \cdot \mathcal{X}_1$
$P_1 = (0 : 0 : 1)$	$x = 0$	$2P_1 + P_2 + P_3$
$P_2 = (0 : 1 : 0)$	$z = 0$	$2P_2 + P_4 + P_6$
$P_3 = (0 : 1 : 1)$	$y + z = 0$	$2P_3 + P_4 + P_7$
$P_4 = (1 : 0 : 0)$	$y = 0$	$P_1 + 2P_4 + P_6$
$P_5 = (1 : 0 : 1)$	$x + z = 0$	$P_2 + 2P_5 + P_7$
$P_6 = (1 : 1 : 0)$	$x + y + z = 0$	$P_3 + P_5 + 2P_6$
$P_7 = (1 : 1 : 1)$	$x + y = 0$	$P_1 + P_6 + 2P_7$

The group $PGL(2, \mathbf{F}_2)$ of $\mathbf{F}_2$ -automorphisms of $\mathbf{P}^2$ acts on the set S , and has order 168. It also acts on the set $\{P_1, \dots, P_7\}$ of $\mathbf{F}_2$ -rational points, and on the set $\{L_1, \dots, L_7\}$ of lines over $\mathbf{F}_2$. Put

$$\mathcal{H} := \{\tau \in PGL(2, \mathbf{F}_2) \mid \tau(P_1) = P_1\}.$$

This is a subgroup of $PGL(2, \mathbf{F}_2)$ of order 24.

Lemma 7 $\mathcal{H}$ acts transitively on S .

Proof: Suppose $\tau \in \mathcal{H}$ is such that $\tau\mathcal{X}_1 = \mathcal{X}_1$. Evidently, $\mathcal{H}$ acts on the group $\text{Div}(\mathcal{X}_1)$ of divisors on $\mathcal{X}_1$, and for every i we have

$$\tau(L_i \cdot \mathcal{X}_1) = \tau L_i \cdot \tau \mathcal{X}_1 = \tau L_i \cdot \mathcal{X}_1 = L_j \cdot \mathcal{X}_1,$$

for some j . Since there is only one line L_i with $v_{P_1}(L_i \cdot \mathcal{X}_1) = 2$, L_1 namely, we must have $\tau L_1 = L_1$. Hence either *i*) $\tau(P_2) = P_2$ and $\tau(P_3) = P_3$, or *ii*) $\tau(P_2) = P_3$ and $\tau(P_3) = P_2$. For similar reasons, in case *i*), $\tau L_2 = L_2$ and $\tau L_3 = L_3$, and in case *ii*) $\tau L_2 = L_3$ and $\tau L_3 = L_2$. In both cases we get

$$\{\tau(P_4)\} = \{\tau(L_2 \cap L_3)\} = \tau L_2 \cap \tau L_3 = L_2 \cap L_3 = \{P_4\}.$$

In case *i*) we now have three non-collinear points P_1, P_2, P_4 fixed by τ , which implies that τ is the identity. Case *ii*) cannot occur, because in this case it follows that $\tau(P_6) = P_7$ and $\tau(P_7) = P_6$, and we get

$$\tau(L_7 \cdot \mathcal{X}_1) = \tau(P_1 + P_6 + 2P_7) = P_1 + 2P_6 + P_7,$$

which is not an intersection divisor $L_j \cdot \mathcal{X}_1$, a contradiction. This proves that the $\mathcal{H}$ -stabilizer of $\mathcal{X}_1$ is trivial, and hence that the $\mathcal{H}$ -orbit of $\mathcal{X}_1$ has order 24. This proves the lemma.

By this lemma, all the 24 curves in the set S are isomorphic (even stronger: they only differ by a projective change of coordinates), and they are all nonsingular and absolutely irreducible, since $\mathcal{X}_1$ is. By the preceding discussion we have the following result.

Lemma 8 *Any absolutely irreducible nonsingular curve defined over $\mathbb{F}_2$, of genus three, having at least seven rational points, is isomorphic to the curve $\mathcal{X}_1$.*

So now we already have a curve $\mathcal{X}_1$ and a divisor

$$D_1 := P_1 + P_2 + \cdots + P_7.$$

The remaining problem is to find a suitable divisor G on $\mathcal{X}_1$, provided it exists.

Remark 25 The following lemma does not only apply to our situation, but it is true for a general triple $(\mathcal{X}, D, G)$. It is an analogue of Lemma 6.

Lemma 9 *If $(\mathcal{X}, D, G)$ is a representation of a q -ary $[n, k]$ code and $m := \deg(G) \geq n$, then $k = n$ or $k \geq (m + n)/2 - g$.*

Proof: (See Definition 1). We have

$$k = l(G) - \dim(\text{kernel } \alpha_L) = l(G) - l(G - D) \leq n,$$

by the Riemann-Roch theorem. If inequality holds, then necessarily

$$\dim \Omega(G - D) > 0,$$

and by Clifford's theorem,

$$l(G - D) \leq 1 + (m - n)/2.$$

Hence

$$k = l(G) - l(G - D) \geq m + 1 - g - 1 - (m - n)/2 = (m + n)/2 - g.$$

Returning to the specific curve $\mathcal{X}_1$, let

$$C := C_L(\mathcal{X}_1, D_1, G)$$

have dimension k . If $k = 4$, then necessarily $\deg(G) = 6$ or $\deg(G) = 7$, by Lemmas 6 and 9. If $\deg(G) = 6$, then indeed $k = l(G) = 6 + 1 - 3 = 4$, by the Riemann-Roch theorem. If $\deg(G) = 7$, then $l(G) = 7 + 1 - 3 = 5$, and we have $k = l(G) - l(G - D_1)$, while $\deg(G - D_1) = 0$. Hence $k = 4$ in this case if and only if $G - D_1$ is a principal divisor, i.e., $G \sim D_1$. We have proved the following lemma.

Lemma 10 *The dimension of $C = C_L(\mathcal{X}_1, D_1, G)$ equals 4 if and only if $\deg(G) = 6$ or $G \sim D_1$ (provided $\text{supp}(G) \cap \text{supp}(D_1) = \emptyset$).*

The thing left to do is to settle the problem that C might have the wrong minimum distance. Any binary $[7, 4]$ code has minimum distance at most 3. Hence $d(C) \leq 3$.

The following lemma applies to the curve $\mathcal{X}_1$.

Lemma 11 *Let $\mathcal{X}$ be a non-hyperelliptic curve of genus $g \geq 3$. If B is an effective divisor on $\mathcal{X}$ of degree at most two, then $l(B) = 1$. Hence, if two effective divisors on $\mathcal{X}$ of degree at most two are equivalent, then they are equal.*

Proof: The case $B = 0$ is trivial. Suppose that $\deg(B) > 0$. Since B is effective, we have

$$1 \leq l(B) = \deg(B) + 1 - g + i(B),$$

by the Riemann-Roch Theorem. Since $\deg(B) \leq 2$ and $g \geq 3$, it follows that $i(B) > 0$. Since

$$0 < \deg(B) < 4 \leq 2g - 2,$$

B is not principal or canonical, hence

$$l(B) < 1 + \deg(B)/2 \leq 2,$$

by Clifford's Theorem, Remark 10, and the assumption that $\mathcal{X}$ is not hyperelliptic. This proves the lemma.

Proposition 15 *If $G \sim D_1$ (and $\text{supp}(G) \cap \text{supp}(D_1) = \emptyset$), then $C = C_L(\mathcal{X}_1, D_1, G)$ is a binary $[7, 4, 3]$ code.*

Proof: The only thing left to prove is that $d(C) \geq 3$. Let $G = D_1 + (f_0)$, with f_0 a nonzero rational function on $\mathcal{X}_1$. Since $\text{supp}(G) \cap \text{supp}(D_1) = \emptyset$,

$$v_{P_i}(f_0) = -1 \quad \text{for } i = 1, 2, \dots, 7.$$

A nonzero codeword of weight ≤ 2 exists if and only if an

$$f \in L(G) \setminus L(G - D_1)$$

exists such that

$$(f) \geq -G + P_a + P_b + P_c + P_d + P_e,$$

for some distinct a, b, c, d, e .

Since $G = D_1 + (f_0)$, this is equivalent to

$$(f) \geq -(f_0) - P_s - P_t, \quad \text{for some distinct } s \text{ and } t,$$

that is to say,

$$ff_0 \in L(P_s + P_t).$$

But this cannot happen, since we would have $ff_0 = 1$, by Lemma 11, and hence

$$v_{P_i}(f) = -v_{P_i}(f_0) = 1,$$

for $i = 1, 2, \dots, 7$, contradicting $f \notin L(G - D_1)$.

Remark 26 As already pointed out in Remark 20, divisors $G \in [D_1]$ with $\text{supp}(G) \cap \text{supp}(D_1) = \emptyset$ exist, by the theorem of independence of valuations. To give the representation explicitly, we need an explicit example of such a divisor. Let f_0 be the rational function on $\mathcal{X}_1$ defined as follows:

$$f_0 = \frac{H_2 H_3}{H_1 L_1 L_6},$$

where

$$H_1 = x^3 z + x^2 y^2 + x^2 z^2 + xy^3 + y^3 z + yz^3,$$

$$H_2 = x^3 + xy^2 + x^2 z + xz^2 + y^2 z + yz^2 + xyz,$$

$$H_3 = x^3 + y^3 + x^2 y + xy^2 + xz^2 + yz^2 + xyz.$$

The equations of the lines L_1 and L_6 are $x = 0$ and $x + y + z = 0$, respectively, see Table III. For convenience, we use the same notation

for a form and its zero set. Let Q be the place of degree 3 on $\mathcal{X}_1$ that corresponds to the orbit

$$\{(\alpha^2 : \alpha : 1), (\alpha^4 : \alpha^2 : 1), (\alpha : \alpha^4 : 1)\}$$

of the $\mathbf{F}_8$ -rational point $\{(\alpha^2 : \alpha : 1)\}$ on $\mathcal{X}_1$, where $\text{Gal}(\mathbf{F}_8/\mathbf{F}_2)$ is the group acting, and $\mathbf{F}_8 = \mathbf{F}_2(\alpha)$ with

$$\alpha^3 + \alpha + 1 = 0.$$

Let T and R be the places of degree 8 on $\mathcal{X}_1$ corresponding to the orbits of the $\mathbf{F}_{256}$ -rational points $(\beta^6 : \beta^7 : 1)$ and $(\beta^{215} : \beta^{87} : 1)$ on $\mathcal{X}_1$, respectively, where $\text{Gal}(\mathbf{F}_{256}/\mathbf{F}_2)$ is the group acting, and $\mathbf{F}_{256} = \mathbf{F}_2(\beta)$ with

$$\beta^8 + \beta^4 + \beta^3 + \beta^2 + 1 = 0.$$

For the intersection divisors with the curve $\mathcal{X}_1$, we have

$$H_1 \cdot \mathcal{X}_1 = D_1 + 3Q,$$

$$H_2 \cdot \mathcal{X}_1 = P_1 + P_2 + P_3 + P_6 + T,$$

$$H_3 \cdot \mathcal{X}_1 = P_1 + P_3 + P_5 + P_6 + R,$$

$$L_1 \cdot \mathcal{X}_1 = 2P_1 + P_2 + P_3,$$

$$L_6 \cdot \mathcal{X}_1 = P_3 + P_5 + 2P_6.$$

The curve H_1 is one of the curves in the set S . Put

$$G_1 := D_1 + (f_0).$$

Then

$$G_1 = T + R - 3Q.$$

We have $G_1 \sim D_1$ and $\text{supp}(G_1) \cap \text{supp}(D_1) = \emptyset$. Consequently, $(\mathcal{X}_1, D_1, G_1)$ is a WAG representation of an $H(3, 2)$.

This settles the case $G \sim D_1$. We shall now investigate the case $\deg(G) = 6$. This case requires more work. For $r := 1, 2, \dots$, let N_r be the number of points (of degree one) on $\mathcal{X}_1$ over $\mathbf{F}_{2^r}$. Let $\mathcal{D}_i$ denote the set of all effective divisors on $\mathcal{X}_1$ of degree i , and let $a_i := \#\mathcal{D}_i$. One computes that $N_1 = 7$, $N_2 = 7$ and $N_3 = 10$. Hence there are no places of degree 2, and there is exactly one place of degree 3 on

$\mathcal{X}_1$. This is the place Q mentioned in Remark 26. From N_1 , N_2 and N_3 the zeta-function of $\mathcal{X}_1$ can be computed. The function $Z(\mathcal{X}_1, t)$, a rational function of t , is defined by

$$Z(\mathcal{X}_1, t) = \sum_{i=0}^{\infty} a_i t^i = \exp \left(\sum_{r=1}^{\infty} \frac{N_r}{r} t^r \right).$$

One computes that the polynomial

$$P(t) = 1 + 4t + 9t^2 + 15t^3 + 18t^4 + 16t^5 + 8t^6$$

satisfies

$$Z(\mathcal{X}_1, t) = \frac{P(t)}{(1-t)(1-2t)}.$$

It follows that

$$h = \#Pic_0(\mathcal{X}_1) = P(1) = 71,$$

and that

$$a_0 = 1, a_1 = 7, a_2 = 28, a_3 = 85.$$

For the underlying theory of zeta functions, see [16, p.66 a.f.], for instance.

Lemma 12 *Let B be a divisor of degree 3 on $\mathcal{X}_1$.*

a) If $B \sim P_a + P_b + P_c$ for three distinct collinear rational points P_a, P_b, P_c , then $l(B) = 2$. Otherwise, $l(B) = 1$.

b) Suppose B is effective, moreover. If $B \leq L_i \cdot \mathcal{X}_1$ for some i , then $l(B) = 2$. Otherwise $l(B) = 1$.

Proof: a) Consider the map

$$\phi: \mathcal{D}_3 \rightarrow Pic_3(\mathcal{X}_1), \quad \phi(B) = [B],$$

where $[B]$ is the linear equivalence class of B . Let B be a divisor on $\mathcal{X}_1$ of degree 3. By the Riemann-Roch theorem, $l(B) \geq 1$, and by Clifford's theorem, $l(B) \leq 2$. The number of inverse images of $[B]$ under ϕ equals

$$\#|B| = \#\mathbf{P}(L(B)) = 2^{l(B)} - 1,$$

which is 1 or 3. In particular, ϕ is surjective. Here $|B|$ is the complete linear system associated to B , that is, the set of effective divisors linearly equivalent to B . Now suppose that a, b, c are distinct numbers

such that P_a, P_b and P_c are collinear. Without loss of generality we may assume that the line through these three points is the line L_a . We can choose d, e, f, g such that

$$\{a, b, c, d, e, f, g\} = \{1, 2, \dots, 7\}$$

and

$$\begin{aligned} L_a \cdot \mathcal{X}_1 &= 2P_a + P_b + P_c, \\ L_d \cdot \mathcal{X}_1 &= P_a + 2P_d + P_e, \\ L_f \cdot \mathcal{X}_1 &= P_a + 2P_f + P_g. \end{aligned}$$

Now $0, 1, L_d/L_a, L_f/L_a$ are four distinct functions in $L(P_a + P_b + P_c)$. Hence

$$L(P_a + P_b + P_c) = \{0, 1, L_d/L_a, L_f/L_a\},$$

$$l(P_a + P_b + P_c) = 2,$$

and

$$\#|P_a + P_b + P_c| = 3.$$

We have

$$\left(\frac{L_d}{L_a}\right) = 2P_d + P_e - P_a - P_b - P_c,$$

$$\left(\frac{L_f}{L_a}\right) = 2P_f + P_g - P_a - P_b - P_c.$$

Hence

$$\begin{aligned} \phi^{-1}([P_a + P_b + P_c]) &= |P_a + P_b + P_c| \\ &= \{P_a + P_b + P_c, 2P_d + P_e, 2P_f + P_g\}. \end{aligned}$$

Since there are seven lines over $\mathbf{F}_2$ in $\mathbf{P}^2$, there are seven possibilities for $P_a + P_b + P_c$. These seven divisors represent seven *distinct* elements of $\text{Pic}_3(\mathcal{X}_1)$, each having three inverse images under ϕ . All the other $h-7 = 71-7 = 64$ elements of $\text{Pic}_3(\mathcal{X}_1)$ must have exactly one inverse image, since

$$a_3 = 85 = 7 \cdot 3 + 64 \cdot 1.$$

So, for every divisor B not equivalent to one of the seven divisors $P_a + P_b + P_c$, we have $\#|B| = 1$, hence $l(B) = 1$. This proves *a*).

b) This follows immediately from the proof of *a*).

Lemma 13 *The group $PGL(2, \mathbf{F}_2)$ has an element τ of order 7 such that $\tau(\mathcal{X}_1) = \mathcal{X}_1$. For any such τ , the subgroup $\langle \tau \rangle$ of $PGL(2, \mathbf{F}_2)$ generated by τ acts transitively on the set $\{P_1, P_2, \dots, P_7\}$ of rational points.*

Proof: The automorphism

$$\tau: (x : y : z) \mapsto (x + y + z : x + y : y + z)$$

has order 7. One easily verifies that $\tau(\mathcal{X}_1) = \mathcal{X}_1$. Let τ be an automorphism, not necessarily this one, of order 7 with $\tau(\mathcal{X}_1) = \mathcal{X}_1$. Then the $\langle \tau \rangle$ -orbit of P_1 has order 1 or 7. But, as we saw in the proof of Lemma 7, from $\tau(\mathcal{X}_1) = \mathcal{X}_1$ and $\tau(P_1) = P_1$, it would follow that τ is the identity, a contradiction. Hence the $\langle \tau \rangle$ -orbit of P_1 has order 7. This proves the lemma.

Proposition 16 *Suppose $\deg(G) = 6$ (and $\text{supp}(G) \cap \text{supp}(D_1) = \emptyset$). Then $C = C_L(\mathcal{X}_1, D_1, G)$ is a binary $[7, 4, 3]$ code if and only if*

$$G \sim 2Q.$$

Proof: A nonzero codeword of weight ≤ 2 corresponds to a

$$g_0 \in L(G) \setminus \{0\}$$

with

$$(g_0) \geq -G + P_a + P_b + P_c + P_d + P_e \quad \text{for some distinct } a, b, c, d, e.$$

Since

$$\deg(-G + P_a + \dots + P_e) = -1,$$

there is a rational point P_f such that

$$(g_0) = -G + P_a + \dots + P_e + P_f.$$

Define the set of divisors Λ by

$$\Lambda := \{P_a + P_b + P_c + P_d + P_e + P_f \mid a, b, c, d, e \text{ distinct}\}$$

It follows that $d(C) = 3$ if and only if

$$G \not\sim E \text{ for all } E \in \Lambda. \quad (4)$$

The set Λ has 112 elements. Let us determine the number t of equivalence classes in Λ under the (induced) linear equivalence relation. For $E \in \Lambda$ we denote by $\overline{E}$ its equivalence class. Write

$$\Lambda = \Lambda^* \cup \Lambda_1 \cup \dots \cup \Lambda_7,$$

(disjoint union), where

$$\Lambda^* = \{P_a + P_b + P_c + P_d + P_e + P_f | a, b, c, d, e, f \text{ distinct}\},$$

and

$$\Lambda_f = \{P_a + P_b + P_c + P_d + P_e + P_f | \\ a, b, c, d, e \text{ distinct and } f \in \{a, b, c, d, e\}\}.$$

Then $\#\Lambda^* = 7$, and

$$\#\Lambda_f = \binom{6}{4} = 15,$$

for every f .

For every $E \in \Lambda^*$ we have $\#\overline{E} = 1$. Namely, if

$$E = P_a + P_b + \dots + P_f \sim E'$$

for an

$$E' = P_{a'} + P_{b'} + \dots + P_{f'} \in \Lambda,$$

with

$$a', b', c', d', e' \text{ distinct,}$$

then

$$\#(\{a, b, c, d, e, f\} \cap \{a', b', c', d', e'\}) \geq 4.$$

Without loss of generality $a = a'$, $b = b'$, $c = c'$, $d = d'$. Hence

$$P_e + P_f \sim P_{e'} + P_{f'}.$$

By Lemma 11,

$$P_e + P_f = P_{e'} + P_{f'},$$

and hence $E = E'$.

For $f := 1, \dots, 7$ and every i , define

$$w_i(f) := \#\{E \in \Lambda_f | \#\overline{E} = i\}.$$

We shall determine these numbers. By Lemma 13,

$$w_i(f) = w_i(1) =: w_i,$$

for all i and f , hence it suffices to consider the case $f = 1$. Suppose that

$$E = P_a + P_b + \cdots P_e + P_1 \in \Lambda_1,$$

$$E' \in \Lambda, \quad E \neq E', \quad \text{and} \quad E \sim E'.$$

Then $E' \in \Lambda_{f'}$ for some f' . Write

$$E' = P_{a'} + P_{b'} + \cdots + P_{f'}.$$

Then

$$\#(\{a, b, c, d, e\} \cap \{a', b', c', d', e'\}) \geq 3.$$

Without loss of generality, $c = c'$, $d = d'$, $e = e'$. Hence

$$P_a + P_b + P_1 \sim P_{a'} + P_{b'} + P_{f'}.$$

These two divisors are unequal, because E and E' are unequal. Hence

$$l(P_a + P_b + P_1) > 1.$$

Using Lemma 12 and Table III, one readily finds out that there are five possibilities for the divisor $P_a + P_b + P_1$. They are listed in the second column of Table IV. For each of them, following the proof of Lemma 12, one easily determines all the possible divisors $P_{a'} + P_{b'} + P_{f'}$. Except in the two cases

$$(P_a + P_b + P_1, P_{a'} + P_{b'} + P_{f'}) = (2P_1 + P_2, P_3 + P_4 + P_7)$$

and

$$(P_a + P_b + P_1, P_{a'} + P_{b'} + P_{f'}) = (2P_1 + P_3, P_2 + P_4 + P_6),$$

there is only one choice of f' and $\{c, d, e\}$, such that both a, b, c, d, e and a', b', c, d, e are five distinct numbers. In each of the two exceptional

cases, there are three such choices.

Table IV

This table is used in the proof of Proposition 16

All $E \in \Lambda_1$ with $\#E > 1$ are listed

$\{a, b\}$	$P_a + P_b + P_1$	$P_{a'} + P_{b'} + P_{f'}$	f'	$\{a', b'\}$	$\{c, d, e\}$	$E =$ $P_a + P_b + P_c + P_d + P_e + P_1$
2, 3	$P_1 + P_2 + P_3$	$2P_4 + P_5$	4	4, 5	1, 6, 7	$2P_1 + P_2 + P_3 + P_6 + P_7$
		$P_6 + 2P_7$	7	6, 7	1, 4, 5	$2P_1 + P_2 + P_3 + P_4 + P_5$
4, 5	$P_1 + P_4 + P_5$	$2P_2 + P_6$	2	2, 6	1, 3, 7	$2P_1 + P_3 + P_4 + P_5 + P_7$
		$2P_3 + P_7$	3	3, 7	1, 2, 6	$2P_1 + P_2 + P_4 + P_5 + P_6$
6, 7	$P_1 + P_6 + P_7$	$2P_3 + P_4$	3	3, 4	1, 2, 5	$2P_1 + P_2 + P_5 + P_6 + P_7$
		$P_2 + 2P_5$	5	2, 5	1, 3, 4	$2P_1 + P_3 + P_4 + P_6 + P_7$
1, 2	$2P_1 + P_2$	$P_3 + P_4 + P_7$	3	4, 7	3, 5, 6	$2P_1 + P_2 + P_3 + P_5 + P_6$
			4	3, 7	4, 5, 6	$2P_1 + P_2 + P_4 + P_5 + P_6$
			7	3, 4	5, 6, 7	$2P_1 + P_2 + P_5 + P_6 + P_7$
1, 3	$2P_1 + P_3$	$P_5 + 2P_6$	6	5, 6	3, 4, 7	$2P_1 + P_2 + P_3 + P_4 + P_7$
		$P_2 + P_4 + P_6$	2	4, 6	2, 5, 7	$2P_1 + P_2 + P_3 + P_5 + P_7$
			4	2, 6	4, 5, 7	$2P_1 + P_3 + P_4 + P_5 + P_7$
			6	2, 4	5, 6, 7	$2P_1 + P_3 + P_5 + P_6 + P_7$
		$2P_5 + P_7$	5	5, 7	2, 4, 6	$2P_1 + P_2 + P_3 + P_4 + P_6$

Of the fifteen elements of Λ_1 , there are four which do not appear in the last column of Table IV, eight which appear once, and three which appear twice. Taking also the column with the values of f' into consideration, we see that $w_1 = 4$, $w_2 = 8$, and $w_3 = 3$, and that $w_i = 0$ for $i > 3$. Let t_i be the number of equivalence classes in Λ which have exactly i elements. We have

$$t_1 = \#\Lambda^* + \sum_{f=1}^7 w_1(f) = \#\Lambda^* + 7w_1 = 7 + 7 \cdot 4 = 35,$$

$$t_2 = \frac{1}{2} \sum_{f=1}^7 w_2(f) = 7w_2/2 = 7 \cdot 8/2 = 28,$$

$$t_3 = \frac{1}{3} \sum_{f=1}^7 w_3(f) = 7w_3/3 = 7 \cdot 3/3 = 7,$$

$$t_i = 0 \text{ for } i > 3.$$

Hence $t = \sum_i t_i = 35 + 28 + 7 = 70$. But $h = \#Pic_6(\mathcal{X}_1) = 71$. Hence there is a unique divisor G of degree 6 (up to linear equivalence) which

satisfies (4). We claim that $2Q$ is such a divisor. To prove this, let τ be the automorphism mentioned at the beginning of the proof of Lemma 13. The group $\langle \tau \rangle$ acts on the set $\text{Div}(\mathcal{X}_1)$ of divisors on $\mathcal{X}_1$. Observe that $\tau Q = Q$, and that for any $E \in \Lambda$, the $\langle \tau \rangle$ -orbit of E is contained in Λ and has order 7. Now suppose that $2Q \sim E$, for some $E \in \Lambda$. Then $2Q = \tau^i 2Q \sim \tau^i E$, for all i . Hence the seven divisors in the $\langle \tau \rangle$ -orbit of E are equivalent elements in Λ . But we have just seen that all equivalence classes in Λ have less than seven elements, a contradiction. This completes the proof of Proposition 16.

Remark 27 We find a basis of $L(2Q)$ and a generator matrix of $C_L(\mathcal{X}_1, D_1, 2Q)$ as follows. Define the forms

$$H_4 := x^2 + y^2 + z^2 + xy,$$

$$H_5 := y^2 + z^2 + xy + xz + yz$$

Then

$$H_4 \cdot \mathcal{X}_1 = P_3 + P_5 + 3P_7 + Q,$$

$$H_5 \cdot \mathcal{X}_1 = P_4 + 3P_5 + P_6 + Q.$$

Define the following rational functions on $\mathcal{X}_1$:

$$f_1 := \frac{H_5 L_1 L_3}{H_1},$$

$$f_2 := \frac{L_3 L_5 L_7^2}{H_4^2},$$

$$f_3 := \frac{L_4 L_5^2 L_6}{H_5^2}.$$

The form H_1 was already defined in Remark 26. The lines L_3, L_5, L_7 and L_4, L_5, L_6 are the three lines through P_7 and P_5 , respectively, see Table III. Now $\{1, f_1, f_2, f_3\}$ is a basis of $L(2Q)$. Indeed, we have

$$(1) = 0,$$

$$(f_1) = P_1 + 2P_3 + P_4 + 2P_5 - 2Q,$$

$$(f_2) = 2P_1 + P_2 + P_4 + 2P_6 - 2Q,$$

$$(f_3) = P_1 + 2P_2 + P_3 + 2P_7 - 2Q.$$

Hence $1, f_1, f_2, f_3$ are in $L(2Q)$. It is easily verified that they are linearly independent. This basis of $L(2Q)$ gives the generator matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

of $C_L(\mathcal{X}_1, D_1, 2Q)$. This should be a generator matrix of a $[7, 4, 3]$ code, and indeed it is.

Remark 28 In Proposition 16 we gave a SAG representation of an $H(3, 2)$: $C_L(\mathcal{X}_1, D_1, 2Q)$ is an $H(3, 2)$. As pointed out in Remark 1, we can also give this code as a $C_\Omega(\mathcal{X}_1, D_1, G)$ code, equivalently. Following (an adjusted version of) the proof of [4, Ch.8, Prop.8, p.207], we find that the differential

$$\omega_1 := \frac{z^3}{y^2z + x^3 + x^2z} d\left(\frac{x}{z}\right)$$

has divisor

$$(\omega_1) = L_2 \cdot \mathcal{X}_1 = 2P_2 + P_4 + P_6.$$

Note that $y^2z + x^3 + x^2z$ is the partial derivative to y of the left-hand side of (3). Define the forms

$$H_6 := y^3 + yz^2 + y^2z + x^2z + xyz,$$

$$H_7 := y^3 + z^3 + x^2z + y^2z + xyz.$$

Then

$$H_6 \cdot \mathcal{X}_1 = P_1 + P_4 + Q + U,$$

$$H_7 \cdot \mathcal{X}_1 = P_4 + P_6 + V,$$

where U is the place of degree 7 on $\mathcal{X}_1$ corresponding to the orbit of the $\mathbf{F}_{128}$ -rational point $(\gamma^{90} : \gamma^{23} : 1)$, with $\text{Gal}(\mathbf{F}_{128}/\mathbf{F}_2)$ acting, and where V is the place of degree 10 on $\mathcal{X}_1$ corresponding to the orbit of the $\mathbf{F}_{1024}$ -rational point $(\delta^{1017} : \delta^{159} : 1)$, with $\text{Gal}(\mathbf{F}_{1024}/\mathbf{F}_2)$ acting. Here $\mathbf{F}_{128} = \mathbf{F}_2(\gamma)$ with

$$\gamma^7 + \gamma^3 + 1 = 0,$$

and $\mathbf{F}_{1024} = \mathbf{F}_2(\delta)$ with

$$\delta^{10} + \delta^3 + 1 = 0.$$

Put

$$\omega := \frac{H_6 H_7}{H_1 L_2 L_4} \omega_1.$$

Then $(\omega) = U + V - 2Q - D_1$. Hence ω has a simple pole at P_i and $\text{res}_{P_i}(\omega) = 1$ for $i = 1, \dots, 7$. Define G'_1 by

$$2Q = (\omega) - G'_1 + D_1.$$

Then

$$G'_1 = U + V - 4Q.$$

By Remark 1, $C_{\mathbf{N}}(\mathcal{X}_1, D_1, G'_1) = C_L(\mathcal{X}_1, D_1, 2Q)$.

Remark 29 We have proved that $H(3, 2)$ has exactly two (WAG) representation classes with $g = 3$. These are

$$[(\mathcal{X}_1, D_1, G_1)] \quad \text{and} \quad [(\mathcal{X}_1, D_1, 2Q)],$$

the latter of which is SAG, moreover. This shows that $H(3, 2)$ has more than one WAG representation class.

Lemma 14 *The (SAG) representation $(\mathcal{X}_1, D_1, 2Q)$ is minimal.*

Proof: As noted already at the beginning, after Remark 24, the genus $g(\mathcal{X}_1) = 3$ is minimal. The divisor $2Q$ is base point free, since its degree is $6 \geq 2g$, see [11, p.308, Cor.3.2]. The result follows by Corollary 2.

If $[(\mathcal{X}, D, G)]$ is a *minimal* AG representation class of $H(3, 2)$, then $(g, m) = (3, 6)$ or $(4, 6)$, by Proposition 12 and Table II. By Lemma 8, Lemma 10 and Proposition 16, there is exactly one AG representation class of $H(3, 2)$ with $(g, m) = (3, 6)$, $[(\mathcal{X}_1, D_1, 2Q)]$ namely, and this representation class is *minimal* by Lemma 14. From the next proposition it follows that there exists no AG representation class of $H(3, 2)$ with $(g, m) = (4, 6)$ (minimal or not). To avoid any misunderstandings: the definitions of P_i, Q , etc., that we used until now do not apply to Proposition 17 and Lemma 15 and their proofs.

Proposition 17 *If $(\mathcal{X}, D, G)$ is an AG representation of an $H(3, 2)$, then $g(\mathcal{X}) \neq 4$.*

Proof: Suppose that $(\mathcal{X}, D, G)$ is an AG representation of a binary $[7, 4, 3]$ code, $D = P_1 + \cdots + P_7$ and $g = 4$. The curve $\mathcal{X}$ is not hyperelliptic, since it has more than $2q + 2 = 6$ $\mathbf{F}_2$ -rational points. If $m = \deg(G) < 6$, then

$$4 = l(G) = m + 1 - 4 + i(G),$$

hence $i(G) > 0$, by the Riemann-Roch theorem. But then

$$l(G) \leq 1 + m/2 < 4,$$

by Clifford's theorem. Hence $m \geq 6$, and because $m < n = 7$, $m = 6$. So G is a divisor of degree $2g - 2$ with $i(G) = 1$. In other words, G is a canonical divisor, that is to say, $G = (\omega_0)$ for a differential ω_0 . Because $C_L(\mathcal{X}, D, G)$ is an $H(3, 2)$, $C_\Omega(\mathcal{X}, D, G)$ is an $S(3, 2)$, see Proposition 1 and Definition 5. By Remark 1, there is a divisor G' on $\mathcal{X}$ such that $G' \sim D$ and

$$C_L(\mathcal{X}, D, G') = C_\Omega(\mathcal{X}, D, G).$$

So $C_L(\mathcal{X}, D, G')$ is a $[7, 3, 4]$ code.

We claim that

$$l(P_a + P_b + P_c) = 1, \quad \text{for all } a, b, c \in \{1, \dots, 7\} \\ \text{with } a \neq b, a \neq c, b \neq c.$$

The proof of this claim is actually more or less the reverse of the proof of Proposition 15. Namely, write $G' = D + (f_0)$. Suppose that $l(P_a + P_b + P_c) > 1$ for some distinct a, b, c . Let $f_1 \in L(P_a + P_b + P_c)$ with $f_1 \neq 0, 1$. Put $f := f_1/f_0$. Then $f \notin L(G' - D) = L((f_0))$, since otherwise $(f) = -(f_0)$, and consequently $f_1 = f f_0 = 1$, which gives a contradiction by the choice of f_1 . On the other hand,

$$f f_0 = f_1 \in L(P_a + P_b + P_c)$$

implies that

$$(f) \geq -(f_0) - P_a - P_b - P_c = -G' + P_r + P_s + P_t + P_u,$$

where

$$P_r + P_s + P_t + P_u = D - P_a - P_b - P_c.$$

It follows that $f \in L(G') \setminus L(G' - D)$, and that $\alpha_L(f)$ is a nonzero codeword in $C_L(\mathcal{X}, D, G')$ of weight at most three, see Definition 1.

This contradicts the fact that $C_L(\mathcal{X}, D, G')$ is a $[7, 3, 4]$ code.

We proceed with the proof of the proposition. Suppose that there exists an effective divisor E of degree three on $\mathcal{X}$ with $l(E) > 1$. Then E is obviously base point free, see Lemma 11. The morphism

$$\varphi_E: \mathcal{X} \rightarrow \mathbf{P}^1$$

has degree three, see Remark 6, and we have $\varphi_E(P_i) \in \mathbf{P}^1(\mathbf{F}_2)$ for $i = 1, \dots, 7$. Since $\#\mathbf{P}^1(\mathbf{F}_2) = 3$, there exists a $Q \in \mathbf{P}^1(\mathbf{F}_2)$ with at least three points in

$$\varphi_E^{-1}(Q) \cap \{P_1, \dots, P_7\},$$

P_a, P_b, P_c say. Since $\deg(\varphi_E) = 3$, the pull back $\varphi_E^*(Q)$ of Q under φ_E is equal to $P_a + P_b + P_c$, see [11, p.138, Prop.6.9]. This implies that $E \sim P_a + P_b + P_c$, and hence that

$$l(P_a + P_b + P_c) = l(E) > 1.$$

But this contradicts the previous claim. We conclude that $l(E) = 1$ for all $E \geq 0$ with $\deg(E) = 3$. By the following lemma, however, this is not true, and hence the assumption that $(\mathcal{X}, D, G)$ is an AG representation of an $H(3, 2)$ is wrong. This proves the proposition.

Lemma 15 *If $\mathcal{X}$ is a nonsingular, absolutely irreducible curve over $\mathbf{F}_2$ of genus 4 with at least seven $\mathbf{F}_2$ -rational points, then there exists an effective divisor E on $\mathcal{X}$ with $\deg(E) = 3$ and $l(E) = 2$.*

Proof: Let K be a canonical divisor and let P_0 be a rational point on $\mathcal{X}$. We have $l(K) = 6 + 1 - 4 + 1 = 4$, by Riemann-Roch. Put $G := K - P_0$. Then $\deg(G) = 5$. Since $\mathcal{X}$ is not hyperelliptic, K is very ample, see [11, p.341, Prop.5.2]. Hence $l(G) = l(K) - 1 = 3$ and G is base point free. Let $\mathcal{X}_0$ be the reduced image of $\mathcal{X}$ under the morphism

$$\varphi_G: \mathcal{X} \rightarrow \mathbf{P}^2,$$

see Remark 6. We have

$$5 = \deg(\varphi_G) \cdot \deg(\mathcal{X}_0).$$

As pointed out in Remark 6, $\mathcal{X}_0$ is not a line, hence $\deg(\mathcal{X}_0) \neq 1$. It follows that $\deg(\mathcal{X}_0) = 5$, and that $\deg(\varphi_G) = 1$, i.e., φ_G is a birational morphism. We have

$$4 = g(\mathcal{X}) = g(\mathcal{X}_0) = \frac{1}{2}(5-1)(5-2) - \sum_{P \in \mathcal{X}_0} \delta_P \deg(P), \quad (5)$$

where δ_P is the delta invariant at P , see [18]. We have

$$\delta_P \geq m_P(m_P - 1)/2,$$

where m_P is the multiplicity of $\mathcal{X}_0$ at P . From (5) it follows that

$$\sum_{P \in \mathcal{X}_0} \delta_P \deg(P) = 2. \quad (6)$$

Hence $\mathcal{X}_0$ has two rational singular points, each with delta invariant 1, or one rational point with delta invariant 2, or one singular point of degree two with delta invariant 1. In every case, the singular point(s) have multiplicity 2 (since if $m_P \geq 3$, then $\delta_P \geq 3(3 - 1)/2 = 3$, contradicting (6)).

We claim that $\mathcal{X}_0$ has a rational singular point. To prove this, suppose $\mathcal{X}_0$ has no such point. Then $\mathcal{X}_0$ has a singular point Q of degree 2. There is exactly one line through Q in $\mathbf{P}^2$, defined over $\mathbf{F}_2$. We call this line L_1 . By Bézout's theorem, L_1 intersects $\mathcal{X}_0$ at 5 points, counted with multiplicities. The intersection multiplicity at Q is even and at least 4, hence equal to 4, and there is exactly one rational point P_1 in $L_1 \cap \mathcal{X}_0$. Let L_2 and L_3 be the other two lines through P_1 in $\mathbf{P}^2$, defined over $\mathbf{F}_2$. Then φ_G maps every rational point of $\mathcal{X}$ to a rational point in

$$(L_1 \cap \mathcal{X}_0) \cup (L_2 \cap \mathcal{X}_0) \cup (L_3 \cap \mathcal{X}_0).$$

But $L_1 \cap \mathcal{X}_0$ contains exactly one rational point, P_1 namely, and $L_2 \cap \mathcal{X}_0$ and $L_3 \cap \mathcal{X}_0$ each contain at most two rational points not equal to P_1 . Hence φ_G maps (at least) 7 rational points of $\mathcal{X}$ to at most 5 rational points of $\mathcal{X}_0$. Thus there are two rational points Q_1, Q_2 on $\mathcal{X}$ such that

$$\varphi_G(Q_1) = \varphi_G(Q_2),$$

and $\varphi_G(Q_1)$ is a rational singular point of $\mathcal{X}_0$, a contradiction. This proves the claim.

Thus $\mathcal{X}_0$ is a plane model of degree 5 of $\mathcal{X}$, with at least one rational singular point, which we call Q_0 . As noted earlier in the proof, the multiplicity of $\mathcal{X}_0$ at Q_0 is 2. Hence there is an effective divisor B of degree 2 such that

$$\mathcal{X}_0 \cdot M \geq B$$

for every line M through Q_0 , defined over $\mathbf{F}_2$. Besides Q_0 , there is at least one other rational point on $\mathcal{X}_0$, since otherwise φ_G would map (at

least) 7 rational points of $\mathcal{X}$ to Q_0 , and $m_{Q_0} \geq 7 > 2$, a contradiction. Let Q'_0 be such another rational point on $\mathcal{X}_0$. Let M_1 be the line through Q_0 and Q'_0 , and let M_2 be one of the other two lines through Q_0 defined over $\mathbf{F}_2$. Then $Q'_0 \notin M_2$. Put

$$R_i := \mathcal{X}_0 \cdot M_i - B$$

for $i = 1, 2$. Then $R_i \geq 0$ and $\deg(R_i) = 3$ for both i . Put

$$f := M_2/M_1.$$

Then

$$\begin{aligned} f &\in L(R_1), \\ (f) &= R_2 - R_1, \end{aligned}$$

and f is not a constant, since it has a pole at Q'_0 . Hence $l(R_1) \geq 2$. In fact, we have equality, by Riemann-Roch and Clifford's theorem. To prove the proposition, choose $E := \varphi_G^*(R_1)$, the pull back of R_1 under φ_G .

We summarize our main results concerning the Hamming codes in the following theorem.

Theorem 6 *a) $H(1, q)$ and $H(2, q)$ are SAG, for every q .*

b) $H(3, 2)$ is SAG.

c) $H(r, q)$ is not AG if $r \geq 3$ and $(r, q) \neq (3, 2)$.

d) $[(\mathcal{X}_1, D_1, 2Q)]$ is a minimal SAG representation class of $H(3, 2)$.

e) $[(\mathcal{X}_1, D_1, 2Q)]$ is the only minimal AG representation class of $H(3, 2)$. Here $\mathcal{X}_1$ is defined by (3), D_1 is defined after Lemma 8, and Q is defined in Remark 26.

B. Another Example

Let C be the binary $[6, 4, 2]$ -code with generator matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

This code was mentioned by Katsman and Tsfasman in [13], where they more or less raised the question whether this code is algebraic-geometric. We shall give the answer to this question here.

Proposition 18 *The code C is SAG.*

Proof: Let $\mathcal{X}_2$ be the plane projective curve of degree four defined over $\mathbf{F}_2$ by the equation

$$xyz^2 + (x^3 + x^2y + y^3)z + x^3y + xy^3 = 0.$$

As is easily verified, this curve has exactly one singularity: the point $P := (0 : 0 : 1)$ is an ordinary double point. The tangents to $\mathcal{X}_2$ at P are $x = 0$ and $y = 0$. It follows, by Bézout's theorem, that $\mathcal{X}_2$ is absolutely irreducible. The curve is a hyperelliptic curve, of genus 2. Besides the singular point P , there are four other rational points on $\mathcal{X}_2$:

$$P_1 := (1 : 0 : 0),$$

$$P_2 := (1 : 1 : 0),$$

$$P_3 := (0 : 1 : 0),$$

and

$$P_4 := (1 : 1 : 1).$$

The singular point P gives two rational points on the nonsingular model of $\mathcal{X}_2$. Or, to put it differently, it corresponds to two places (=discrete valuation rings) of degree one in the function field of $\mathcal{X}_2$ over $\mathbf{F}_2$. We call these places P_5 and P_6 . Let L and M be the lines $z = 0$ and $x + y + z = 0$, respectively. The line L is the tangent to $\mathcal{X}_2$ at P_2 . We have

$$L \cdot \mathcal{X}_2 = P_1 + 2P_2 + P_3.$$

The only rational point in $M \cap \mathcal{X}_2$ is P_2 , and the intersection at this point is transversal. Define the divisor G_2 by

$$M \cdot \mathcal{X}_2 = P_2 + G_2.$$

Then $G_2 \geq 0$, $\deg(G_2) = 3$ and $\text{supp}(G_2) \cap \text{supp}(D_2) = \emptyset$. Here we have put $D_2 := P_1 + P_2 + \cdots + P_6$. (Although we do not need this, it follows that G_2 is a place of degree three. This place turns out to be

$$\{(1 : \alpha : 1 + \alpha), (1 : \alpha^2 : 1 + \alpha^2), (1 : \alpha^4 : 1 + \alpha^4)\},$$

where $\mathbf{F}_8 = \mathbf{F}_2(\alpha)$ with $\alpha^3 + \alpha + 1 = 0$).

By the Riemann-Roch theorem, $l(G_2) = 3 + 1 - 2 = 2$. The rational functions 1 and $L/M (= z/(x + y + z))$ are in $L(G_2)$, and they (obviously) form a basis. This basis of $L(G_2)$ gives

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

as a generator matrix of the binary $[6,2,3]$ code $C_L(\mathcal{X}_2, D_2, G_2)$. Since

$$2 = 2g - 2 < 3 = \deg(G_2) < 6 = n,$$

$C_L(\mathcal{X}_2, D_2, G_2)$ is SAG. Since

$$C_L(\mathcal{X}_2, D_2, G_2)^\perp = C,$$

C is SAG too, by Corollary 1. This completes the proof of the proposition.

Acknowledgment

We would like to thank A.B. Sørensen, for the use of his zeta function computer program, and C. Faber for an indication of the result of Lemma 7.

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NEDERLANDSE SAMENVATTING

Overdekkende Codes, Perfecte Codes en Codes van Algebraïsche Krommen

Dit proefschrift bestaat uit een achttal artikelen en een inleidend hoofdstuk. De in deze artikelen behandelde onderwerpen vallen binnen de wiskundige onderzoeksgebieden die bekend staan als de combinatoriek en de algebraïsche coderingstheorie. De gebruikte methoden en technieken zijn voornamelijk van combinatorische, algebraïsche of algebraïsch-meetkundige aard.

Het eerste artikel gaat over ondergrenzen voor het aantal codewoorden in overdekkende codes. Door middel van telargumenten komen we tot een significante en tamelijk algemene verbetering van de bolbedekkingsgrens, welke een bekende ondergrens is voor $K(n, R)$, het minimale aantal codewoorden in een binaire code van lengte n met overdekkingsstraal R . Als gevolg hiervan verkrijgen we een nieuwe oneindige rij van exact bepaalde waarden van $K(n, R)$:

$$K(2^r, 1) = 2^{(2^r - r)} \quad \text{voor } r = 1, 2, \dots$$

Hiermee samenhangend wordt een vermoeden over perfecte codes uitgesproken.

In het tweede artikel generaliseren we de begrippen normaal en subnormaal naar codes over willekeurige alfabetten. We onderzoeken welke q -aire codes normaal of subnormaal zijn. We bewijzen dat een q -aire perfecte code normaal is dan en slechts dan als hij binair of triviaal is. We leiden een algemene ondergrens af voor het aantal codewoorden in een ternaire normale code.

Deels gebruikmakend van de in het eerste artikel geïntroduceerde technieken wordt in het derde artikel de non-existentie van zekere zogenaamde 'gemengde' perfecte codes bewezen.

In het vierde artikel wordt voor het geval $R = 1$ het vermoeden van Cohen, Lobstein en Sloane, dat zich onder de codes van lengte n met overdekkingsstraal R en $K(n, R)$ codewoorden een normale code bevindt, opgelost. Er wordt zelfs bewezen dat *alle* dergelijke codes normaal zijn als $R = 1$. Tevens bewijzen we dat iedere binaire code met minimumafstand tenminste tweemaal de overdekkingsstraal normaal is, en generaliseren we de constructie van Frankl van abnormale binaire codes met overdekkingsstraal 1 naar willekeurige overdekkingsstralen.

In het vijfde en zesde artikel onderzoeken we hoe de resultaten uit het eerste artikel uit te breiden zijn tot codes over willekeurige alfabetten (q -air of gemengd). Bovendien komen nuttige variaties op het basisidee naar voren, en wordt gekeken naar toepassingen op foutenverbeterende codes.

Het zevende artikel gaat over zogenaamde 'perfect multiple coverings' (PMC's). Dit zijn codes die perfect zijn in de volgende zin: voor vaste getallen r en μ zijn er voor ieder woord precies μ codewoorden binnen Hamming afstand r . Het betreft hier een generalisatie van de gewone perfecte codes, welke we krijgen in het geval $\mu = 1$. We geven een korte analyse van het bestaan van PMC's. Hierbij beperken we ons voornamelijk tot het geval $r = 1$.

Tenslotte onderzoeken we in het achtste artikel welke lineaire codes verkregen kunnen worden via algebraïsche krommen over eindige lichamen door middel van Goppa's constructie. Het blijkt dat deze vraag slechts zinvol is als we een zekere beperking opleggen aan de graad van de betreffende divisor op de kromme, daar anders iedere lineaire code op deze wijze geconstrueerd kan worden. Met deze restrictie bewijzen we dat de q -aire Hamming codes met redundantie 1 of 2 en de binaire $[7,4,3]$ code de enige q -aire Hamming codes zijn die in deze zin algebraïsch meetkundig zijn. In het geval van de binaire $[7,4,3]$ code bewijzen we bovendien dat de constructie, dat wil zeggen het drietal (kromme, punten, divisor), in zekere zin uniek is.

CURRICULUM VITAE

De auteur van dit proefschrift werd geboren op 1 augustus 1963 te Leiderdorp. Het V.W.O.-diploma behaalde hij in 1981 aan het Bonaventuracollege te Leiden. Vervolgens studeerde hij één jaar Elektrotechniek aan de toenmalige Technische Hogeschool te Delft. Hierna, vanaf 1982, studeerde hij twee jaar Wiskunde en Informatica aan de Rijksuniversiteit te Leiden. Zijn studie Wiskunde (oude stijl) zette hij voort en rondde hij af met het candidaatsexamen in januari 1986, en het doctoraalexamen (cum laude) in december 1986. Hij studeerde af bij Prof. dr. R. Tijdeman op een combinatorisch probleem uit de coderingstheorie. Tijdens zijn studie was hij van 1 september 1984 tot 1 januari 1985 student-assistent. Sinds 1 juli 1987 is hij onder leiding van Prof. dr. J. H. van Lint als onderzoeker werkzaam binnen de vakgroep Discrete Wiskunde van de Faculteit Wiskunde en Informatica van de Technische Universiteit te Eindhoven, bezoldigd door de Nederlandse Organisatie voor Zuiver-Wetenschappelijk Onderzoek (ZWO), thans de Nederlandse Organisatie voor Wetenschappelijk Onderzoek (NWO).

STELLINGEN

behorende bij het proefschrift

'Covering Codes, Perfect Codes, and Codes from Algebraic Curves'

door G. J. M. van Wee

4 juni 1991, Technische Universiteit Eindhoven

I

Let $K_q(n, R)$ and $V_q(n, R)$ be defined as on pages 5 and 49 of this thesis. Let $A_q(n, d)$ be the maximum cardinality of a q -ary code of length n with minimum distance d . For every integer n with $n \equiv 5 \pmod{6}$ we have

$$K_2(n, 1) \geq \frac{2^n(5 + V_2(n, 2)) - 3(A_2(n, 3) - 1)(n + 1)}{(n + 1)V_2(n, 2)}.$$

This yields

$$K_2(11, 1) \geq 177, \quad K_2(17, 1) \geq 7416, \quad K_2(23, 1) \geq 352641, \\ \text{and} \quad K_2(29, 1) \geq 17985484,$$

where the lower bounds on $A_2(n, d)$ are taken from

A. E. Brouwer, J. B. Shearer, N. J. A. Sloane, and W. D. Smith, A new table of constant weight codes, *IEEE Trans. Inform. Theory* **36** (1990), 1334-1380.

II

Analogous to Theorem 1 in

G. D. Cohen, A. C. Lobstein, and N. J. A. Sloane, Further results on the covering radius of codes, *IEEE Trans. Inform. Theory* **32** (1986), 680-694,

we have

$$K_3(n, R) \geq \frac{3^n - A_3(n, 2R + 1) \binom{2R}{R}}{V_3(n, R) - \binom{2R}{R}}.$$

As a consequence,

$$K_3(6, 1) \geq 58, \quad K_3(7, 1) \geq 147, \quad K_3(8, 1) \geq 393, \quad K_3(9, 1) \geq 1048, \\ K_3(10, 1) \geq 2812, \quad K_3(11, 1) \geq 7767, \quad \text{and} \quad K_3(12, 1) \geq 21395,$$

where the upper bounds on $A_3(n, d)$ are taken from

R. J. M. Vaessens, E. H. L. Aarts, and J. H. van Lint, Genetic algorithms in coding theory - A table for $A_3(n, d)$. Preprint, 1990.

III

Let $n, R \in \mathbb{Z}$ with $n \geq 2R + 1 \geq 1$. Define ε as on page 10 of this thesis. Let

$$\begin{aligned}\beta_1 &:= (2n + 1 - 2R)(n - R + 2\varepsilon - 1), & \gamma_1 &:= \varepsilon(2n - R + \varepsilon - 1) \binom{n}{R}, \\ \beta_2 &:= (n + \varepsilon)(n - R) + \varepsilon, & \gamma_2 &:= \varepsilon n \binom{n}{R}, \\ \beta_3 &:= (n - R + 1)(n - R - 1 + 2\varepsilon), & \gamma_3 &:= \gamma_2, \\ \beta_4 &:= (n - R)(R - \varepsilon + 1) + \varepsilon, & \gamma_4 &:= \varepsilon(R - \varepsilon + 1) \binom{n}{R}, \\ \beta_5 &:= n - 2R + 2\varepsilon, & \gamma_5 &:= \varepsilon \binom{n}{R}, \\ \beta_6 &:= (n - 2R + 1 + \varepsilon)(n + \varepsilon - R - 1) + (\varepsilon - 1)(n - \varepsilon), & \gamma_6 &:= \varepsilon(n + \varepsilon - R - 1) \binom{n}{R}.\end{aligned}$$

Then

$$K_2(n, R) \geq \frac{2^n \beta_i}{V_2(n, R) \beta_i - \gamma_i},$$

for $i = 1, 2$, and for

$$i = \begin{cases} 3, 4 & \text{if } \varepsilon < (R + 1)/2, \\ 5 & \text{if } \varepsilon \geq (R + 1)/2, \\ 6 & \text{if } \varepsilon > (R + 1)/2, \end{cases}$$

improving several of the lower bounds of Table I in the first paper in this thesis. These improvements are listed in the following table*, where the marks a, b, c, d, e, f correspond to $i = 1, \dots, 6$, respectively.

n	$R = 2$	$R = 3$	$R = 4$	$R = 5$	$R = 6$	$R = 7$	$R = 8$	$R = 9$	$R = 10$
13	97 a	26 e							
14									
15			22 a						
16	511 c	112 a							
17		177 e			8 c				
18	1693 a	282 c		27 b					
19	2898 c								
20		891 a	202 a	56 e					
21	9897 b	1475 e	318 e	84 c	33 b				
22	17298 c	2454 c	506 c		47 a		9 a		
23			808 c		68 e				
24	60368 b	8108 b		363 a	100 a	40 a			
25	107210 c	13832 e	2540 b	569 f	149 c	57 a		11 c	
26		23675 c	4188 f	899 e	224 c	82 e		14 a	
27	380298 b		6942 c	1428 c		121 e	49 a		
28	683945 c	80629 b	11551 c		647 a	178 c	69 a		13 a
29		140567 e			1009 f		100 f		17 a
30	2457439 b	245672 c	38039 b	7152 b	1588 f	398 c	145 e	59 a	
31	4464778 c		64695 f	11751 f	2513 e		213 c	84 a	
32		855357 b	110442 c	19413 e	3998 e	1147 b	315 c	120 e	
33	16209225 b	1516050 e	189050 e	32191 c	6386 c	1783 f	470 c	174 e	72 a

* With thanks to F. C. Bussemaker for computing the entries.

IV

For all integers q and n with $q \geq 2$, $n \geq 2$, and $(n-2)(q-1) \not\equiv 0 \pmod{3}$, the following relation between $K_q(n, 1)$ and $K_q(n, 2)$ holds:

$$K_q(n, 2) \geq \frac{q^n + K_q(n, 1)}{V_q(n, 2) + 1}.$$

V

Let $n, q \in \mathbb{N}$ with $q \geq 2$. For $d, R_1, R_2 \in \{0, 1, \dots, n\}$ define

$$I(R_1, R_2, d) := |B_{R_1}(\mathbf{x}) \cap B_{R_2}(\mathbf{y})|,$$

where $\mathbf{x}, \mathbf{y} \in \mathbb{Z}_q^n$ such that $d(\mathbf{x}, \mathbf{y}) = d$. We have $I(R_1, R_2, d) =$

$$\sum_{r_1=0}^{R_1} \sum_{r_2=0}^{R_2} \sum_{a=0}^{r_1} \binom{d}{a} \binom{d-a}{r_1-r_2-2a+d} (q-2)^{r_1-r_2-2a+d} \binom{n-d}{r_2-d+a} (q-1)^{r_2-d+a}$$

(with the convention that $\binom{n}{k} = 0$ if $n < 0$ or $k < 0$ or $k > n$).

VI

Let C be a subnormal q -ary code of length n with covering radius R . Suppose that $n > R + q - 1$. Then

$$|C| \geq \frac{2q^n}{2V_q(n, R) - I(R, R, \lfloor qR/(q-1) \rfloor + 1)},$$

where the function I in the denominator is as previously defined.

VII

The question raised by Janwa whether the $[6, 4, 3]$ Hamming code over $\mathbb{F}_5$ can be constructed from an algebraic curve of genus 0, and more generally, whether a code from a curve of genus 0 with covering radius smaller than its redundancy exists at all, has been answered in the affirmative in the eighth paper of this thesis (Proposition 14).

H. Janwa, Some optimal codes from algebraic geometry and their covering radii, *Europ. J. Combin.* **11** (1990), 249-266.

VIII

Let C be a linear $[n, k, d]$ code over $\mathbb{F}_q$ with covering radius R . Let $C^\perp$ be the dual code of C and let $d^\perp$ be the minimum distance of $C^\perp$. Suppose that $d^\perp = q$ and $k < n - 1$. If $C^\perp$ attains the Griesmer bound, then $R < n - k$.

IX

If C is a linear $[n, k, d]$ code over F_q that is not MDS, then there exists a linear $[n - d^\perp, k - d^\perp + 1, d]$ code, where $d^\perp$ is the minimum distance of $C^\perp$, the dual code of C .

X

For $m \geq 4$, let $C(m)$ be the binary linear code of length $m(m-1)(m-2)/2$ that is generated by the characteristic vectors of the cutsets of the triangular graph $T(m)$ on $m(m-1)/2$ vertices. The proof given by Basart and Huguet that the minimum distance of $C(m)$ equals $2(m-2)$ is incomplete. Their proof that the number of codewords of weight $2(m-2)$ in $C(m)$ equals $m(m-1)/2$ is incorrect. Nevertheless, both statements are true.

J. M. Basart and L. Huguet, From $T(m)$ triangular graphs to single-error-correcting codes, in "Applicable Algebra, Error-Correcting Codes, Combinatorics and Computer Algebra" (Th. Beth and M. Clausen, Eds.), Proc. 4th Int. Conf. AAECC-4, Karlsruhe, Germany, 1986, Lecture Notes in Comp. Sc., vol. 307, pp. 5-12, Springer-Verlag, Berlin, 1988.

XI

Indien de aannames in het hoofdstuk "2. Waar de prestatie van afhangt" van het hieronder genoemde boek, met name in het deelhoofdstuk "2.4. Afzetmechanika," juist zijn, dan geldt het volgende: Een schaatser die tijdens een langebaanwedstrijd op een 400 meter ijsbaan onder constante omstandigheden en met een constante techniek (d.w.z. een constante hoeveelheid effectieve arbeid per afzet, een constante zit, een constante romphouding, etc.) achter elkaar steeds exact dezelfde rondetijden realiseert, rijdt, afgezien van fluctuaties tijdens het doorlopen van achtereenvolgens inzet-, glij- en afzetfase, niet met een constante snelheid.

H. Gemser en G. J. van Ingen Schenau, "Handboek Wedstrijdschaatsen," Eisma B.V., Leeuwarden, 1987.

XII

Gezien het feit dat het ter vervanging van het oude stelsel van wetenschappelijk assistenten in 1986 aan de Nederlandse universiteiten ingevoerde stelsel van assistenten- en onderzoekers in opleiding (het AIO/OIO-stelsel) gebaseerd is op de verkorting tot vier jaar van de nominale duur van op of na 1 september 1982 aangevangen universitaire studies, gelet op de nominale duur van minstens vijf jaar van studies die vóór genoemde datum aanvingen, gelet in het bijzonder op de nominale afstudeerdatum van alumni van de lichting 1981 in relatie tot de datum van invoering van het AIO/OIO-stelsel, en gelet op het aanzienlijke verschil tussen de arbeidsvoorwaarden van de AIO of OIO en die van hun voorgangers, kan men niet anders concluderen dan dat onrecht is gedaan aan de belangen van eerder genoemde groep alumni.